Desarrollo de una herramienta en Python para el postprocesado y comparación de múltiples casos transitorios pertenecientes a un modelo térmico diseñado en ESATAN-TMS

Trabajo de Fin de Grado

ESPECIALIDAD: Propulsión Aeroespacial

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Acronyms

1. Abstract

This Bachelor's Thesis focuses on the development of a Python-based post-processing tool for the thermal analysis of space systems using results generated with ESATAN-TMS.

Building upon a previous study limited to the post-processing of steady-state radiative cases, this work extends the analysis to fully transient thermal simulations. In addition, the tool has been designed to handle and compare multiple simulation cases associated with the same thermal model, allowing a more complete assessment of the system response under different operating conditions and orbital scenarios.

Since ESATAN-TMS presents limitations regarding integrated data visualization and case-to-case comparison, this project implements custom post-processing routines in Python, using libraries such as NumPy, Pandas, and Matplotlib. These routines automate the extraction, organization, and visualization of nodal results, including temperature evolution curves, transient heat flows, and comparative plots between different cases.

The resulting methodology improves the interpretation of complex transient thermal results and provides a more flexible framework for evaluating the thermal performance of a system throughout realistic orbital thermal cycles. By enabling the simultaneous analysis of several cases belonging to the same model, the developed tool supports a more efficient and systematic validation of thermal behaviour in space applications.

2. Introduction

Thermal management is an essential part of spacecraft design, as the temperature of each component must remain within allowable limits throughout the mission.

Since both the space environment and the operating conditions of the spacecraft evolve across different mission phases, steady-state cases cannot fully represent the time-dependent thermal response. Transient simulations are therefore needed to examine the evolution of temperatures and heat flows.

The design of a spacecraft thermal control system relies on numerical analysis tools to predict system behaviour under different mission and operating conditions. These tools are used to represent the spacecraft through a thermal model and to calculate variables such as nodal temperatures and heat flows. Some of the best-known tools include THERMICA, Thermal Desktop together with SINDA/FLUINT, Simcenter 3D Space Systems Thermal (SST) and ESATAN-TMS, which provides a specialised environment for the development and solution of spacecraft thermal models.

The amount of information generated by these simulations can become considerable, particularly when transient analyses or multiple cases of the same model are considered. In such situations, the results must be organised and represented in a way that allows their evolution to be examined and case-dependent effects to be identified. Dedicated post-processing tools can therefore support the interpretation of the simulation results.

This Bachelor's Thesis builds on an existing Python-based post-processing tool initially intended for steady-state analysis. The work combines the adaptation of the tool to transient and multi-case analysis with the development of a representative spacecraft thermal model in ESATAN-TMS. Selected thermal results are also used to perform a simplified electrical power analysis and provide the inputs for a separate battery model. The specific objectives of the project are presented in the following section.

2.1. Objectives

The main objective of the project is to enable the existing application to process, analyse, and compare several transient cases associated with the same ESATAN-TMS thermal model within a common environment.

Achieving this objective requires the coordinated development of the thermal model and the post-processing tool. The ESATAN-TMS model provides a consistent set of transient results, while the application is adapted to manage and compare them.

The thermal results are also used to estimate the electrical power generated by the solar panels and to calculate the electrical power balance. These calculations provide the inputs for a separate simplified battery model.

Accordingly, the specific objectives of the project are:

- Develop a representative spacecraft thermal model and define transient cases under different environmental and operating conditions.
- Adapt the existing application to load and manage several transient result files while retaining a common model hierarchy.
- Implement methods for representing and comparing temperatures, thermal loads, heat flows and exchanged thermal energy.
- Generate graphical outputs and summary tables, and allow the corresponding analyses and configurations to be saved, restored and exported.
- Estimate solar-panel electrical generation, calculate the electrical power balance and use the resulting power profiles as inputs to a separate simplified battery model.
- Apply the complete analysis process to the selected transient cases and compare the thermal responses obtained under different environmental and operating conditions.

3. State of the Art

Spacecraft thermal analyses are commonly performed using specialised software tools, including THERMICA, Thermal Desktop (TD) coupled with SINDA/FLUINT, SST and ESATAN-TMS [5–7]. The result files generated by ESATAN-TMS contain a considerable amount of data that must be organised, examined, and interpreted to evaluate the thermal behaviour of the model. Two thermal regimes are considered in the analysis. A steady-state analysis provides a time-independent thermal solution for a predefined set of boundary conditions, whereas a transient analysis describes the evolution of the calculated results throughout the simulated period [8, 9].

An interactive Python-based post-processing tool facilitates the interpretation of these results by extracting and organising the data generated by ESATAN-TMS and presenting them through numerical and graphical representations [1]. The initial implementation was tested primarily with steady-state results from HELIOS, a thermal model developed in ESATAN-TMS, and therefore focuses on a single steady-state case. The extended application processes transient results and compares several cases associated with the same thermal model.

3.1. Post-Processing Capabilities in ESATAN-TMS

ESATAN-TMS supports the numerical solution of spacecraft thermal models under both steady-state and transient conditions. For each analysis case, the resulting dataset contains nodal temperatures, applied thermal loads, and conductive and radiative heat flows. The software provides numerical and graphical tools for examining these results [9, 10].

The structure of the exported information depends on the selected outputs and the type of analysis. Transient results associate each variable with successive output times, and heat-flow data identify the nodes or surfaces involved in each exchange. In every case, numerical identifiers must remain linked to the physical elements they represent [9, 10].

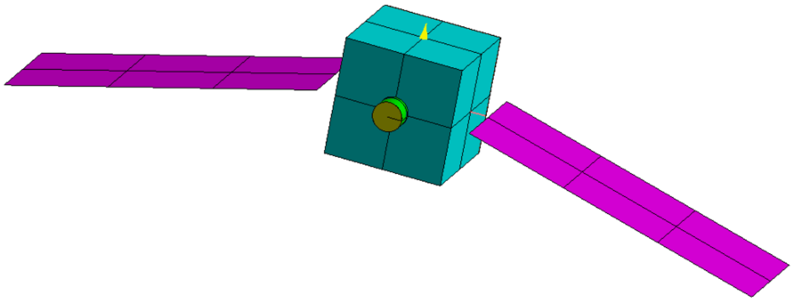
These datasets provide the information required to evaluate the thermal behaviour of the model. When transient analyses or several simulation cases are considered, however, direct inspection can become less practical. Relevant results must therefore be selected, organised, and compared consistently.

The post-processing functions available in ESATAN-TMS may not cover every requirement of a particular analysis. The Python implementation complements these functions by reorganising the exported thermal data according to an engineering hierarchy and providing additional methods for selecting, grouping, and representing the results [9].

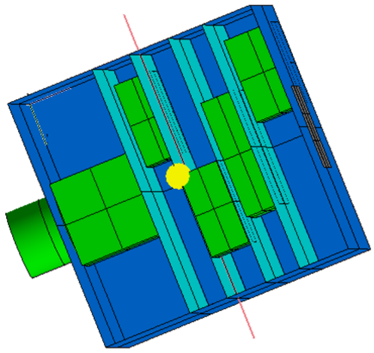
3.2. Steady-State Thermal Model: HELIOS

HELIOS is a thermal model developed in ESATAN-TMS with a single steady-state analysis case. The model supports the development and testing of the initial Python-based tool. As a first approximation of a spacecraft thermal system, it provides sufficient detail to generate a consistent set of thermal results. The model contains the spacecraft structure, solar panels, batteries, radiating surfaces, and multi-layer insulation. The different components are connected through the conductive and radiative couplings that define the thermal network of the model [1].

The complete spacecraft configuration and a view of the internal components are shown in Figure 3.1.



(a) External view of the satellite model.



(b) Internal view of the satellite model.

Figure 3.1: External and internal views of the HELIOS thermal model. Taken from [1].

The steady-state solution of HELIOS contains the thermal data that are extracted and organised by the initial Python code before being examined through the graphical interface. The application developed for this thesis uses a separate spacecraft thermal model with several cyclic transient analysis cases.

3.3. Python-Based Post-Processing of Steady-State Cases

The initial Python-based implementation uses the steady-state results generated for the HELIOS model in ESATAN-TMS [1]. In this type of analysis, a single temperature is calculated for each node under the selected boundary conditions, while the corresponding thermal loads and heat flows remain independent of time.

The imported results are linked to the hierarchy defined for the thermal model, so that each node can be located within the corresponding component and subsystem. Instead of working directly with the complete list of node identifiers contained in the ESATAN-TMS files, the user can examine the results through the spacecraft structure. This relationship between the thermal data and the physical model helps identify the origin of each value and interpret it in an engineering context.

The main modules of this codebase and their connections are shown in Figure 3.2.

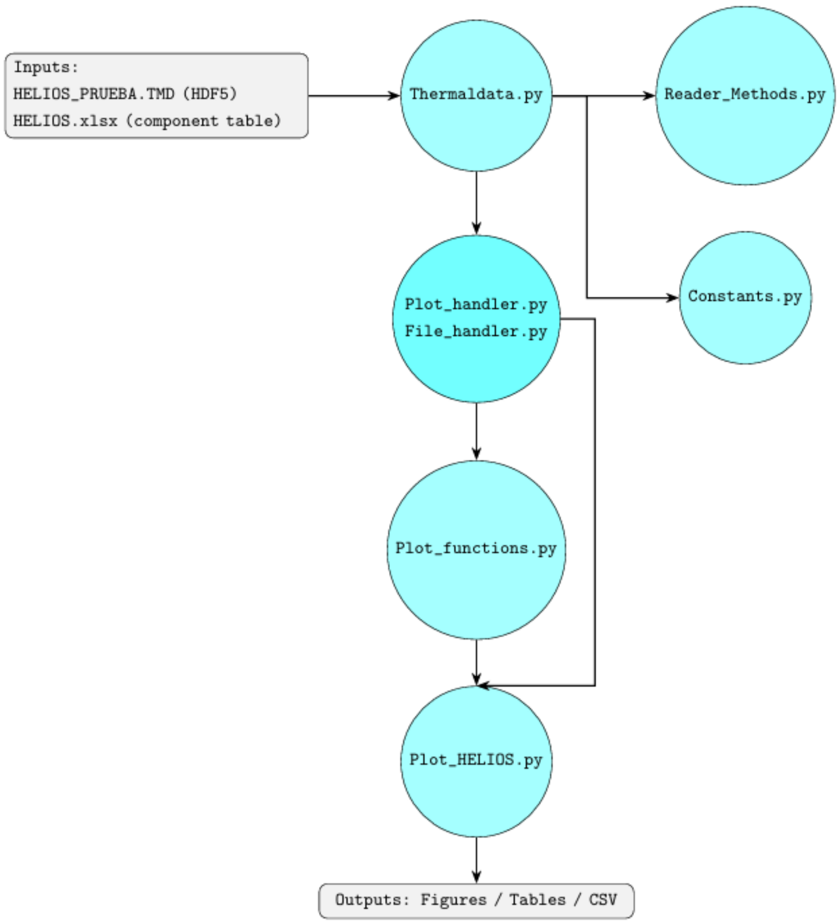


Figure 3.2: Architecture of the initial post-processing implementation. Taken from [1].

The architecture separates result reading, data organisation and representation. `Thermaldata.py` imports the HELIOS result file and the external component table, while the supporting modules handle data selection, file management and plotting. The interface is provided by `Plot_HELIOS.py`, from which figures, tables and data files can be generated [1].

This implementation is suitable for examining the thermal results of an individual steady-state case. However, it does not provide complete post-processing of transient results or allow several simulation cases to be loaded and compared simultaneously. Extending the same structure to time-dependent results and multiple cases requires changes to both data management and representation.

3.4. Motivation for the Present Work

Transient simulations associate each calculated value with a specific simulation time. The post-processing tools must preserve this correspondence when results are selected, represented, and compared, particularly when several components or interfaces are considered together.

During an orbital cycle, transitions between solar illumination and eclipse and changes in internal dissipation produce variations in the thermal response of the spacecraft. These variations must therefore be examined throughout the orbital cycle to determine when they occur and which elements are most affected.

Comparing simulation cases introduces an additional requirement. Spacecraft thermal analyses may include several cases representing different environmental and operating conditions. Processing them separately requires equivalent elements to be selected again for every case and prevents their direct comparison within the same representation. Managing cases associated with the same thermal model together allows equivalent selections to be applied and the differences between them to be calculated consistently.

For these reasons, the application is extended to process transient results and manage several simulation cases associated with the same thermal model. This allows equivalent components and interfaces to be compared throughout the simulated period and the intervals in which the differences between cases are most significant to be identified.

4. Theoretical Background

Spacecraft temperatures are determined by the interaction between internal heat dissipation, external radiative loads and heat transfer through the structure. These contributions change throughout the mission, leading to different temperature conditions in each component. Their analysis requires an understanding of the space environment, the main heat-transfer mechanisms and the mathematical methods used to represent the physical system [2, 8, 11].

4.1. Spacecraft Thermal Control

Thermal control is required in many engineering systems because temperature affects equipment performance, material behaviour and reliability. Heat generated within a system and energy exchanged with the external environment must be managed so that the components remain within acceptable temperature limits [3].

In spacecraft, the Thermal Control System (TCS) maintains onboard components within their specified temperature limits throughout the mission, despite changes in environmental and operating conditions. Excessive temperatures, large thermal gradients and repeated thermal cycles can shorten component life, reduce measurement accuracy or prevent equipment from functioning correctly. The design must also satisfy the restrictions imposed by the available mass, volume and electrical power [2, 11, 12].

Thermal requirements are defined by the function and operating state of each component. They include allowable temperature ranges, limits on thermal gradients and requirements for temperature stability. For example, electrical and electronic equipment, including batteries and onboard computers, operate within specified temperature ranges. Optical instruments such as telescopes often require tighter control of temperature gradients to limit dimensional changes and maintain alignment [11, 12].

The temperature of each component is determined by the balance between internally dissipated power and the energy exchanged with the external environment. This balance is influenced by the amount of energy absorbed by the external surfaces, the heat transferred through the internal structure and the energy emitted towards space. Since these contributions change during the mission, the required control must remain effective under different operating and environmental conditions [2, 11].

Different thermal-control measures are combined to meet these requirements. Surface coatings modify solar absorption and infrared emission, radiators provide suitable surfaces for heat rejection, and Multi-Layer Insulation (MLI) limits unwanted radiative exchange. Conductive links distribute heat between components and structural elements, while heaters provide additional energy when passive measures are insufficient under cold conditions [2, 11, 13].

The selection of these measures is governed by the permitted temperature ranges, expected heat loads and mission phases. Thermal control is therefore not limited to maintaining a single uniform temperature. It must also control gradients, reduce excessive variations and meet the individual requirements of each component [11, 12].

4.2. Space Thermal Environment

The space environment is one of the main factors affecting heat exchange through the external surfaces of a spacecraft. In orbit, energy is exchanged with space mainly by radiation because external convective transport is absent. Conduction remains important within the structure and between connected components, but it does not provide a direct mechanism for rejecting heat to space [2, 11].

For a spacecraft orbiting the Earth, the main environmental heat inputs are direct solar radiation, solar radiation reflected by the Earth, known as albedo, and infrared radiation emitted by the planet. At the same time, the external surfaces emit thermal radiation towards deep space. These exchanges are represented in Figure 4.1 [2, 11, 13].

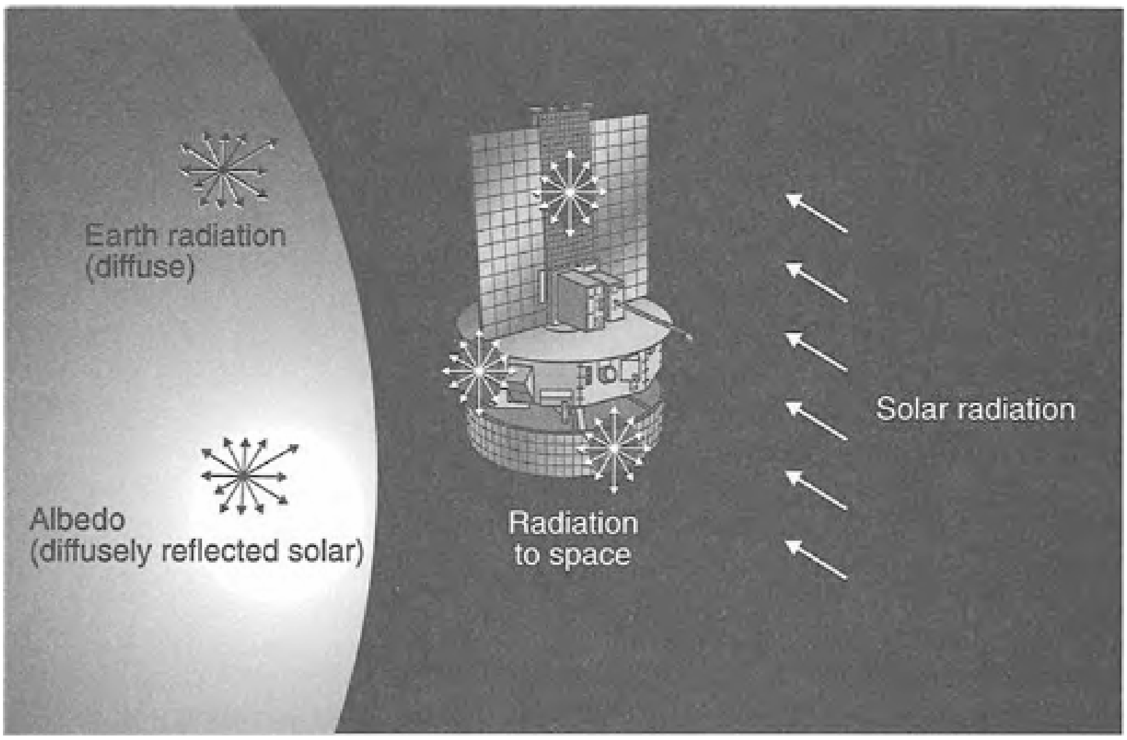


Figure 4.1: Main radiative exchanges between an orbiting spacecraft and the external environment. Taken from [2].

Direct solar radiation generally provides the largest environmental heat input for an Earth-orbiting spacecraft. The power absorbed by a flat surface depends on the solar irradiance, surface area, orientation and solar absorptivity. It can be expressed as [11]:

$$\dot{Q}_{\mathrm{sun}} = \alpha_{\mathrm{s}} G_{\mathrm{s}} A \cos\theta \tag{4.1}$$

where α_{s} is the solar absorptivity, G_{s} is the solar irradiance, A is the surface area and θ is the angle between the surface normal and the solar rays. Since G_{s} is expressed in $\mathrm{W\,m^{-2}}$, the resulting absorbed power is expressed in W [11].

Albedo is solar radiation reflected by the Earth's surface and atmosphere. The albedo contribution is governed by the illuminated portion of the planet visible from the spacecraft and the fraction of reflected energy absorbed by each exposed surface. Planetary infrared radiation instead originates from the Earth's thermal emission and is influenced by the infrared properties of the exposed surfaces. In both cases, the incoming energy also varies with the portion of the planet visible from each surface [2, 11].

The contribution made by each source varies with the orbit, spacecraft attitude, exposed area and thermo-optical properties. During a total eclipse, the direct-solar term becomes zero, and rapid variations occur when the spacecraft enters or exits the eclipse region. Albedo follows the visible illuminated region of the Earth, whereas planetary infrared radiation remains present during eclipse [11, 13].

The periodic sequence of illuminated and eclipse phases exposes the spacecraft to alternating heating and cooling during each orbit. Component temperatures adjust over a finite interval because energy is stored and transferred through the structure. A time-dependent analysis is required to represent this delayed response to the changing environmental loads [2, 11].

4.3. Heat Transfer and Thermal Energy Storage

The temperature response of a spacecraft is governed by the amount of heat transferred and stored over time. Heat transfer occurs by conduction, convection or radiation. Conduction and radiation are the main mechanisms in spacecraft thermal analysis, while thermal capacitance accounts for the energy stored by each component [3, 11].

4.3.1. Conduction Heat Transfer

Conduction transfers thermal energy through a material or between bodies in contact without bulk movement of matter. The conductive heat flow is determined by the temperature gradient, the thermal conductivity of the material and the area normal to the direction of transfer. For one-dimensional conduction, Fourier's law is written as [3, 4]

$$\dot{Q}_x = -kA\frac{\mathrm{d}T}{\mathrm{d}x} \tag{4.2}$$

where $\dot{Q}_x$ is the conductive heat flow, k is the thermal conductivity, A is the area normal to the heat flow and $\mathrm{d}T/\mathrm{d}x$ is the temperature gradient. The negative sign indicates that heat flows in the direction of decreasing temperature.
Figure 4.2 shows the one-dimensional plane-wall configuration described by Fourier's law.

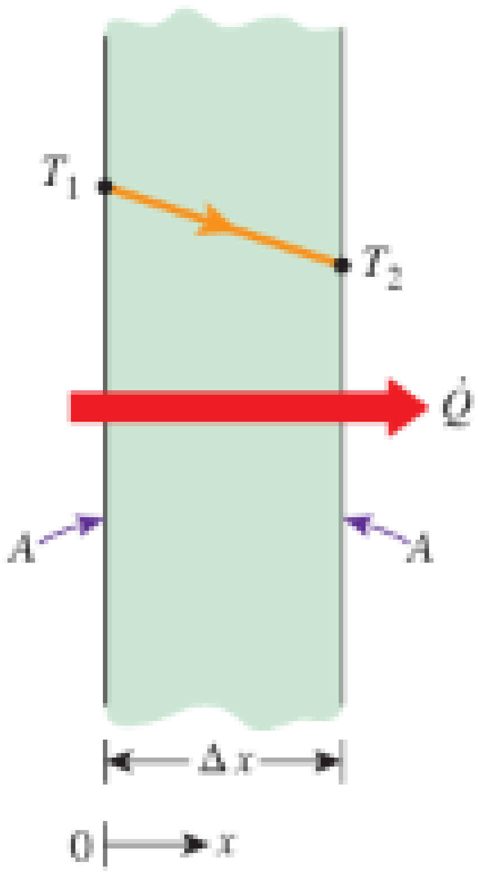


Figure 4.2: One-dimensional heat conduction through a plane wall. Taken from [3].

For a plane layer of thickness L, constant area and constant conductivity, Fourier's law can also be expressed in terms of thermal resistance [3,4]:

$$R_{\text{cond}} = \frac{L}{kA} \tag{4.3}$$

$$\dot{Q}_{12} = \frac{T_1 - T_2}{R_{\text{cond}}} \tag{4.4}$$

The thermal-resistance formulation allows conductive paths to be combined in series or parallel. At an interface between two components, the effective resistance also depends on surface roughness, contact pressure and any intermediate materials. These effects are represented by the contact resistance, while the linear thermal conductance is defined as the inverse of that resistance [2,3]:

$$G^L_{12} = \frac{1}{R_{12}} \tag{4.5}$$

where G^L_{12} is the linear thermal conductance between components 1 and 2, and R_{12} is the corresponding resistance. This relation is particularly useful in nodal models because each conductive connection is represented by a conductance between two nodes. When temperature changes with position or time, Fourier's law is combined with conservation of energy to obtain the heat-diffusion equation [3,4,8].

4.3.2. Radiation Heat Transfer

Unlike conduction, thermal radiation transfers energy through electromagnetic waves, which propagate through vacuum. Radiative exchange takes place between spacecraft surfaces, while external surfaces also emit energy towards space. The power emitted by a real surface is determined by the absolute temperature, area and emissivity, as expressed by the Stefan–Boltzmann law [3,11]:

$$\dot{Q} = \varepsilon\sigma A T^4 \tag{4.6}$$

where ε is the surface emissivity, σ is the Stefan–Boltzmann constant, A is the emitting area and T is the absolute surface temperature. The value of the Stefan–Boltzmann constant is
$\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$ [3].
For an external surface facing deep space without obstruction, the net radiative heat flow is expressed as [11]:

$$\dot{Q}_{\text{rad}} = \varepsilon\sigma A\left(T^4 - T^4_{\text{space}}\right) \tag{4.7}$$

where T_{space} is the effective temperature assigned to deep space. The term T^4_{space} is generally small because the deep-space temperature is much lower than the spacecraft surface temperature [11].

The Stefan–Boltzmann law gives the total power emitted over all wavelengths. The spectral distribution, however, varies with temperature. As temperature increases, the emitted power rises and the maximum moves towards shorter wavelengths. This behaviour explains why radiation received from the Sun is mainly treated as short-wave radiation, whereas the Earth and spacecraft surfaces emit primarily within the infrared range, as illustrated in Figure 4.3 [3, 11].

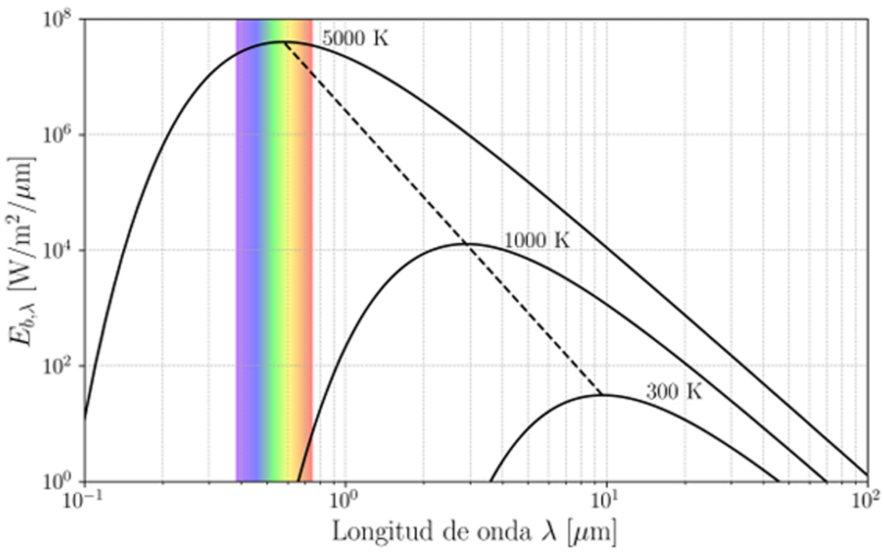


Figure 4.3: Blackbody spectral emissive power at different temperatures and displacement of the emission maximum. Taken from [4].

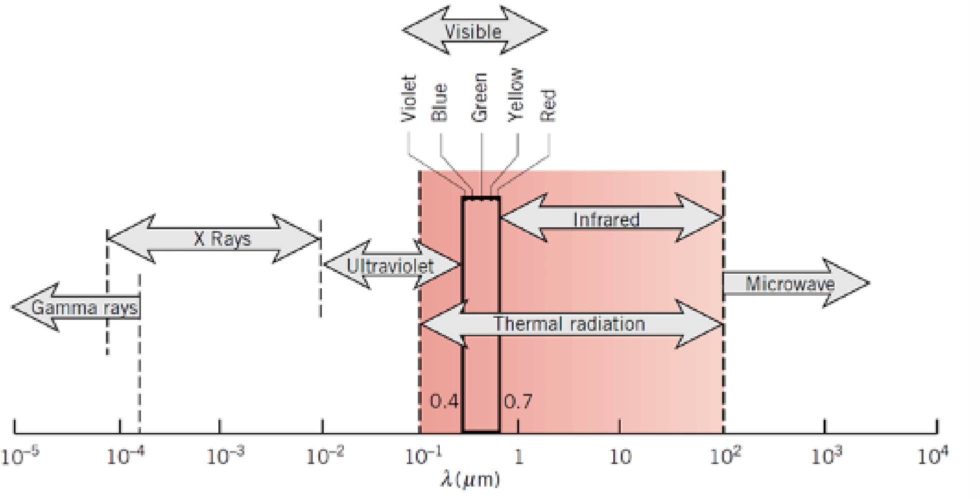
When radiation reaches a surface, part of it is absorbed, part is reflected and the remainder, if any, is transmitted. Conservation of radiative energy requires the corresponding fractions to satisfy [3]

$$\alpha + \rho + \tau = 1 \tag{4.8}$$

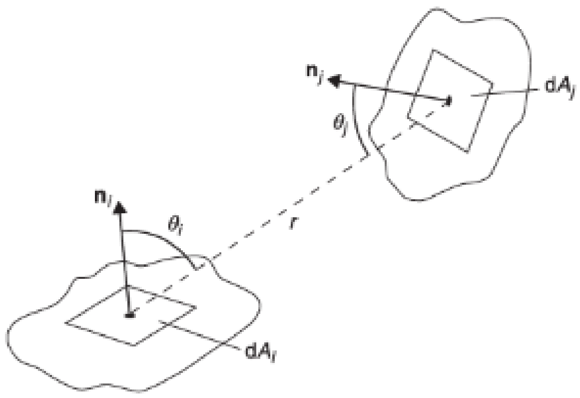
where α, ρ and τ are absorptivity, reflectivity and transmissivity, respectively. For an opaque surface, no radiation is transmitted, and Equation 4.8 reduces to $\alpha + \rho = 1$ [3, 11].

Since these properties vary with wavelength, the solar and infrared bands are treated separately in spacecraft thermal analysis. Solar absorptivity represents the fraction of short-wave solar radiation absorbed by a surface, while infrared emissivity governs the exchange of long-wave thermal radiation. The same surface can therefore have different properties in the two bands [11,13,14]. The wavelength range associated with thermal radiation is shown in Figure 4.4a.

The view factor F_{ij} represents the fraction of radiation leaving surface i that reaches surface j. The value of F_{ij} is determined by the relative position, orientation and area of the two surfaces, as illustrated in Figure 4.4b [11].



(a) Wavelength range of thermal radiation. Taken from [14].



(b) Geometry used to define a view factor. Taken from [11].

Figure 4.4: Wavelength dependence and geometrical definition of thermal radiation.

View factors satisfy the reciprocity relation. When a set of N surfaces forms a closed enclosure, the factors from each surface also satisfy the following rule [3, 11]:

$$A_i F_{ij} = A_j F_{ji} \tag{4.9}$$

$$\sum_{j=1}^{N} F_{ij} = 1 \tag{4.10}$$

where A_i and A_j are the areas of surfaces i and j, respectively, and N is the number of surfaces forming the enclosure.

Radiative exchange is governed by the surface temperatures and thermo-optical properties, as well as by the areas and relative geometry of the surfaces. Analytical view factors are available for simple arrangements, whereas spacecraft radiative models generally calculate them numerically [11, 13].

4.3.3. Thermal Capacitance and Transient Response

Thermal capacitance measures the energy required to change the temperature of a body. For a body with uniform temperature and constant properties, it is given by [2, 3]:

$$C = mc_p \tag{4.11}$$

$$\Delta E = C\Delta T \tag{4.12}$$

where C is the thermal capacitance, m is the mass, c_p is the specific heat capacity and ΔE represents the change in the stored thermal energy of a body.

For a transient process, the energy balance is expressed as [11]:

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\mathrm{net}} \tag{4.13}$$

where $\dot{Q}_{\mathrm{net}}$ is the net heat flow into the body. For the same net heat flow, a larger thermal capacitance results in a slower temperature change [3].

For a single body connected to a constant-temperature boundary through a linear thermal resistance R, the thermal time constant and the corresponding temperature response are expressed as [3]:

$$\tau_{\mathrm{th}} = RC \tag{4.14}$$

$$T(t) = T_\infty + (T_0 - T_\infty)\exp\left(-\frac{t}{\tau_{\mathrm{th}}}\right) \tag{4.15}$$

where T_0 is the initial temperature and T_∞ is the equilibrium temperature approached as time increases. A larger time constant corresponds to a slower response [3].

During a transient process, heat flow changes with time, and the net energy transferred between two instants is obtained by integrating the heat flow over the corresponding interval [3]:

$$E_{12} = \int_{t_1}^{t_2} \dot{Q}(t)\,\mathrm{d}t \tag{4.16}$$

The sign of E_{12} follows the sign convention adopted for $\dot{Q}$. In orbital analysis, this integral quantifies the energy absorbed or released during a selected part of the orbit. For example, a component may store energy during an illuminated interval and release it after eclipse entry. The corresponding temperature change is delayed by the thermal capacitance of the component and by its conductive and radiative connections with the rest of the system [2, 11].

4.4. Mathematical Modelling of Spacecraft Thermal Systems

Spacecraft thermal models combine the equations governing heat transfer and energy storage. The principle of conservation of energy is applied to calculate temperatures and heat flows under specified loads and boundary conditions. Analytical methods provide exact solutions when the geometry and boundary conditions are simple. More complex configurations, involving multiple materials, interfaces and time-dependent inputs, are formulated as a finite system of equations for numerical analysis [11, 13].

4.4.1. Spatial Discretisation Approaches

The numerical solution of heat-transfer equations requires a discrete representation of the geometry and temperature field. The geometry is divided into a finite number of regions, each associated with one or more temperature values. This process is known as spatial discretisation [11, 13].

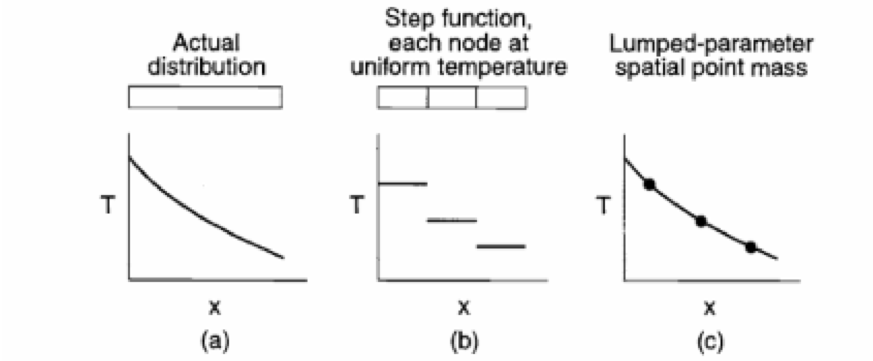
Several numerical methods are available to discretise the thermal problem. Finite-difference methods express spatial derivatives through temperature differences between adjacent grid points. The Finite-Element Method (FEM) divides the geometry into connected elements and describes the temperature variation within each element using interpolation functions. Finite-volume methods apply an energy balance to each control volume using the heat flows across the control-volume boundaries. The three formulations differ, but all of them reduce the thermal problem to a finite system of equations expressed in terms of discrete temperatures [2, 13].

At system level, spacecraft thermal models commonly use a network of lumped nodes. Each node represents a particular region of the model with a single temperature value. Regions where small temperature gradients are expected can be represented by one node, whereas regions with larger gradients or relevant heat-transfer paths are divided into several nodes [11, 13].

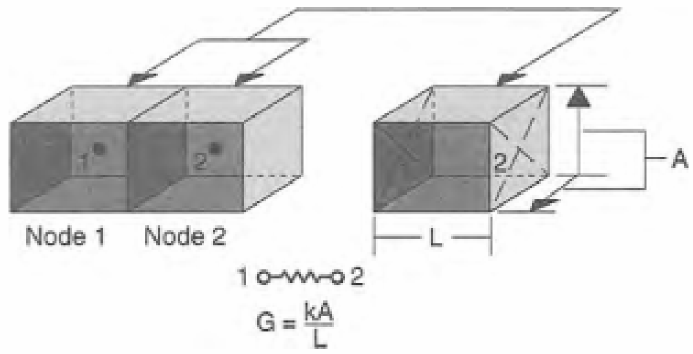
A finer division provides greater spatial resolution, but it also increases the number of conductive and radiative couplings, the calculation time and the volume of output data. The selected nodalisation must resolve the temperatures, gradients and heat paths relevant to the analysis without introducing unnecessary detail [2, 13].

4.4.2. Lumped Parameter Method (LPM)

In the LPM, a thermal system is represented as a network of discrete nodes linked by conductive and radiative couplings. Each node represents a region whose temperature is assumed to be spatially uniform, while the energy stored in that region is represented by a thermal capacitance. Figure 4.5 illustrates the approximation of a continuous temperature distribution by discrete nodal values and the conductive connection between two adjacent nodes [2, 8].



(a) Approximation of a continuous temperature distribution by discrete nodal values.



(b) Conductive connection between two adjacent nodes.

Figure 4.5: Temperature approximation and conductive coupling in a lumped-parameter model. Taken from [2].

The principle of conservation of energy is applied to each of the N capacitive nodes to establish the corresponding energy balance. This converts the continuous temperature field into a coupled system of N first-order ordinary differential equations, one for each nodal temperature. The corresponding balance for a general node i is written as [8, 11]:

$$C_i \frac{\mathrm{d}T_i}{\mathrm{d}t} = \dot{Q}_i(t) + \sum_{\substack{j=1 \\ j\neq i}}^{N} G_{ij}^{L} \left(T_j - T_i\right) + \sigma \sum_{\substack{j=1 \\ j\neq i}}^{N} G_{ij}^{R} \left(T_j^4 - T_i^4\right) \tag{4.17}$$

where C_i is the nodal thermal capacitance, $\dot{Q}_i(t)$ is the applied heat load on the node, G_{ij}^{L} is the linear conductance, G_{ij}^{R} is the radiative exchange coefficient, σ is the Stefan–Boltzmann constant and N is the total number of nodes in the network.

Conductive exchange is proportional to the temperature difference $T_j - T_i$, whereas radiative exchange depends on $T_j^4 - T_i^4$, with the temperatures in the radiative term expressed in kelvin. The conductance coefficients may be constant or depend on the thermal conditions defined in the model. This T^4 dependence makes the resulting system non-linear [8, 11].

The nodal equations are completed by assigning the applied loads and boundary conditions. The loads represent internally dissipated power, external radiative heating or heat supplied by temperature-controlled devices. Boundary conditions specify temperatures, heat inputs or other interactions that are not calculated as unknown nodal temperatures [8, 10].

A transient calculation also requires an initial temperature for each capacitive node. This initial distribution affects the beginning of the solution and is obtained from an estimated condition, a previous transient interval or a steady-state calculation. The influence of this initial distribution must be considered when the required result is a periodic orbital response [8, 10].

4.4.3. Steady-State and Transient Formulations

In a steady-state solution, the nodal temperatures remain constant. As a result, $\mathrm{d}T_i/\mathrm{d}t$ is zero, and the storage term in Equation (4.17) disappears. The resulting non-linear algebraic equations determine the steady-state temperatures for fixed loads and boundary conditions. However, they contain no information about the transition towards this equilibrium state [8, 11].

In a transient solution, the storage term is retained and an initial temperature is assigned to each capacitive node. The coupled differential equations are integrated over time, allowing the loads and boundary conditions to change during the selected interval. The resulting solution describes temperatures and heat flows as functions of time and includes the energy stored in the capacitive nodes [8, 11].

A transient calculation can therefore identify temporary maximum and minimum temperatures, the time at which they occur and the delay between a change in load and the corresponding temperature response. The selected formulation is determined by whether only an equilibrium condition is required or whether the evolution of the system must also be represented [2, 13].

4.4.4. Periodic Thermal Response and Cyclic Convergence

When the orbital geometry and operating sequence repeat after a period P, the applied loads are also periodic. The thermal response is periodic when the nodal temperatures repeat after the same period. To approach this condition, the final temperature distribution from each calculated cycle is used to initialise the next one [10, 11, 15].

$$\dot{Q}_i(t+P) = \dot{Q}_i(t) \tag{4.18}$$

$$T_i(t+P) = T_i(t) \tag{4.19}$$

In numerical calculations, small differences generally remain between the temperatures calculated at the beginning and end of a complete period. Cyclic convergence is accepted when the maximum of these differences is below a selected tolerance [8, 10].

$$\max_i |T_i(t_0 + P) - T_i(t_0)| \leq \varepsilon_{\text{cyc}} \tag{4.20}$$

where t_0 is the initial instant of the evaluated cycle and ε_{cyc} is the convergence criterion. Once this condition is satisfied, the calculated cycle represents the response to the periodic loads rather than the initial numerical adjustment of the model [8, 10, 15].

4.5. Thermal Modelling and Solution Framework in ESATAN-TMS

ESATAN-TMS applies a lumped-parameter nodal formulation through connected geometrical, radiative and thermal models. The physical description is converted into a network containing thermal capacitances, conductive and radiative couplings, boundary conditions and applied loads [8, 9, 16].

The modelling process links the physical configuration with the equations solved during the analysis. Geometrical and surface information is used in the radiative calculations, while material properties and physical connections determine thermal storage and conduction. These contributions are combined in the thermal network before the selected numerical solution is performed [8, 9].

4.5.1. Geometrical and Radiative Models

The geometrical model defines the dimensions, positions and orientations of the modelled elements, together with their materials and thermo-optical properties. The geometrical discretisation also establishes the regions that can later be associated with individual thermal nodes [9, 16].

The corresponding surfaces form the radiative model. View factors and radiative exchange factors between surfaces are calculated from their geometry and properties. During an orbital analysis, the model also determines the solar, albedo and planetary infrared loads absorbed at the selected orbital positions [9, 13, 16].

4.5.2. Thermal Network Representation

The thermal network consists of capacitive nodes linked by conductors. Each capacitive node represents a region with an associated temperature and heat-storage capacity. The nodal capacitance is calculated from the mass and specific heat assigned to the corresponding part of the model [8, 9].

Linear conductors represent solid conduction, contact conductance and other couplings proportional to temperature difference. Radiative conductors represent energy exchange between surfaces and are obtained from the radiative calculation. Environmental inputs, internal dissipation and boundary conditions complete the nodal energy balances [8–10].

Nodes and conductors are generated from the geometry or defined explicitly when a physical element or connection is not represented directly by the geometrical model. This allows equipment units, dedicated conductive links and user-defined thermal interactions to be included without adding unnecessary geometrical detail [8, 9].

4.5.3. Implementation of Steady-State, Transient and Cyclic Solutions

The steady-state solver determines the nodal temperatures for fixed loads and boundary conditions. The transient solver retains the nodal capacitances and initial temperatures and integrates the energy-balance equations over time. Calculation and output times are selected according to the temporal resolution required for the analysis [8, 10].

For periodic cases, the cyclic solver repeats the defined interval and uses the final temperature distribution of one cycle to initialise the next. The process continues until the difference between consecutive cycles satisfies the selected convergence criterion or the permitted number of cycles is reached [8, 10, 16].

The numerical solution provides the nodal temperatures and the contributions included in their energy balances. Depending on the selected outputs, these results may include applied loads and conductive and radiative heat flows, which can subsequently be examined as steady values or as functions of time [9, 10].

5. Thermal Model Development in ESATAN-TMS and Transient Analysis Cases

The thermal model developed in ESATAN-TMS generates the results analysed throughout this work. It does not reproduce a particular flight design in full detail. Instead, it represents the main heat-transfer interactions with sufficient geometrical and physical detail to generate transient results and compare different cases. The same geometry, node hierarchy and thermal network are retained in every simulation, while the environmental and operating conditions are modified between cases.

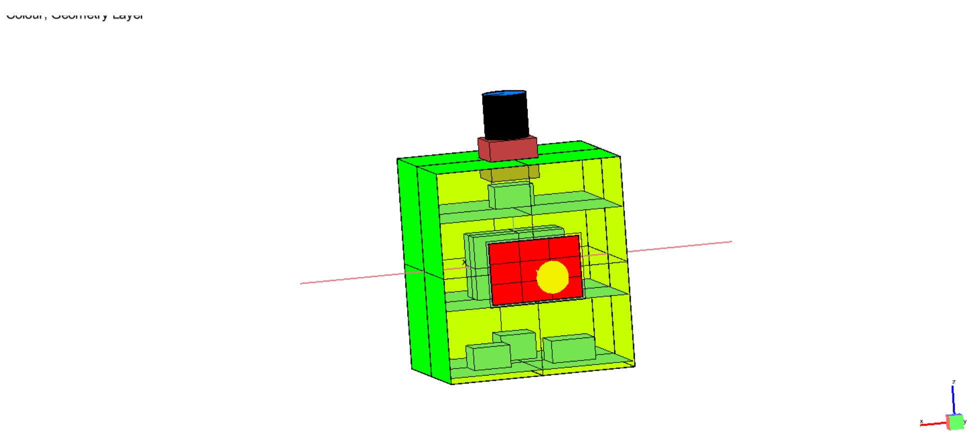
The model contains the main body, two solar-panel wings, a camera, several internal electronic units, a battery, a radiator and Multi-Layer Insulation (MLI). Conductive and radiative couplings connect these elements, allowing temperatures and heat flows to be calculated throughout each simulation. The model follows the lumped-parameter formulation presented in Chapter 4 and implemented in ESATAN-TMS [8,9].

5.1. Model Development and Geometrical Configuration

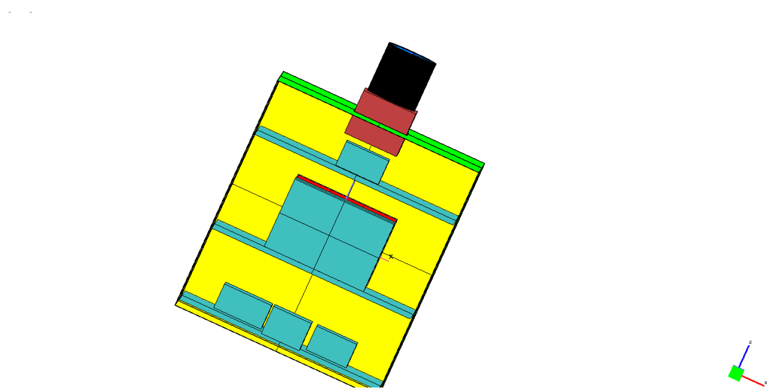
The geometrical model is refined to represent the arrangement of the internal plates and equipment and the main conductive connections between them. This organisation also maintains a direct link between the physical elements and the node hierarchy used by the post-processing application.

The final internal configuration is shown from the $+Y$ and $-Y$ sides in Figure 5.1. Both views show the distribution of the equipment across the structural plates and the different levels of the main body. The dissipative components represented in the model are the camera electronics, the battery, the On-Board Computer (OBC), the communications unit and the Attitude Determination and Control System (ADCS). Each unit is represented separately, allowing an individual internal heat load to be assigned and the corresponding temperature and heat exchanges to be analysed independently.

The external configuration is presented in Figure 5.2. The main body presents a rectangular geometry and is divided into several surface nodes. Two solar-panel wings extend from opposite sides and are discretised independently. The camera assembly extends from the main body. Heat is rejected to space through the radiator, while the MLI-covered surfaces limit heat exchange with the external environment.



(a) Internal configuration viewed from the $+Y$ side.



(b) Internal configuration viewed from the $-Y$ side.

Figure 5.1: Internal configuration of the spacecraft thermal model viewed from opposite sides. The colours correspond to the geometry layer and do not represent temperatures.

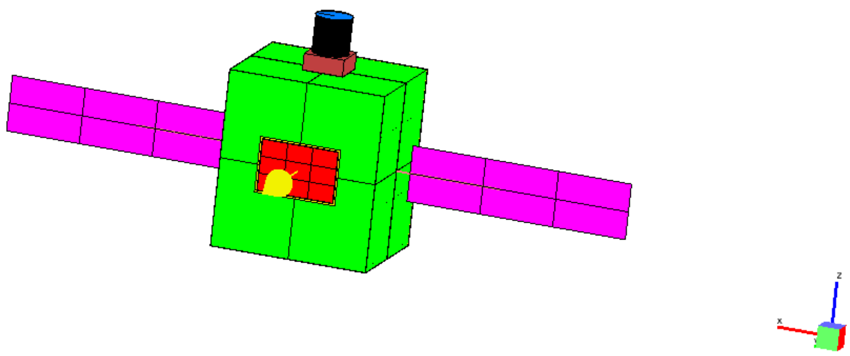


Figure 5.2: Final external configuration of the spacecraft thermal model in ESATAN-TMS. The colours correspond to the geometry layer and do not represent temperatures.

5.2. Thermal Model Hierarchy

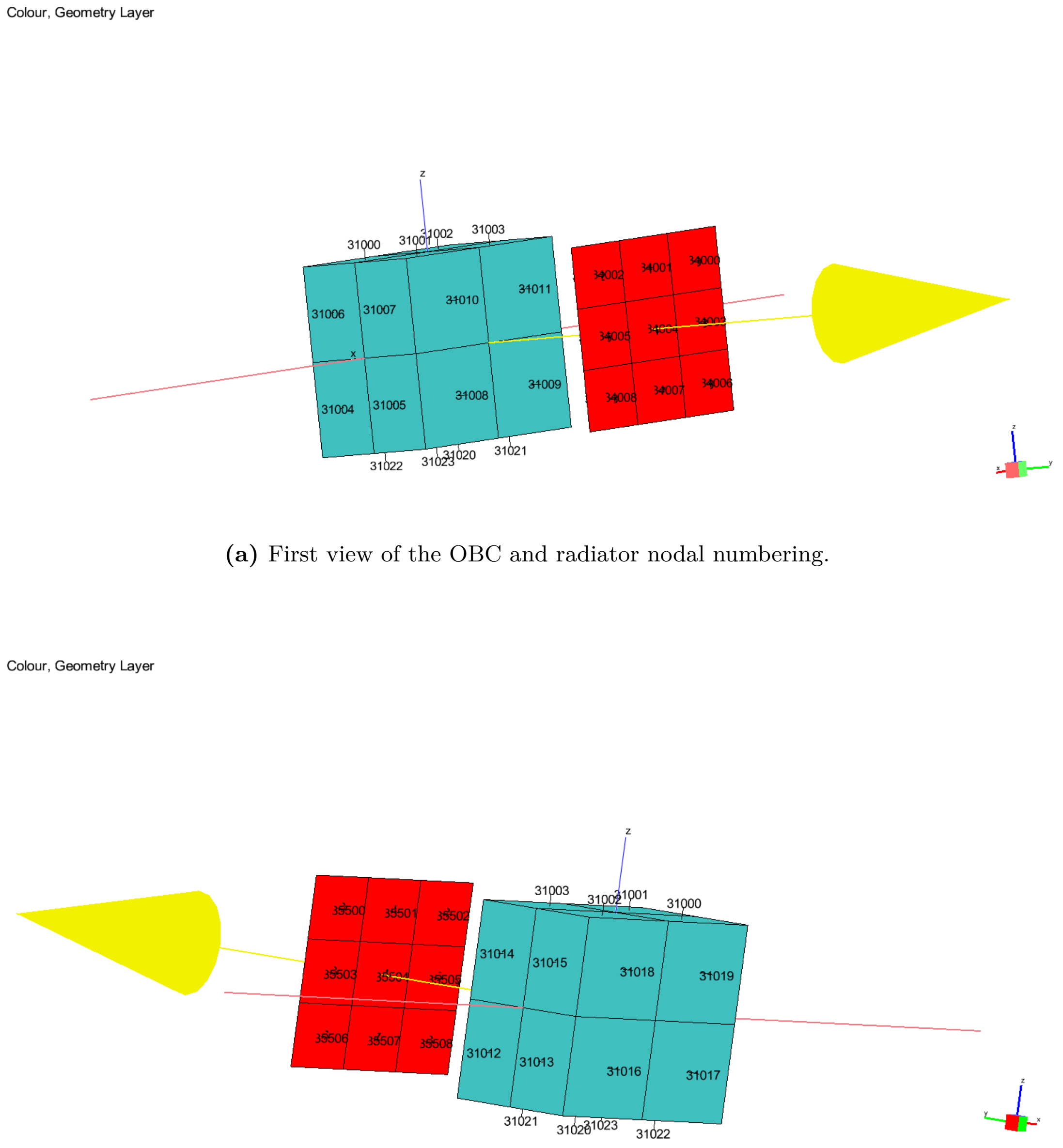
The numerical nodes are grouped according to the physical elements they represent. This hierarchy consists of four levels:

- **Level 0:** includes the complete `SAT_AFM` model and the `SPACE` boundary.
- **Level 1:** separates the main structure from the solar-panel assembly.
- **Level 2:** divides the structure into the body, camera, avionics and MLI groups.
- **Level 3:** contains the individual panels, structural plates, camera elements and equipment units.

Each group is identified by a node range and a short label, giving a consistent naming system throughout the model. The *Power* field identifies the equipment groups to which an internal heat load is applied. Table 5.1.

Table 5.1: Thermal-model hierarchy and node identifiers.

Group	**Level**	**Parent**	**ID**	**Label**	**Power**
Space	0	None	#99999	SPACE	False
SAT_AFM	0	None	#10000–75250	SAT_AFM	False
STRUCTURE	1	SAT_AFM	#10000–51150	STR	False
SOLAR_PANELS	1	SAT_AFM	#70000–75250	SP	False
BODY	2	STRUCTURE	#10000–13650	STR_BOD	False
CAMERA	2	STRUCTURE	#40000–43750	STR_CAM	False
AVIONICS	2	STRUCTURE	#30000–35550	STR_AV	False
MLI	2	STRUCTURE	#50000–51150	STR_MLI	False
SOLAR_PANEL_1	3	SOLAR_PANELS	#70000–72550	SP_1	False
SOLAR_PANEL_2	3	SOLAR_PANELS	#73000–75250	SP_2	False
BODY_PANEL	3	BODY	#10000–11150	STR_BOD_P	False
BODY_PLATE_INF	3	BODY	#12000–12550	STR_BOD_PLI	False
BODY_PLATE_MED	3	BODY	#12600–13050	STR_BOD_PLM	False
BODY_PLATE_SUP	3	BODY	#13100–13650	STR_BOD_PLS	False
CAMERA_BASE	3	CAMERA	#40000–40450	STR_CAM_BASE	False
CAMERA_ELEC	3	CAMERA	#40500–41050	STR_CAM_ELEC	True
CAMERA_LENS	3	CAMERA	#41500–42150	STR_CAM_LENS	False
CAMERA_CYL	3	CAMERA	#42500–43550	STR_CAM_CYL	False
BATTERY	3	AVIONICS	#30000–30550	STR_AV_BAT	True
OBC_BOX	3	AVIONICS	#31000–31550	STR_AV_OBC	True
COMMS_BOX	3	AVIONICS	#32000–32550	STR_AV_COM	True
ADCS_UNIT	3	AVIONICS	#33000–33550	STR_AV_ADCS	True
RADIATOR	3	AVIONICS	#34000–35550	STR_AV_RAD	False
MLI_PANELS	3	MLI	#50000–51150	STR_MLI_P	False



(a) First view of the OBC and radiator nodal numbering.

(b) Opposite view of the OBC and radiator nodal numbering.

Figure 5.3: Example of nodal numbering for the OBC and adjacent radiator surfaces, shown from opposite sides. The figure illustrates the node ranges listed in Table 5.1.

5.3. Material and Surface Properties

The temperature response of each geometrical element depends on the assigned bulk material and the thermo-optical properties of the surfaces. Thermal conductivity, density and specific heat capacity determine heat conduction and energy storage, while the surface properties govern the interaction with solar and infrared radiation. These definitions are common to all analysis cases.

Consequently, changes in the calculated results are associated with the environmental and operating inputs rather than with modifications to the model.

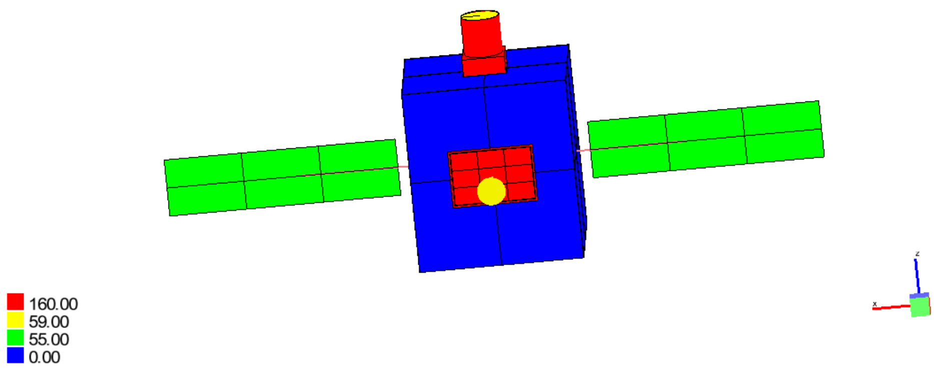
5.3.1. Bulk Properties

The geometrical elements use four material definitions. Aluminium 6061 is assigned to the main structural panels, internal plates and equipment housings. Gallium arsenide represents the solar-cell elements, while a separate optical material is used for the camera lens. The MLI foil has an independent bulk definition whose conductivity is set to zero. Through-thickness heat transfer is represented separately by temperature-dependent effective conductance properties assigned to the MLI surfaces.

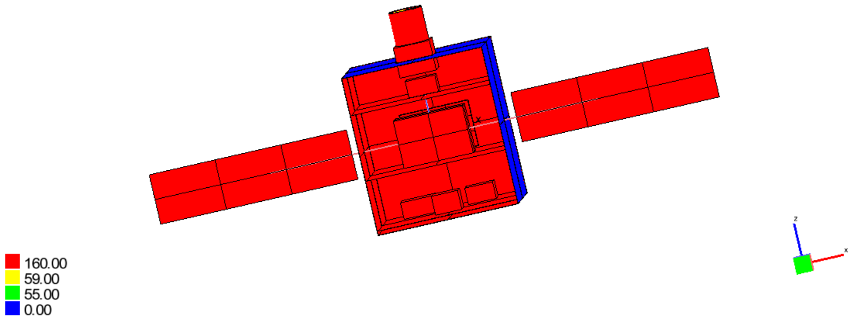
The conductivity, specific heat capacity and density assigned to each material are listed in Table 5.2. Figure 5.4 shows only the conductivity distribution, using two opposite views to cover the complete geometry. This allows the assignments to the structure, solar cells, camera and MLI surfaces to be checked.

Table 5.2: Material properties used in the thermal model.

Name	$\boldsymbol{k}$ [W m^{-1} K^{-1}]	$\boldsymbol{c_p}$ [J kg^{-1} K^{-1}]	$\boldsymbol{\rho}$ [kg m^{-3}]
Al_6061	160	900	2700
GaAs	55	1000	5300
Glass	59	310	5330
MLI_foil	0	900	300



(a) View from the $+Y$ side.



(b) View from the $-Y$ side.

Figure 5.4: Thermal conductivity distribution over the final spacecraft model, viewed from opposite sides.

5.3.2. Thermo-optical Properties

The surface definitions are assigned according to the function of each element. Anodised aluminium is used on selected structural surfaces, while black paint is applied to the internal equipment. An Optical Solar Reflector (OSR) defines the radiator surface, and the solar cells use a separate coating definition. The external and internal MLI surfaces use the `MLI_ext` and `MLI_int` definitions, respectively, while Kapton coated with Indium Tin Oxide (ITO) is assigned to the rear surfaces of the solar panels. An independent property is assigned to the infrared lens.

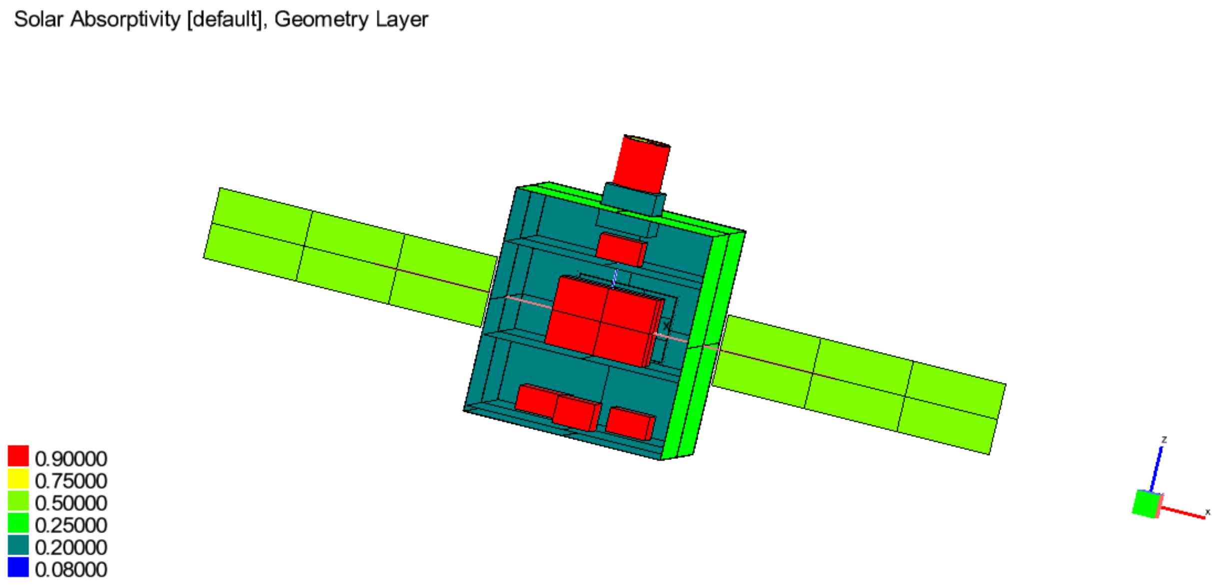
The solar and infrared bands are defined separately. For each surface, the model specifies absorptivity or emissivity, transmissivity and the specular and diffuse components of reflectivity. Within each band, these coefficients are defined so that the corresponding energy fractions sum to unity.

The adopted solar absorptivity and infrared emissivity values are consistent with representative measurements compiled for black coatings, anodised aluminium, solar cells, optical solar reflectors and Kapton films [17]. These data are used as reference values, since a single nominal definition is assigned to each surface in the model. The complete set of properties is summarised in Table 5.3.

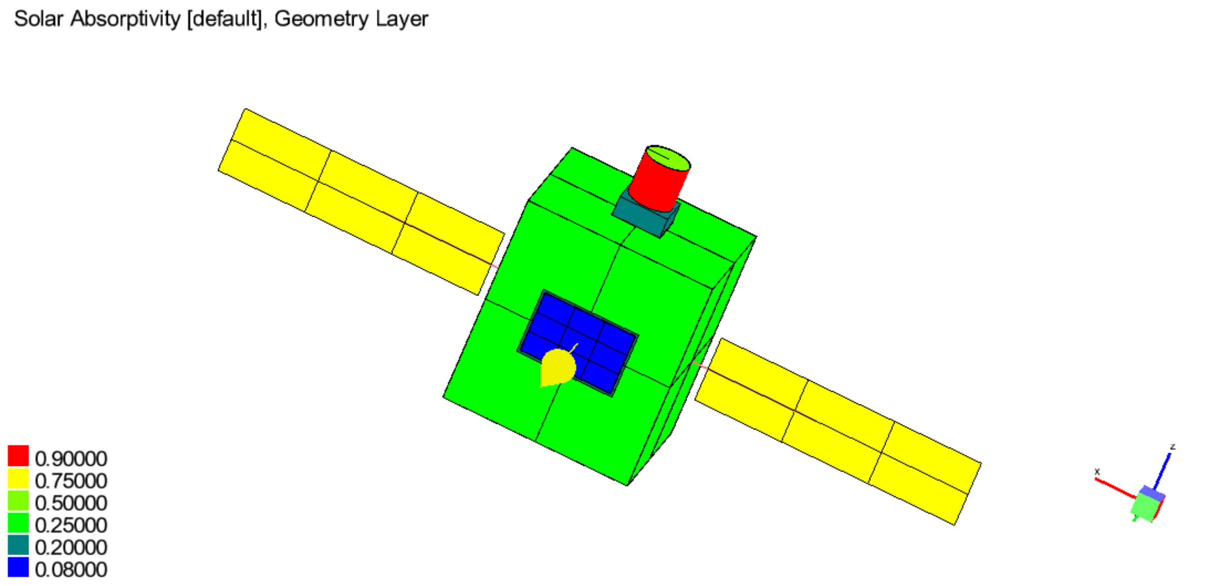
Table 5.3: Thermo-optical coating properties used in the thermal model.

Name	**Infrared**				**Solar**			
	ε	τ	ρ_{spec}	ρ_{diff}	α	τ	ρ_{spec}	ρ_{diff}
Al anodized	0.6	0	0	0.4	0.2	0	0	0.8
Black paint	0.9	0	0	0.1	0.9	0	0	0.1
OSR	0.8	0	0	0.2	0.08	0	0.92	0
Solar cells	0.75	0	0	0.25	0.75	0	0	0.25
IR lens	0.1	0.9	0	0	0.5	0	0	0.5
Kapton ITO	0.8	0	0	0.2	0.5	0	0	0.5
MLI_ext	0.2	0	0	0.8	0.25	0	0	0.75
MLI_int	0.2	0	0	0.8	0.2	0	0	0.8

The solar absorptivity and reflectivity distributions are presented in Figures 5.5 and 5.6. Higher absorptivity is assigned to the solar cells and black-painted surfaces, whereas the radiator OSR has the lowest value. The reflectivity maps show the corresponding distribution of reflected solar energy. Two opposite views are included for each property so that the assignments can be checked over the complete geometry.

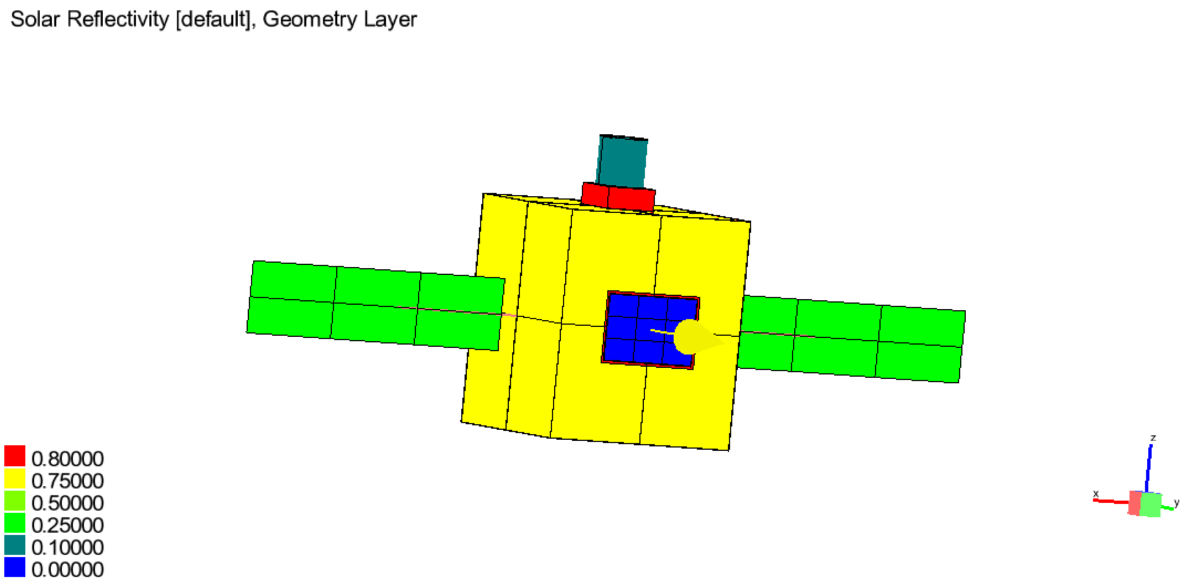


(a) View from one side.

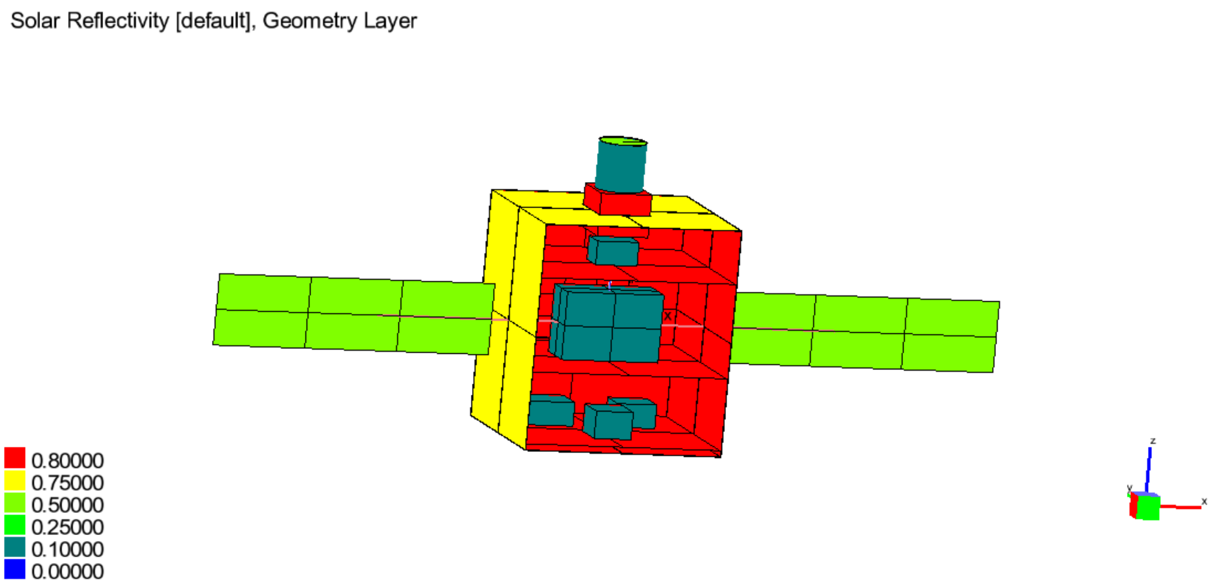


(b) View from the opposite side.

Figure 5.5: Solar absorptivity distribution over the final spacecraft model, viewed from opposite sides.



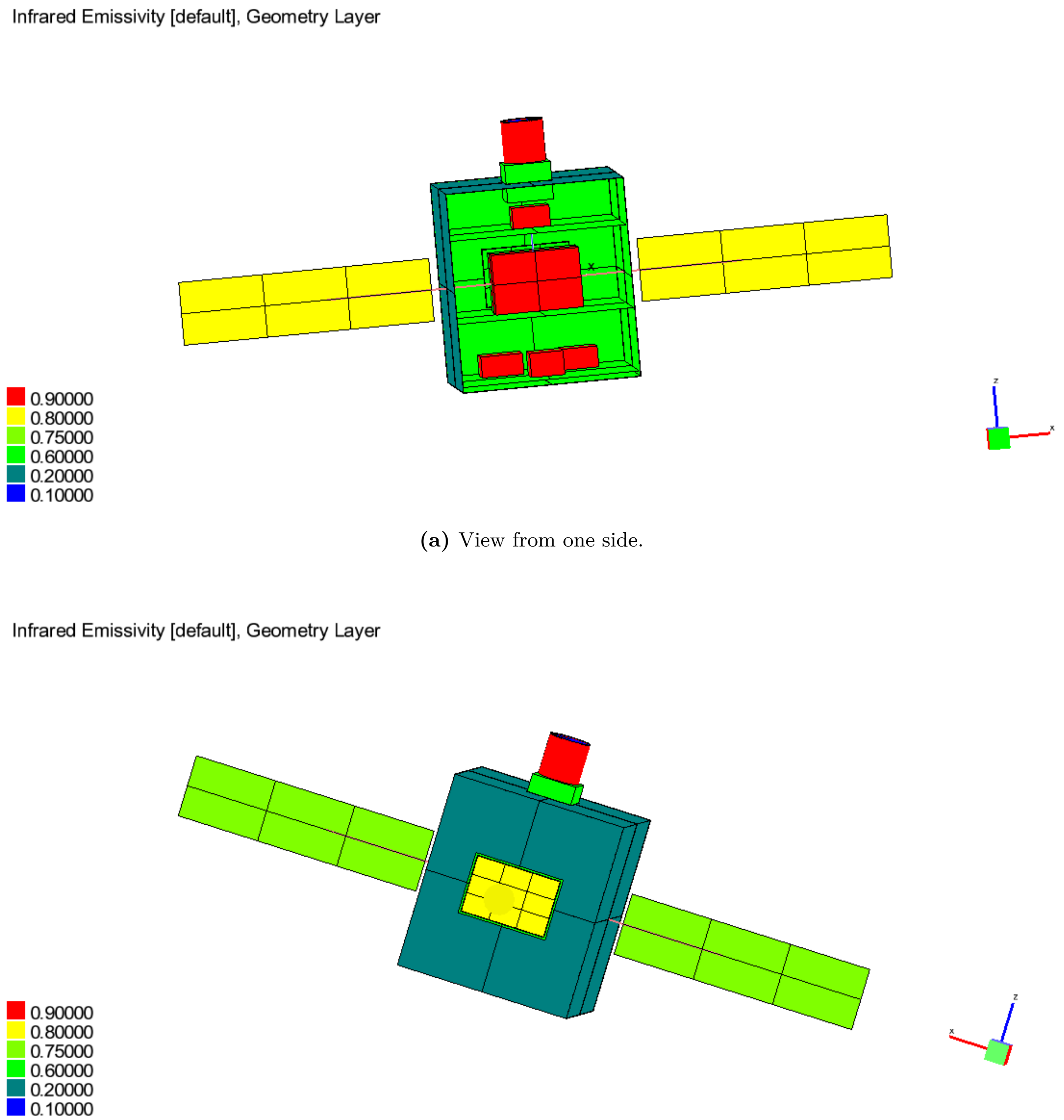
(a) View from one side.



(b) View from the opposite side.

Figure 5.6: Solar reflectivity distribution over the final spacecraft model, viewed from opposite sides.

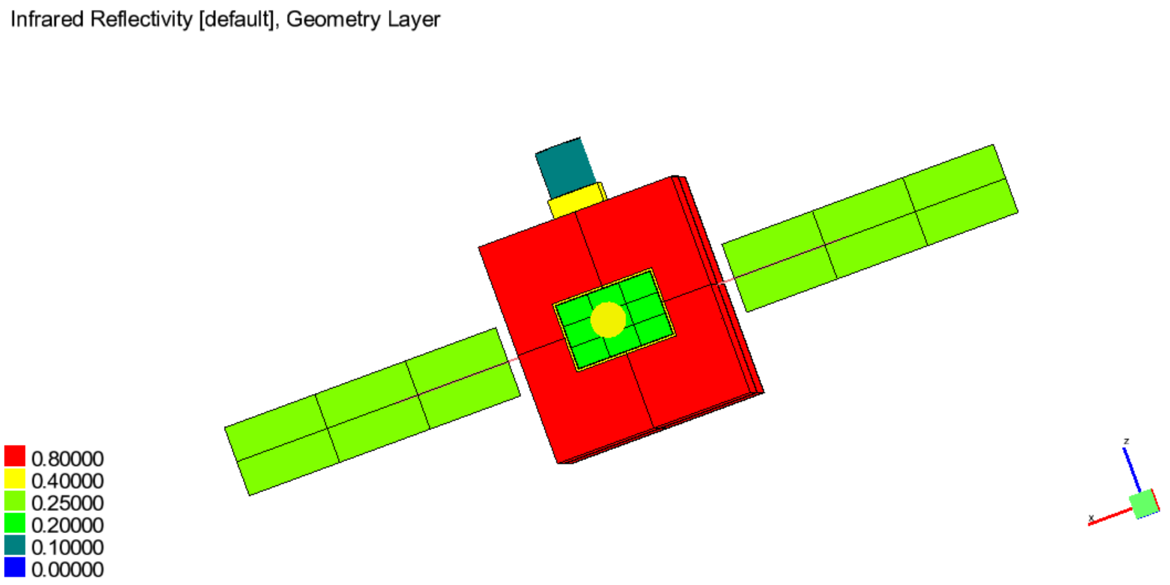
The infrared emissivity and reflectivity distributions are shown in Figures 5.7 and 5.8. High emissivity is assigned to the black-painted surfaces and the radiator, promoting radiative heat exchange. Lower values are used for the internal MLI and the camera lens, which also includes infrared transmissivity. These assignments determine the radiative exchange between internal elements and the rejection of heat from the external surfaces.



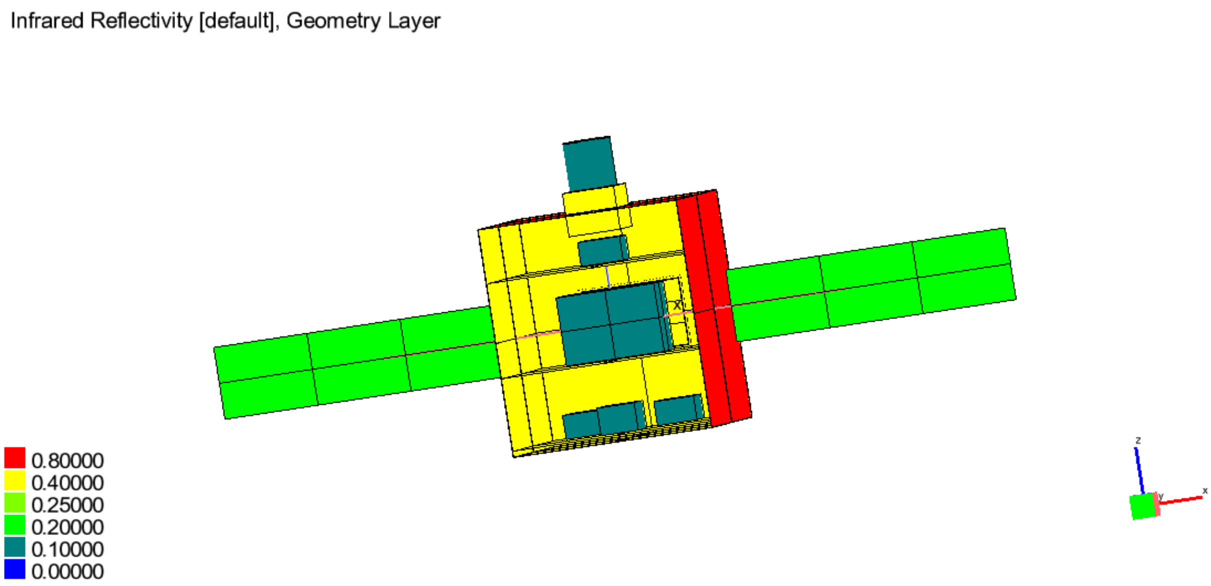
(a) View from one side.

(b) View from the opposite side.

Figure 5.7: Infrared emissivity distribution over the final spacecraft model, viewed from opposite sides.



(a) View from one side.



(b) View from the opposite side.

Figure 5.8: Infrared reflectivity distribution over the final spacecraft model, viewed from opposite sides.

5.4. Nodal Discretisation and Thermal Network Definition

The geometrical model is divided into nodes according to the spatial resolution required for the analysis. Large external elements, including the body panels and solar-panel wings, are represented by several nodes to capture local temperature differences and variations in the absorbed environmental loads. Internal plates and equipment units are also discretised separately to reproduce the main heat-transfer paths through the structure.

Each capacitive node is assigned a material, mass and specific heat capacity. Following this nodal division, the conductive network represents heat transfer through the structure and between the internal equipment and supporting plates. Most conductive paths correspond to physical contacts included in the geometry. When a physical connection is not represented geometrically, an additional conductor is defined. This is the case for the thermal link between the OBC and the radiator, which is included without adding unnecessary geometrical detail.

The radiative network represents exchange between the relevant surfaces and with the deep-space boundary.

Internal heat loads are applied to the nodes representing the dissipative equipment. The values assigned in each transient case are summarised in Table 5.4. The battery load is defined separately for eclipse and non-eclipse intervals.

Table 5.4: Internal heat loads applied in the selected transient cases.

Component	**Applied power (W)**		
	Nominal	**Hot**	**Cold**
Camera electronics	2.00	3.50	0.00
Battery, outside eclipse	1.50	2.06	0.375
Battery, during eclipse	10.00	13.75	2.50
OBC	4.00	5.00	2.00
Communications unit	3.00	4.50	0.50
ADCS	2.00	3.50	0.50

The node definitions and thermal couplings remain unchanged in all transient cases. Only the environmental conditions and internal heat loads are modified. Consequently, each node and thermal coupling represents the same physical element in every exported result file, allowing direct comparison between cases.

5.5. Orbital Environment and Radiative Cases

The orbital environment defines the external radiative conditions applied to the model. Direct solar radiation, planetary albedo and planetary infrared emission are included as external heat inputs [18]. Energy emitted by the external surfaces towards deep space is also considered. These exchanges are represented through the orbital and radiative models of ESATAN-TMS and provide the boundary conditions used in the transient simulations [8].

Three radiative cases are selected: `rad_nominal`, `rad_hot_operational` and `rad_cold_operational`. The same geometry and surface properties are retained in all three cases, while the environmental parameters and eclipse duration differ between them. For the radiative calculation, each orbital revolution is divided into 24 positions separated by 15°. This common angular sampling allows the incident loads to be evaluated consistently throughout the orbit and included in the transient solution.

Figure 5.9 shows the orbital configuration of the nominal case. The spacecraft positions illustrate the discretisation used for the radiative calculation. The green segment represents the portion outside eclipse, whereas the red segment identifies the eclipse interval.

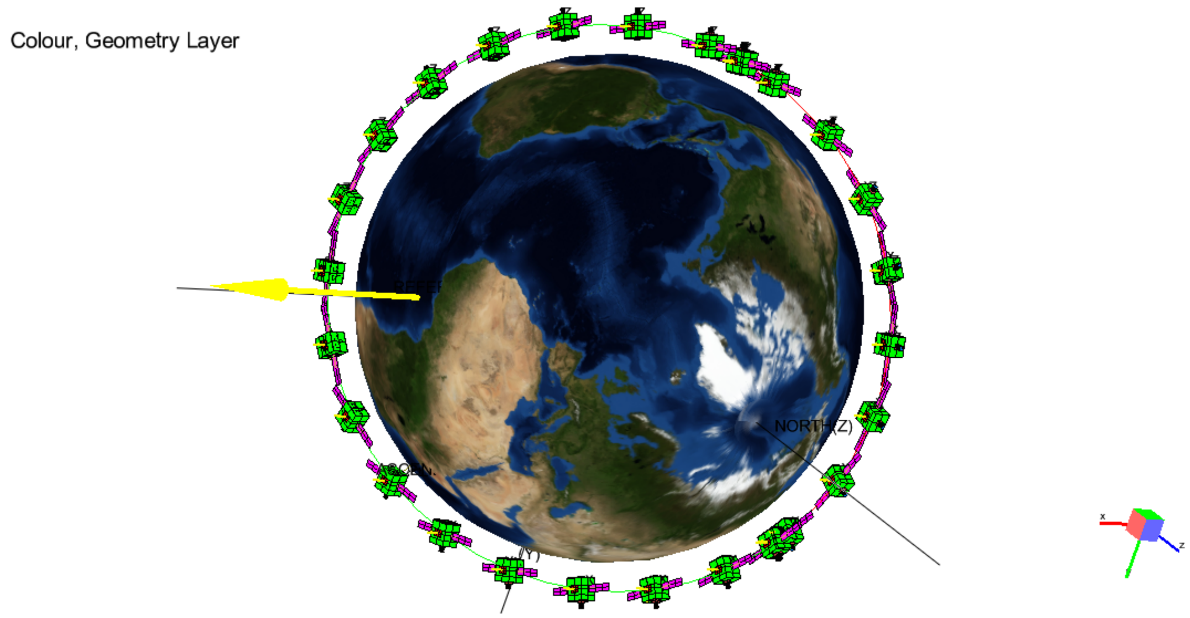


Figure 5.9: Nominal orbital configuration and the 24 positions used for the radiative calculation. The green segment represents the portion outside eclipse, while the red segment indicates the eclipse interval.

The `rad_nominal` case provides the reference environment. `rad_hot_operational` combines a higher external heat input with a shorter eclipse, whereas `rad_cold_operational` combines a lower input with a longer eclipse.

These environments provide clearly differentiated conditions for comparing the hot and cold responses with the nominal case without modifying the physical model. They are selected for the analyses performed in this work and do not cover every condition expected during a flight mission.

5.6. Transient Analysis Cases

5.6.1. Definition of Analysis Cases

Three transient simulations are selected for the final analysis: nominal, hot and cold. The nominal case provides the reference response, while the hot and cold cases represent higher and lower external heat inputs, respectively. Different equipment loads are also assigned according to the operating conditions defined for each simulation. Their purpose is summarised in Table 5.5.

Table 5.5: Transient cases selected from the common thermal model.

Case	**Radiative case**	**Purpose**
Nominal	`rad_nominal`	Reference response used for comparison with the hot and cold cases.
Hot	`rad_hot_operational`	Response under higher external heat input and a shorter eclipse.
Cold	`rad_cold_operational`	Response under lower external heat input and a longer eclipse.

All three simulations use the common model configuration defined above. Only the radiative environment and internal heat loads change between cases. Equivalent nodes and exported result fields therefore retain the same physical meaning, allowing direct comparison with the nominal case.

5.6.2. Solver Configuration and Temporal Sampling

The *Solution Control* configuration applied to the three analysis cases is shown in Figure 5.10.

```
C
      TIMEND=6050.2473
      OUTINT=60.502473
      NLOOP=100
      RELXCA=0.01
      DTIMEI=1.0
      CALL SOLCYC('SLCRNC',0.01D0,0.1D0,6050.2473D0,1000,' ','NONE')
C
      CALL SLCRNC
C
```

Figure 5.10: *Solution Control* execution block used to obtain cyclic convergence and calculate the final transient orbital period in ESATAN-TMS.

The same numerical settings are used in all three simulations. The transient solution is calculated using the Crank–Nicolson method. Before the final results are exported, `SOLCYC` repeats the orbital period until the maximum difference between the initial and final nodal temperatures is below 0.01 °C. A maximum of 1000 cycles is allowed. The converged temperature field is then used as the initial condition for the orbital period retained for analysis.

Each exported simulation covers 6050.247 s and contains 101 output instants separated by approximately 60.50 s. This common time basis records the main changes in the applied loads and allows values from different cases to be compared at equivalent points of the orbit.

5.7. Thermal Model Verification

Verification is performed throughout model development using visual and numerical checks. The material and surface-property maps are inspected to identify missing or incorrectly assigned faces. The node hierarchy is compared with the physical arrangement to confirm that the external panels, internal plates and equipment units belong to the correct model groups.

The conductive and radiative networks are reviewed to identify isolated nodes, missing links and unintended heat-transfer paths. Nodal energy balances are examined to detect inconsistent loads or exchanges. Finally, the exported results are checked to confirm that every selected case satisfies the required cyclic-convergence criterion.

These checks confirm the internal consistency of the model but do not validate it against experimental data. No corresponding flight hardware or thermal-vacuum test data are available for comparison. The conclusions are therefore limited to the behaviour predicted by the model under the defined environmental and operating conditions.

5.8. Simulation Results Export in TMD Format

After each simulation, the corresponding result set is exported from ESATAN-TMS as a separate `.TMD` file. Each file stores the time vector and the nodal results selected for post-processing, including temperatures, applied loads, absorbed environmental loads, and conductive and radiative heat flows. Node numbers and labels are preserved during export, allowing each variable to be associated with the corresponding group in the external hierarchy spreadsheet.

The structural and temporal dimensions shared by the three result files are summarised in Table 5.6. The nominal, hot and cold files retain the same node and conductor structure and cover a common time interval.

Together, the three files form the dataset used for the transient and multi-case analyses. Their common structure allows equivalent variables to be represented at the same orbital instants and compared directly with respect to the nominal case.

Table 5.6: Structural and temporal characteristics of the exported `.TMD` files.

Total thermal nodes per file	284
Capacitive nodes	282
Boundary nodes	1
Inactive nodes	1
Linear conductors, G^L	597
Radiative conductors, G^R	3295
Time interval	6050.247 s
Output instants per file	101
Approximate output time step	60.502 s

The 101 output instants correspond to the temporal sampling of the transient solution and should not be confused with the 24 orbital positions used in the radiative calculation. For each nodal variable, one file therefore contains $284 \times 101 = 28\,684$ values before the conductor results and the remaining nodal result fields are considered.

The absorbed solar loads calculated for the solar panels are also retained in the exported data. These values are used as inputs for the electrical-generation and battery calculations. This processing is performed independently and does not modify the ESATAN-TMS solution.

The three `.TMD` files and the hierarchy spreadsheet form the input dataset used by the Python application to read, organise and compare the results.

6. Development of the Multi-Case Post-Processing Tool

The original post-processing application is extended to handle several transient cases associated with a single ESATAN-TMS model. The hierarchy-based selection and heat-flow representation functions already available are adapted to analyse selected transient instants, while the workflow is expanded to support multi-case loading and comparison, together with the saving and restoration of analyses.

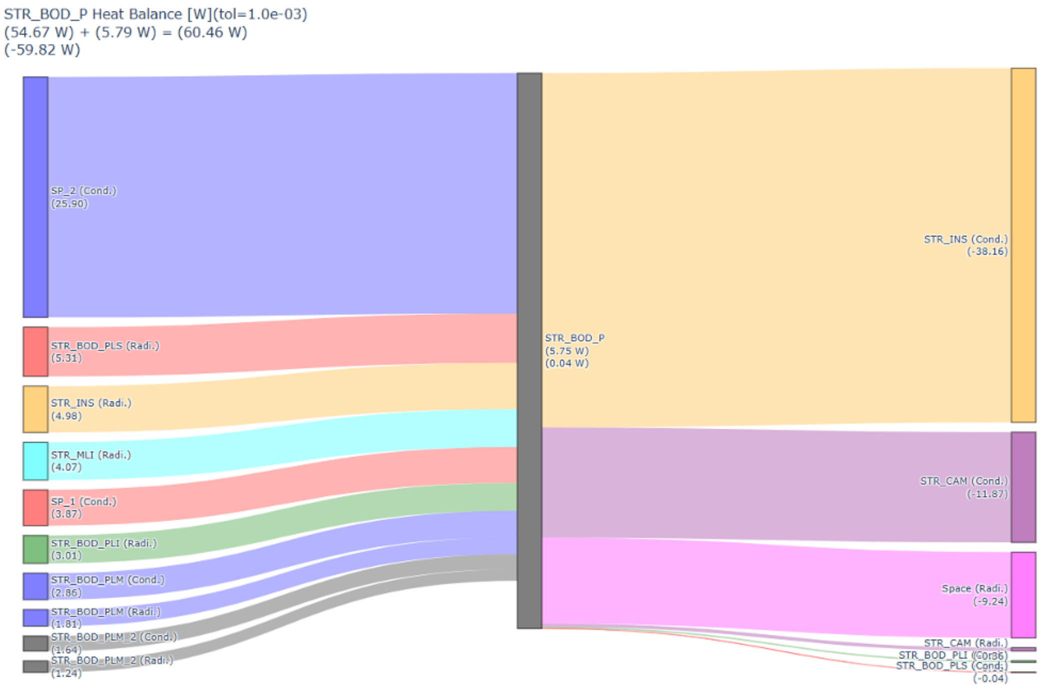
Each `.TMD` file is treated as an independent case and linked to the model hierarchy defined in a common Excel spreadsheet for all cases. This arrangement allows equivalent elements to be examined using consistent selection, time and representation criteria across all loaded simulations.

The nodal model is solved exclusively in ESATAN-TMS. The post-processing tool neither repeats the thermal calculation nor modifies the imported data. Its purpose is to organise these results and provide the functional controls required for inspection, transient post-processing, multi-case comparison and electrical assessment.

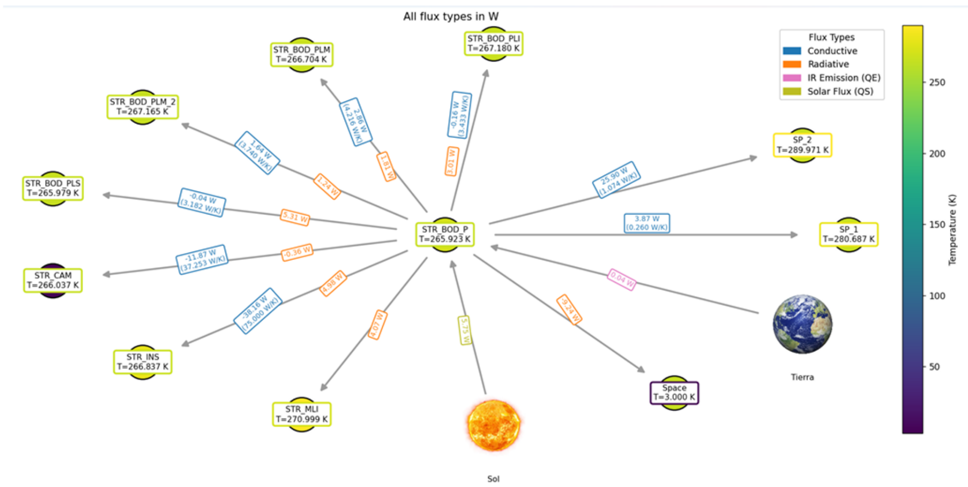
6.1. Evolution of the Original Post-Processing Application

The starting point is the Python-based code developed for the steady-state post-processing of ESATAN-TMS models. Its workflow combines the data contained in a `.TMD` file with an Excel spreadsheet that associates node identifiers with physical components and subsystems. This mapping supports result filtering and the generation of temperature views, group-attribute and conductor tables, and heat-flow representations.

The result reader, hierarchical selection routines and main heat-flow views form the functional basis for the expanded application. These capabilities can also be applied to transient inspection, since the state of a solved model can be examined at a chosen output instant. Figure 6.1 illustrates two representative visualisations generated for the HELIOS steady-state case [1].



(a) Sankey representation of the HELIOS heat balance.



(b) NX diagram of HELIOS including all heat-flow types.

Figure 6.1: Representative heat-flow visualisations generated for the HELIOS steady-state case with the original application. Extracted from [1].

The original interface is mainly designed for the inspection of a single result file. Although some routines already allow access to several output instants when they are present in the source file, most of the interface is oriented towards steady-state analysis. In addition, saved figures and tables do not record the active case, involved elements or output instant used to re-create them, so the associated configuration cannot be restored directly.

The current implementation incorporates several new functions that expand the capabilities of the application. Multiple compatible cases remain accessible within a single session, equivalent elements are selected through a shared hierarchy, and instantaneous outputs are generated at a single selected transient time. Saved figures and tables also store the case, element and representation settings required to restore the saved configuration.

Accordingly, the existing reading, filtering and representation functions are integrated into an expanded workflow. The resulting tool accepts multiple `.TMD` files associated with a shared model structure and incorporates the transient and electrical-balance modules required to meet the objectives of the project. Nominal, hot and cold simulations serve as representative cases, but the implemented workflow is not limited to these specific conditions.

6.2. Application Architecture and Data Flow

The software follows a modular organisation that separates data interpretation, interface coordination and the individual functional areas. The architecture is described through the path followed by the imported data rather than as a detailed inventory of internal storage or execution mechanisms.

Execution starts in `app_thermalviewer/app_interactiva.py`, which receives the shared hierarchy, the internal analysis identifier and the selected `.TMD` files. This entry point builds the collection of loaded cases and distributes it among the tabs forming the user interface.

`core/thermaldata.py` acts as the data layer. It interprets each result file and gives the interface access to nodal, group, time-dependent and conductor information. Individual `.TMD` files are processed separately, while the Excel mapping defines the component organisation required to select equivalent elements.

Shared operations are grouped in `app_thermalviewer/app_helpers.py` and `app_thermalviewer/viewer_utils.py`. These modules handle hierarchy ordering, selection validation, unit management, time-dependent access, and direct or difference-based representations across several cases.

The files contained in `tabs/` apply these operations to the controls, tables and figures associated with each functional area. Each file implements one of the main interface tabs. For example, `tab_case_info.py` displays general information about the loaded simulations, `tab_flow_explorer.py` handles heat-flow inspection, `tab_transient.py` implements the transient post-processing module, and `tab_sp.py` contains the electrical-balance and battery calculations. Further modules, including `tab_group_attributes.py`, `tab_gl_conductors.py`, `tab_gr_conductors.py`, `tab_nodes.py` and `tab_group_study.py`, cover the remaining model-inspection and conductor-analysis areas.

A final group of modules handles the saving, restoration and export of analyses throughout the interface. Each generated figure or table is linked to the main settings needed to reproduce or export it. Table 6.1 summarises the functional responsibilities of the principal software groups.

Table 6.1: Main software groups and their functional responsibilities.

Module or directory	**Main responsibility**
`app_interactiva.py`	Coordinates file inputs, loaded cases and execution of the analysis tabs.
`core/thermaldata.py`	Reads each result file and provides access to nodes, attributes, time histories and conductor data.
`app_helpers.py`, `viewer_utils.py`	Provide hierarchy management, selection validation, unit handling, case-comparison operations and common output preparation.
`tabs/`	Contains one module for each analysis tab, including case inspection, heat-flow exploration, conductor analysis, transient post-processing, and the electrical balance and battery model.
`scene_manager.py`, `scene_ui.py`, `restored_scenes.py`	Provide the controls required to save, list, restore and export complete analysis configurations and their generated outputs.

Figure 6.2 outlines the resulting data flow. The input files are interpreted separately, coordinated through a central application layer and passed to the relevant processing modules. This structure links the generated figures and tables to the saving and export controls available in each area.

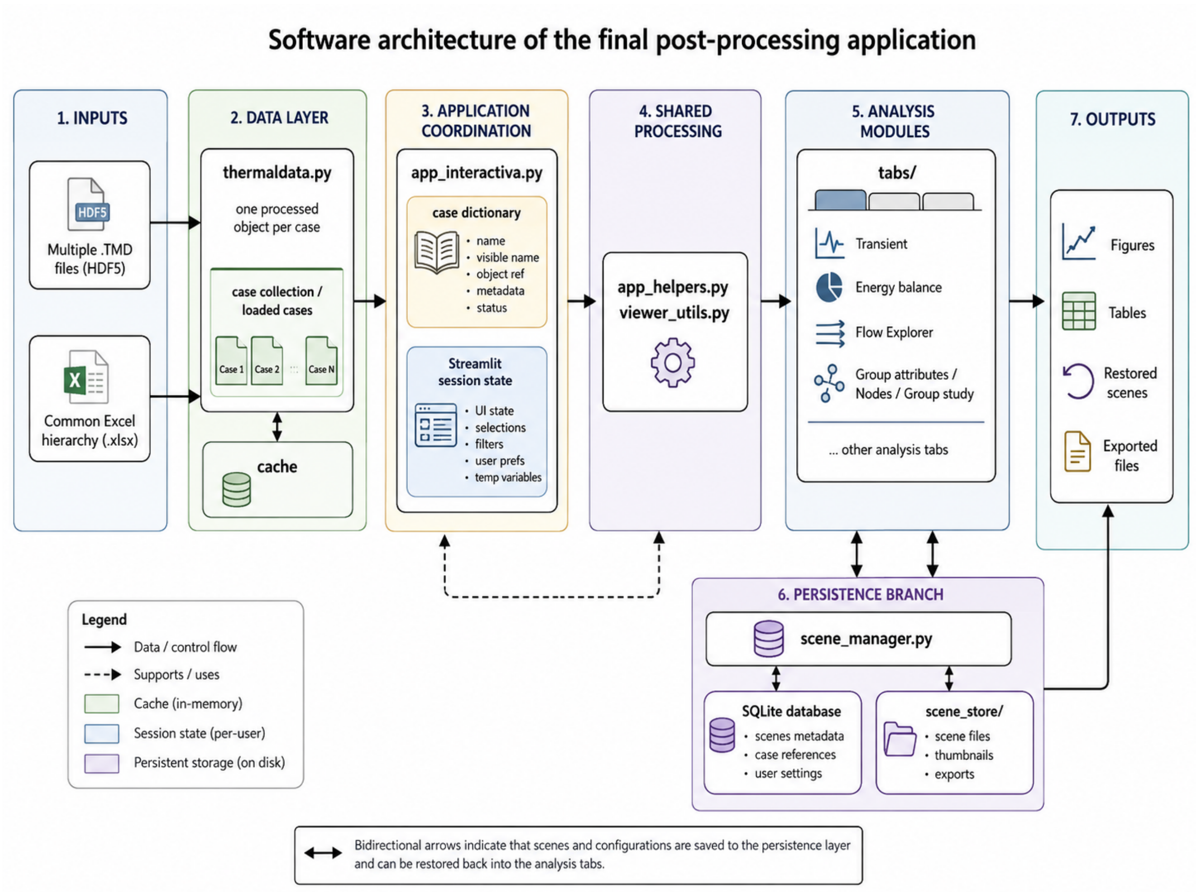


Figure 6.2: Main data flow and software architecture of the current post-processing application.

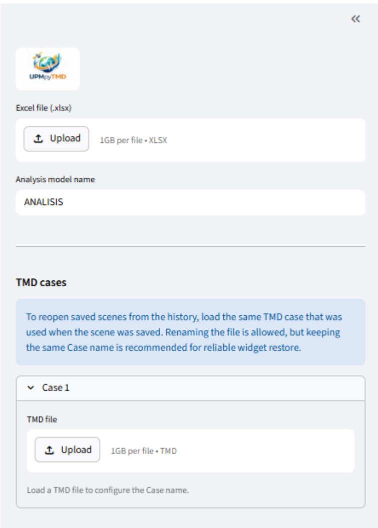
Imported simulation data are kept separate from the values calculated during post-processing. Changes to the active groups, cases, units, output time or representation mode affect only the current analysis and leave the imported `.TMD` files unchanged. Consequently, every tab can access the loaded case collection and apply consistent selections without creating modified copies of the source data.

6.3. Multi-Case Loading and Case Organisation

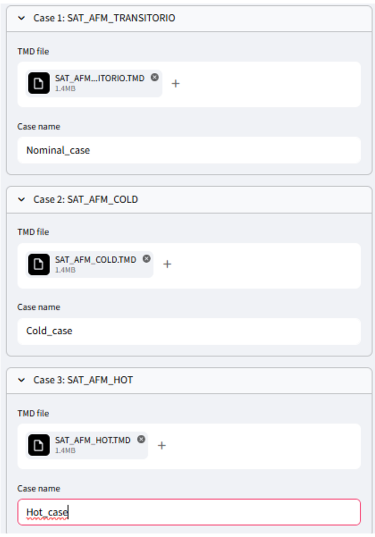
The loading stage combines a single Excel hierarchy with multiple transient `.TMD` files. One spreadsheet serves all simulations because they share identical node identifiers, physical groups and parent–child relationships. Environmental and operating conditions change between simulations, but the model organisation remains unchanged.

Each `.TMD` file contains an internal analysis identifier, while the interface assigns a visible name to the loaded simulation. Different files may therefore use an identical internal identifier, such as `ANALISIS`, but appear as *Nominal*, *Hot* or *Cold* in selectors, legends, tables and saved analyses.

Before loading a file, the interface provides an editable field for its visible name. If two entries receive identical names, a suffix is added to keep every key in the case collection unique. Figure 6.3 shows the initial inputs and the configuration of multiple result files.



(a) Initial hierarchy and result-file inputs.



(b) Configuration of several files through editable case names.

Figure 6.3: Multi-case loading interface with one shared hierarchy and multiple `.TMD` result files.

The loader processes every file separately and stores it as an independent case. Any compatible tab can then request the corresponding variable or component selection from one or more simulations without combining or modifying their original numerical data.

The Excel hierarchy does more than replace node identifiers with readable names. It maps each physical group to its node range and defines its position within the parent–child structure of the model. Since all loaded simulations use this mapping, one component selection can be applied directly to each of them.

For example, after the user selects `BODY`, the tool resolves its node range once. Each simulation then supplies the values of those nodes at the requested time, without requiring the numerical identifiers to be entered again for the nominal, hot and cold simulations. The resulting tables and curves therefore refer to the corresponding region before any overlay or difference calculation is performed.

The hierarchy also controls the ordering of the available model elements. Selectors follow the hierarchical parent–child sequence defined in Excel rather than a purely alphabetical list.

In hierarchy-aware selectors, when a parent group and one of its descendants are chosen together, the application removes the descendant because the parent already contains its nodes. A visible warning identifies the corrected selection. This safeguard prevents repeated nodes from altering aggregated results.

Some modules intentionally retain a direct additive selection rule and display all hierarchy levels. In these modules, selecting both a parent and one of its descendants would add both contributions, so the user must choose mutually disjoint groups. The electrical-load groups selected in Chapter 8 are Level-3 equipment groups with non-overlapping node ranges and therefore do not introduce double counting.

6.4. Functional Adaptation of the Application Interface

The final version brings the inherited inspection tabs and the newly developed modules together within a single interface. The sidebar contains the inputs required throughout the session, while every tab receives the loaded case collection and shared Excel hierarchy. The user can therefore move between model inspection, transient and electrical post-processing without loading the source files again or redefining the model structure.

Figure 6.4 presents the resulting layout after loading the Excel hierarchy spreadsheet and three transient cases. The sidebar groups the model inputs, whereas the main area provides access to the different tabs and their case, element, time and representation controls.

The existing group of tabs covers model inspection and the heat-flow functions offered by the original tool. `Case info` summarises the datasets, groups and couplings contained in the loaded simulations. `Group attributes`, `Nodes` and `Group study` provide different levels of detail for the chosen parts of the model. `GL conductors` and `GR conductors` examine conductive and radiative connections, respectively. `Flow Explorer: Multilevel Comparison` uses equivalent component selections to generate instantaneous Sankey and network representations.

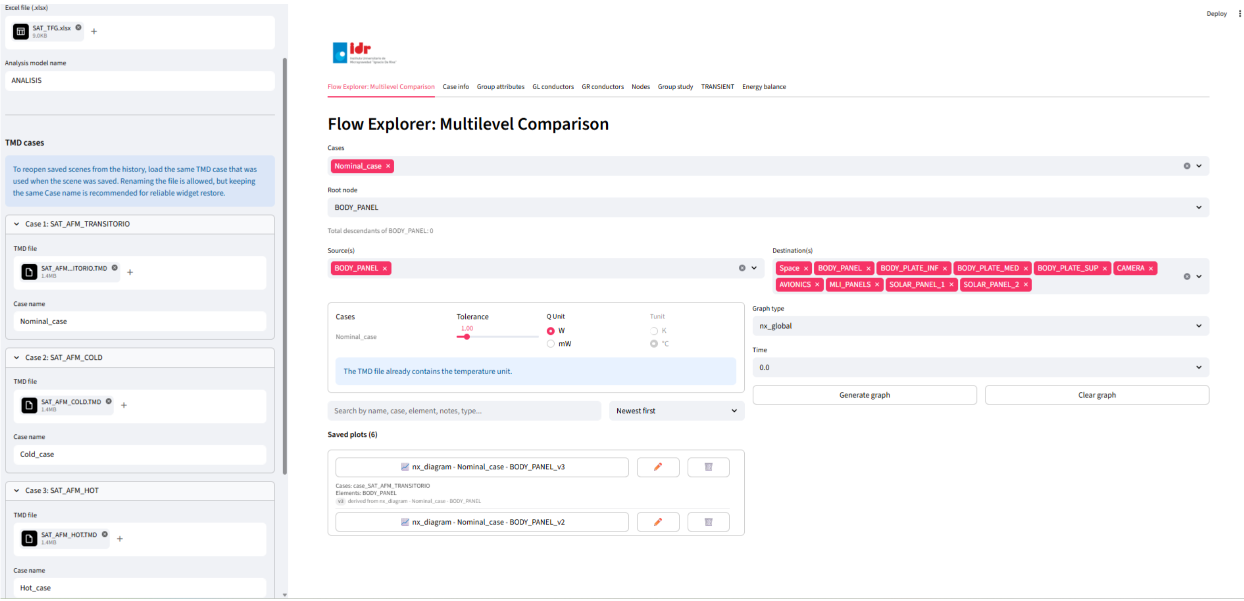


Figure 6.4: Final post-processing interface after loading three transient cases. The sidebar groups the shared inputs, while the main area provides access to the model-inspection, transient and electrical-analysis modules.

Most changes introduced in these tabs affect their operation rather than their appearance. Each compatible module now reads from the loaded case collection, applies a consistent component selection and, when transient data are available, evaluates the requested information at a chosen output instant.

A visual comparison between the initial and final versions of these inherited modules is provided in Appendix A. The comparison focuses on the transient and multi-case adaptations while retaining equivalent physical selections in both interfaces.

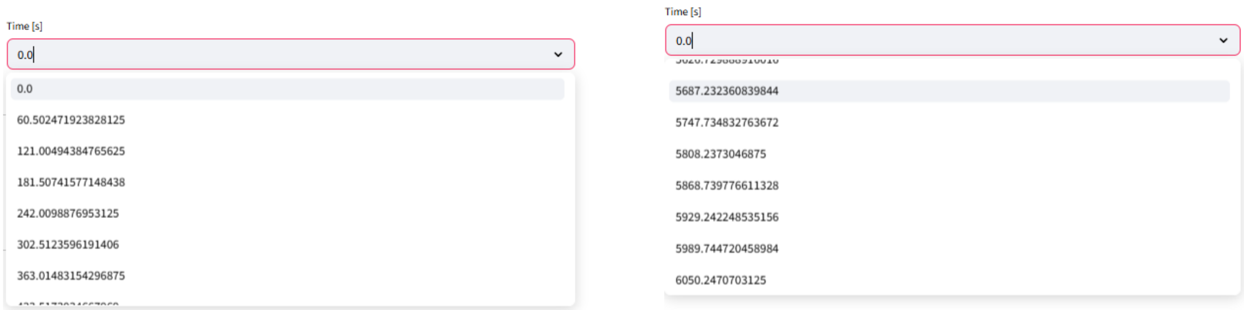
The controls shown by each tab depend on the information present in the active case. Temperature units already defined in the result file remain fixed, whereas compatible heat-flow and display units can be changed when required. Before generating a figure or table, the tool checks for empty case selections, unsupported combinations and missing data. Clear messages identify the problem and prevent the calculation from continuing with incomplete inputs. These checks give all compatible modules a consistent operating logic.

Further secondary implementation details, including hierarchy safeguards, temporal evaluation, conductor inspection, validation feedback and persistent-storage integrity checks, are documented in Appendix B.

6.4.1. Time Selection

The inherited inspection tabs represent the state of the model at a selected output instant, even when the source file contains a complete transient solution. When multiple output steps are present, each compatible tab adds a time selector. The module reads the stored time vector and applies the chosen instant to group attributes, nodal tables, conductor summaries and instantaneous heat-flow representations.

This adaptation extends the original steady-state views without changing their interpretation. Instead of replacing them with complete time series, the user can inspect any stored output instant with the defined component selection and graphical tools. Figure 6.5 illustrates the selector at the beginning and end of a stored time vector.



(a) Initial output instants. **(b)** Final output instants.

Figure 6.5: Output-time selector used by the instantaneous inspection modules when transient data are available.

When multiple simulations are active, the module applies the chosen instant to each compatible simulation. Saving the analysis also records this setting, so restoring it returns the time control to its previous position.

6.4.2. Multi-Case Representation Options

The case selector allows each compatible tab to work with one or more loaded simulations. In direct representation mode, the tool applies consistent selection and temporal settings to every chosen case. Curves can be superimposed when a shared graph provides a clear comparison, whereas tables and complex network representations remain separated when combining them would reduce readability.

The additional *Case difference* mode isolates the numerical change between simulations. The user chooses one reference case and one or more comparison cases. The tool checks that the requested variables are compatible and generates one difference result for each reference–comparison pair. Figure 6.6 presents the representation mode and the associated case controls.

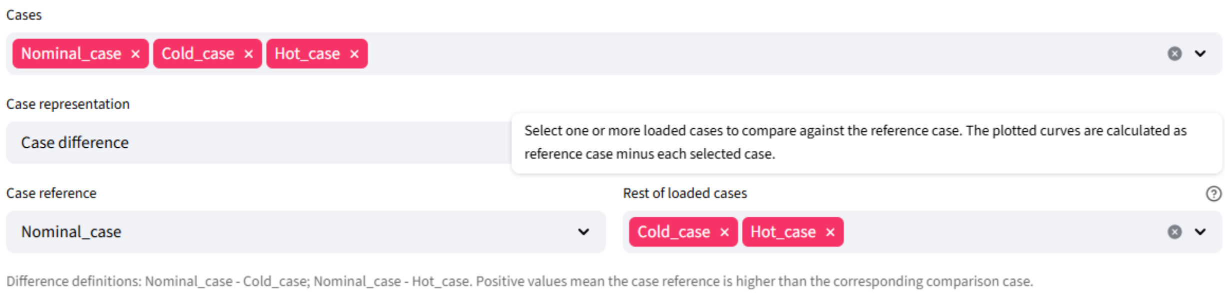


Figure 6.6: Configuration of the case-difference representation.

For any compatible result x, the adopted convention is

$$\Delta x_{R-C} = x_R - x_C \tag{6.1}$$

where R denotes the reference case and C one comparison case. A positive value indicates that the reference result is greater, while a negative value indicates that the comparison result is greater. The reference–comparison order is used consistently in legends, table headings, captions and saved analyses.

Direct representation and case difference serve different purposes. An overlay keeps the absolute response of each simulation visible and makes shared trends easier to identify. The difference removes the common level and highlights the effect of a change in environmental or operating conditions. Both modes use the defined component selections, units and time settings, so the user can change the representation without entering the complete configuration again.

The last two tabs contain the main modules added to the application. `TRANSIENT` processes complete time series of temperatures, thermal loads, interface temperature differences and heat flows. `Energy balance` estimates solar generation, electrical load demand and battery behaviour from the chosen simulations.

6.5. Saving, Restoring and Exporting Analyses

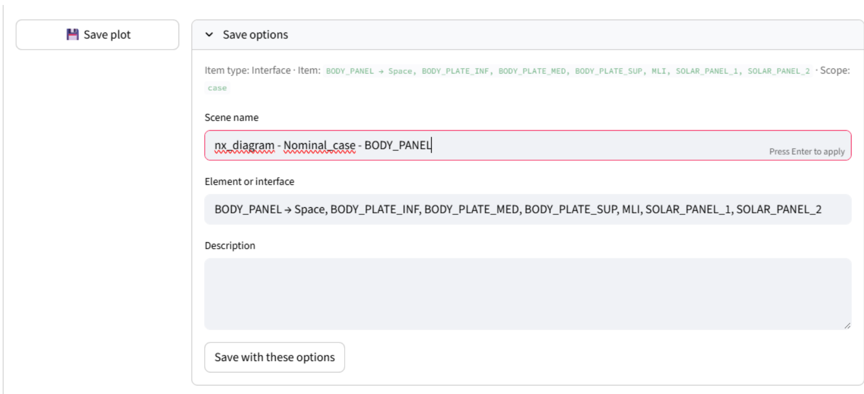
Combining several simulations, involved elements, variables, time instants and representation settings can produce a large number of valid outputs. Exporting a figure or table preserves the final result, but not the complete configuration needed to reproduce it inside the interface. For this reason, the compatible tabs include an area for saving and restoring their generated outputs.

Once a figure or table has been created, the corresponding saving control becomes active. Depending on the result type, the option appears as *Save plot*, *Save table* or as a combined control when both outputs belong to a single analysis. The user assigns a descriptive name and may add a short explanation before saving it.

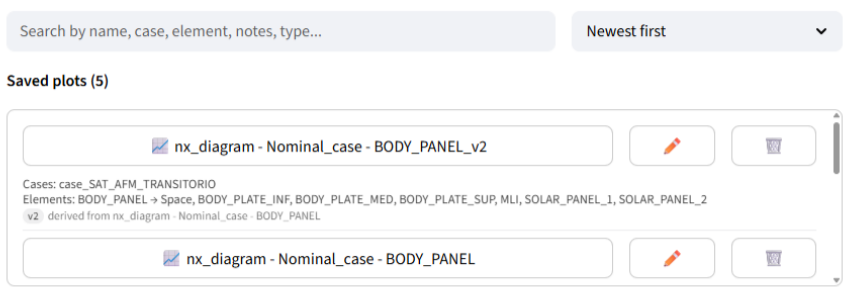
Each saved analysis, referred to as a scene within the interface, records the source tab together with the cases, involved elements, variables, units, time instant and representation mode used to produce the result. These settings allow the stored configuration to be recovered without repeating the complete selection process.

Saved analyses use two coordinated local storage levels. The SQLite database `scene_memory.db` records the model, cases, source tab, configuration settings and paths associated with each entry, while the `scene_store/` directory contains the generated files. Each scene stores its metadata and configuration in `scene.json`; Plotly figures are saved in JSON and HTML formats, Matplotlib figures in PNG and SVG, and tables in CSV and XLSX. The database entry and its associated local folder must remain consistent for the complete configuration to be restored.

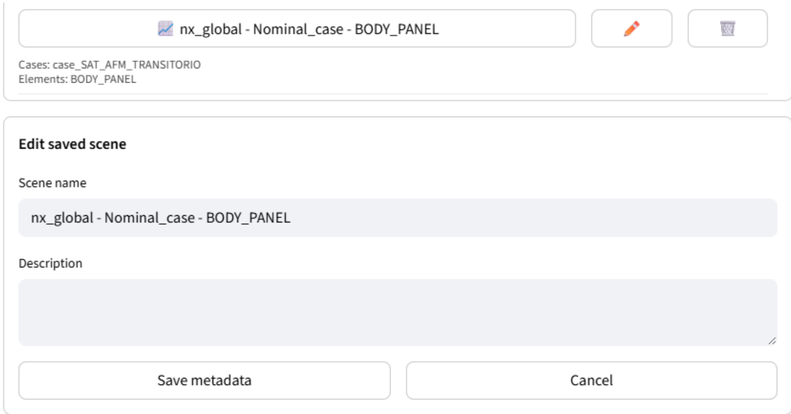
Figure 6.7 illustrates the main operations available after generating an output. The first view contains the information entered before saving it. The second shows an original analysis and a later version derived from it, while the third illustrates the editing of its identifying information.



(a) Definition of the scene name, involved elements and optional description before saving the output.



(b) Saved-analysis list containing an original result and a subsequently derived version. Selecting the displayed name restores its configuration, while the adjacent controls allow the entry to be edited or deleted.



(c) Editing of the name and description associated with an existing scene.

Figure 6.7: Sequence used to save, restore and manage a generated output.

Selecting the name of a stored entry returns its main settings to the tab from which it was created. The source files and required cases must already be loaded, but the physical selections, variables, units, output time and representation options are recovered automatically. The controls placed next to each entry allow its name and description to be modified or the complete entry to be removed.

Restoring an analysis does not overwrite the saved result. If the recovered configuration is modified and stored again, the tool creates a new entry and identifies it as a derived version. The previous output remains accessible, while the new one can use another case, time instant, variable or representation setting.

Search and ordering controls simplify the management of larger collections. Saved analyses can be located through their name, case, model element, description or output type. Although all compatible tabs follow a consistent saving procedure, each area displays the entries and files associated with its own calculations.

Figures and tables can also be exported using the formats offered by their source module. These files provide conventional outputs for reports, engineering documentation or later processing outside the interface.

Scene storage and file export fulfil complementary purposes. The saved-analysis system preserves the configuration and relationships required to continue the work inside the application. Download controls create external copies of figures or tables for reports and engineering documentation. Together, both functions connect interactive exploration with reproducible post-processing and final result presentation.

Overall, the application loads multiple compatible cases, associates them with one hierarchy, applies consistent selections and time settings, generates direct or difference outputs, and stores the complete configuration whenever it must be recovered. This architecture provides a common foundation for the transient and electrical modules.

7. Transient Thermal Analysis

The multi-case structure developed in Chapter 6 provides the model hierarchy, case collection, time controls and representation options used by the analysis modules. Within this framework, the `TRANSIENT` tab processes the time-dependent temperatures, thermal loads and conductor data stored in the imported ESATAN-TMS `.TMD` files. The original simulation results remain unchanged, and all additional calculations are performed during post-processing.

The nominal case `SAT_AFM_TRANSITORIO.TMD` is used to illustrate temperature and thermal-load evolution, interface temperature differences, variations over selected time windows, summary values, conductive and radiative heat flows, and accumulated thermal energy. Comparisons between different environmental and operating cases are reserved for the results chapter.

7.1. Temperature and Thermal-Load Evolution

The first area provides a general view of the transient response through the evolution of temperatures, thermal loads and selected group attributes. To build these time-dependent representations, the application uses the Excel hierarchy to associate each physical group g with its corresponding set of nodes $\mathcal{N}_g$.

At every output instant, the minimum, mean and maximum temperatures of the group are calculated as:

$$T_{g,\text{mín}}(t) = \text{mín}[T_n(t)]_{n\in\mathcal{N}_g} \tag{7.1}$$

$$T_{g,\text{mean}}(t) = \frac{1}{|\mathcal{N}_g|}\sum_{n\in\mathcal{N}_g} T_n(t) \tag{7.2}$$

$$T_{g,\text{máx}}(t) = \text{máx}[T_n(t)]_{n\in\mathcal{N}_g} \tag{7.3}$$

Nodal thermal-load contributions are additive, so they are combined through a different aggregation rule. For any available contribution Q_x, the corresponding group evolution is obtained from:

$$Q_{x,g}(t) = \sum_{n\in\mathcal{N}_g} Q_{x,n}(t), \qquad x \in \{A, E, S, I, R\} \tag{7.4}$$

Here, Q_A, Q_E and Q_S denote the absorbed albedo, planetary infrared and direct-solar loads, respectively, while Q_I and Q_R represent the internal and additional applied loads exported with the solved model. Together with the temperature statistics defined above, these aggregated series form the basis for the subsequent analyses available in the tab.

The configuration begins with the selection of one or more cases and model elements. For each element, the user then chooses the attributes to be represented. The available options are $T_{\text{máx}}$, T_{mean}, $T_{\text{mín}}$, Q_A, Q_E, Q_S, Q_I, Q_R, C and A. The three temperature statistics are activated by default, although the user can select any other combination.

The temperature statistics describe the thermal state of the selected element. T_{mean} represents its overall evolution, while $T_{\text{máx}}$ and $T_{\text{mín}}$ identify the highest and lowest nodal temperatures at each output instant. The difference between these limits defines the instantaneous temperature range within the corresponding group.

The thermal-load terms can be represented separately or together with the temperature statistics. Their curves show how the environmental and internally applied loads acting on the selected element vary throughout the simulation. When an element contains several nodes, the displayed value corresponds to the group sum defined in Equation (7.4).

Thermal capacitance C and surface area A can also be displayed. Unlike the temperatures and loads, these properties remain constant throughout the transient simulation and are therefore not used in the time-dependent examples presented below.

The display units are configured independently from the attribute selection. The temperature unit is fixed to the value stored in the loaded `.TMD` file, while thermal loads can be represented in watts or milliwatts and elapsed time in seconds, minutes, hours, days, months or years. Further controls modify the horizontal-axis label, graph width and height, axis tick font size and legend font size.

The interface also provides the multi-case representation alternative introduced in Section 6.4.2. The examples presented here use a single case, since this is sufficient to explain the functionality of this area of the tab.

Figure 7.1 shows the complete configuration area used to define the cases, elements, attributes, units and graphical settings of the resulting plot.

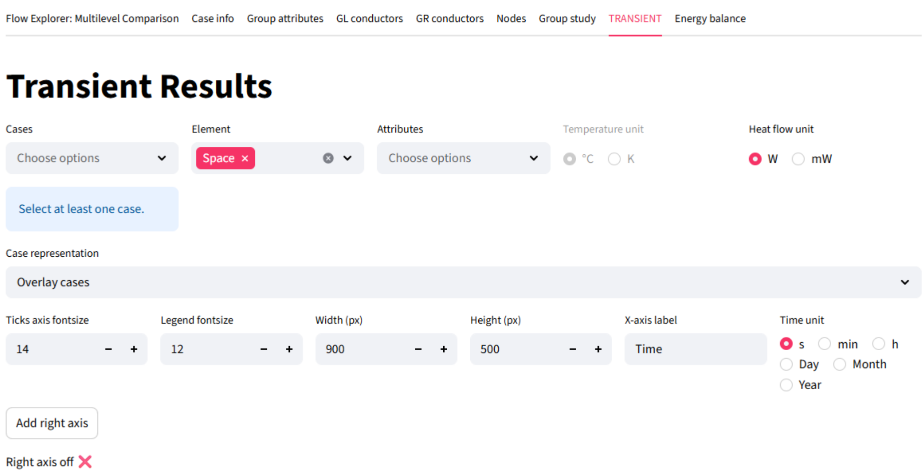


Figure 7.1: Configuration area for selecting cases, model elements, represented attributes, display units and graphical settings.

Figure 7.2 illustrates the single-axis representation using the nominal temperature response of `SOLAR_PANEL_1`. The curves show the heating and cooling of the panel throughout the simulated orbital period, while their separation indicates the temperature range among its nodes.

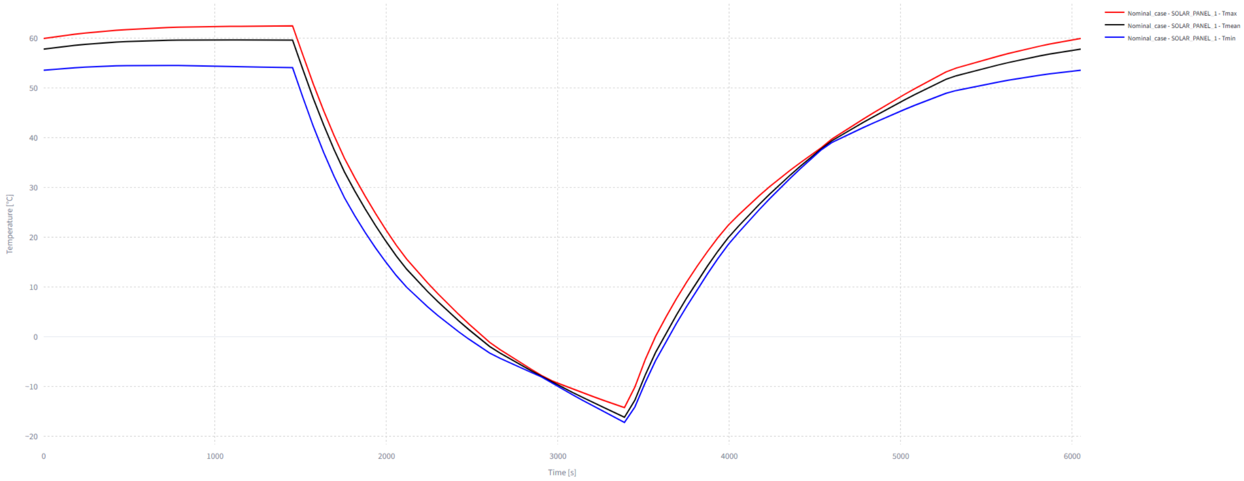


Figure 7.2: Minimum, mean and maximum temperatures of `SOLAR_PANEL_1` for the nominal transient case.

A second vertical axis can be added when attributes with different units are combined in one plot. The attributes assigned to both axes retain a common time coordinate, allowing their evolution to be examined simultaneously.

Figure 7.3 illustrates a dual-axis representation for the battery. Its temperatures are displayed on the left axis, while the internal load Q_I is shown on the right axis. The values assigned during and outside eclipse are given in Table 5.4.

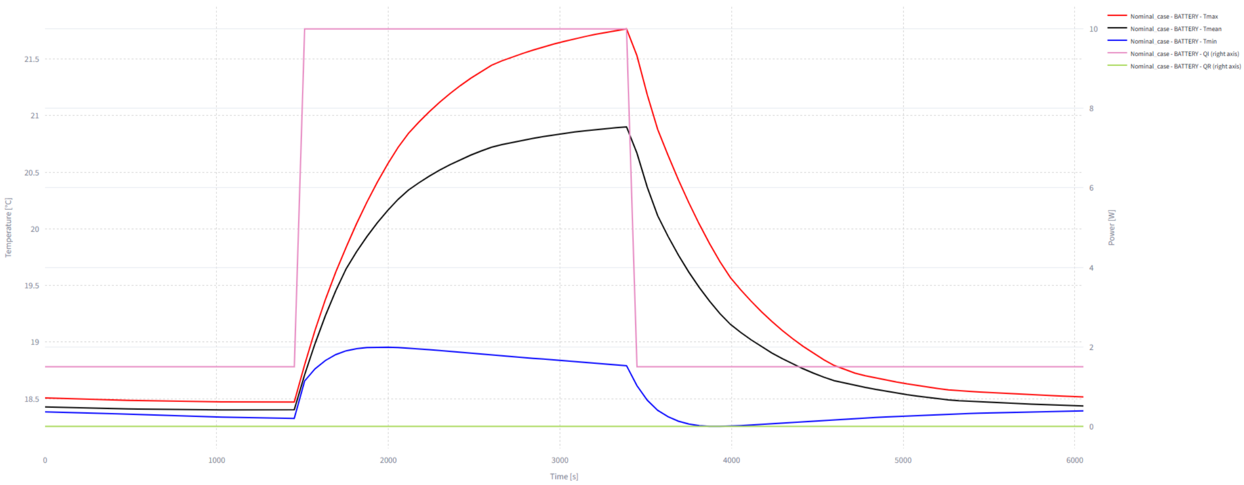


Figure 7.3: Battery temperatures and eclipse-dependent internal load Q_I for the nominal transient case.

7.2. Interface Temperature-Difference Analysis

The second area examines the temperature difference between two selected groups. The interface is defined by a source group and a target group, and several source–target pairs may be added to the same analysis. One or more cases can then be evaluated using the resulting interface selection.

The target list may be restricted to groups connected to the source through explicit GL or GR conductors. This filter limits the selection to interfaces represented directly in the thermal network. The user can add one target at a time or include all the connections detected for the chosen source.

Two calculation bases are available, together with an optional limit for the maximum permitted temperature difference. Figure 7.4 shows the controls used to select the cases, source and target groups, calculation method and acceptance limit.

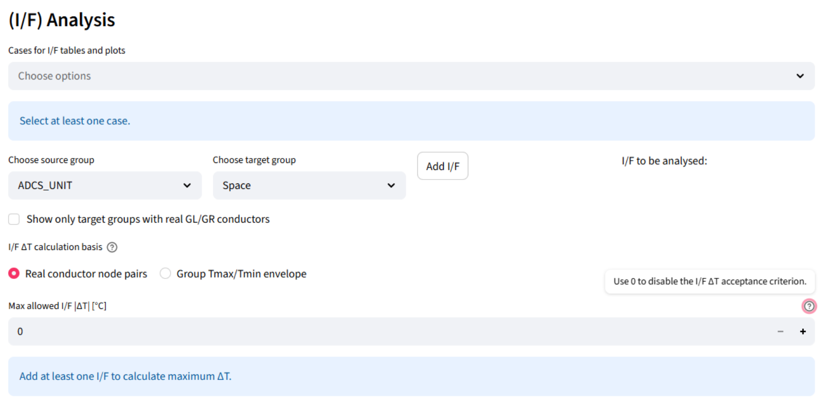


Figure 7.4: Configuration area for selecting cases, source and target groups, the interface calculation basis and the optional temperature-difference limit.

The default calculation basis uses the node pairs connected through explicit GL or GR conductors. The set $\mathcal{P}_{AB}$ contains the pairs (i, j) linking source group A with target group B. The absolute temperature difference associated with each pair is

$$\Delta T_{ij}(t) = |T_i(t) - T_j(t)| \tag{7.5}$$

At every output instant, the interface value corresponds to the largest difference among the connected pairs:

$$\Delta T_{A,B}^{\mathrm{I/F}}(t) = \text{máx}[\Delta T_{ij}(t)]_{(i,j)\in\mathcal{P}_{AB}} \tag{7.6}$$

The maximum difference reached throughout the complete simulation is then

$$\Delta T_{A,B,\text{máx}}^{\mathrm{I/F}} = \text{máx}\left[\Delta T_{A,B}^{\mathrm{I/F}}(t)\right]_t \tag{7.7}$$

At each output instant, the tool represents the hotter and colder temperatures of the connected pair producing the largest absolute difference. This pair may change throughout the simulation. A vertical marker identifies the global maximum, and the summary output records the nodes and coupling associated with that event.

The example presented below uses `OBC_BOX` as the source and `RADIATOR` as the target. This interface is representative because the thermal network contains explicit thermal couplings between both elements, as described in Section 5.4. The critical node pair identified for the nominal case is connected through a radiative conductor, GR.

Figure 7.5 shows that the maximum temperature difference reaches 20.8 °C at 3388.1 s.

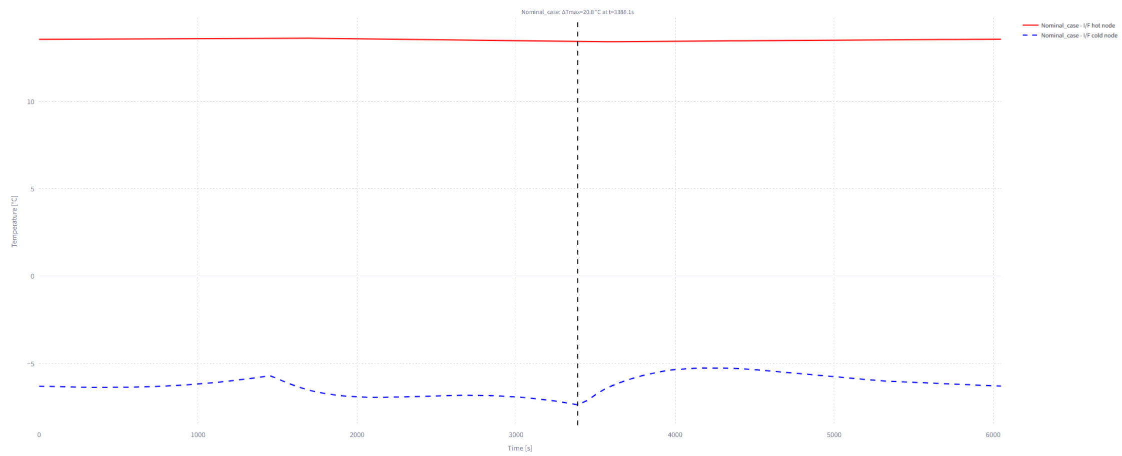


Figure 7.5: Hot and cold temperatures of the connected pair producing the largest interface difference at each output instant between `OBC_BOX` and `RADIATOR`. The vertical marker identifies the global maximum in the nominal transient case.

All targets connected to the same source group through explicit conductors were then added. For each source–target pair, Table 7.1 records the maximum absolute temperature difference, the event time, the hot and cold temperatures, the critical node pair and the coupling type or types defined for that pair.

Table 7.1: Maximum interface temperature differences detected from `OBC_BOX` in the nominal transient case.

Target group	$\lvert\Delta T\rvert_{\text{máx}}$ [°C]	Time [s]	T_{hot} [°C]	T_{cold} [°C]	Hot–cold nodes	Coupling
`RADIATOR`	20.81	3388.1	13.44	-7.37	31023–35506	GR
`BODY`	7.71	6050.2	12.87	5.16	31014–11102	GR
`BODY_PLATE_MED`	2.41	3388.1	13.70	11.30	12602–31011	GR
`BODY_PANEL`	7.71	6050.2	12.87	5.16	31014–11102	GR
`BODY_PLATE_SUP`	2.95	1512.6	13.67	10.73	31022–13603	GR
`Space`	283.74	2238.6	13.74	-270.00	31022–99999	GR
`BODY_PLATE_INF`	6.01	1875.6	18.18	12.17	12000–31008	GR

Note: `Space` is included because GR links to the environmental boundary are also detected. Equal values may occur when target groups overlap within the model hierarchy.

The alternative calculation basis uses the complete temperature envelopes of the source and target groups. It does not require the nodes producing the result to be connected by an explicit conductor. The corresponding difference is defined as

$$\Delta T^{\text{env}}_{A,B}(t) = \text{máx}[|T_{A,\text{máx}}(t) - T_{B,\text{mín}}(t)|\,, |T_{B,\text{máx}}(t) - T_{A,\text{mín}}(t)|] \tag{7.8}$$

This method provides a more conservative comparison between the complete groups, but the nodes defining the result do not necessarily represent a physical interface. The conductor-based calculation therefore offers greater traceability whenever a direct GL or GR connection is available.

A maximum permitted difference ΔT_{lim} can also be introduced. The available margin is calculated from

$$M_{\text{I/F}} = \Delta T_{\text{lim}} - \Delta T_{\text{máx}} \tag{7.9}$$

where $\Delta T_{\text{máx}}$ denotes the maximum value obtained with the selected calculation basis. A non-negative margin produces a `PASS` result, whereas a negative value produces `FAIL`. Setting ΔT_{lim} to zero disables this criterion, as in the nominal example presented above.

7.3. Temperature Variation over Selected Time Windows

The third area evaluates how much the temperature of each node changes over a selected time window. This representation complements the absolute temperatures examined in Section 7.1, since a component may remain within its operating limits while still undergoing a significant increase or decrease during a shorter interval.

The configuration starts with the selection of one or more cases, model elements and time windows. Several windows can be processed simultaneously, and the resulting variations may be represented through their maximum and minimum nodal values or supplemented with the corresponding mean curve. The elapsed-time unit, graph dimensions, axis labels and font sizes can also be modified. Two optional limits evaluate the maximum absolute temperature change and its average magnitude over the selected interval.

Figure 7.6 shows the controls used to define the cases, elements, time windows, representation method and acceptance criteria.

For a requested window τ, the temperature variation of node n is calculated from

$$\Delta T_n(t;\tau) = T_n(t) - T_n(t - \tau^*) \tag{7.10}$$

where τ^* is the effective interval reproduced by the output-time vector. When the simulation uses a uniform output step Δt_m, the number of samples required for the calculation is

$$k_\tau = \left\lceil \frac{\tau}{\Delta t_m} \right\rceil \qquad \tau^* = k_\tau \Delta t_m \tag{7.11}$$

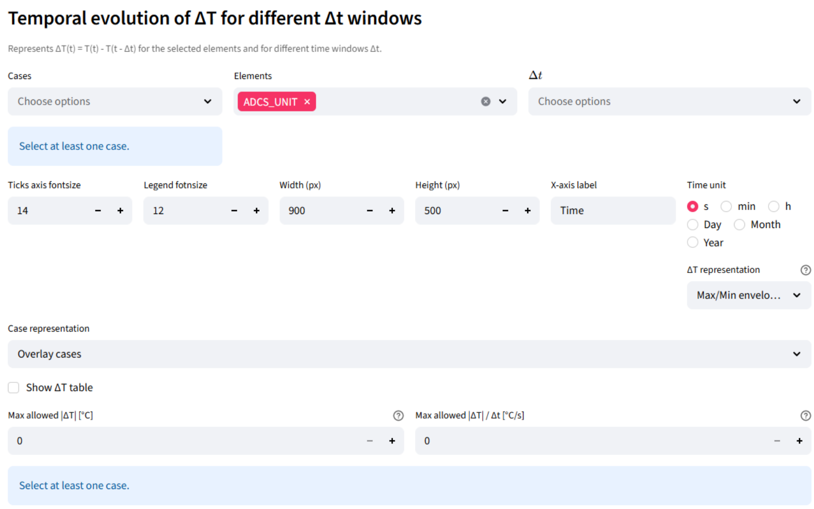


Figure 7.6: Configuration area for selecting the cases, model elements, time windows, representation method and optional temperature-variation limits.

The ceiling operation prevents the effective interval from being shorter than the requested value. If both values differ, the interface reports τ^* together with the selected window.

For a group g, the maximum–minimum representation is obtained from the nodal variations:

$$\Delta T_{g,\text{mín}}(t;\tau) = \text{mín}[\Delta T_n(t;\tau)]_{n\in\mathcal{N}_g} \tag{7.12}$$

$$\Delta T_{g,\text{máx}}(t;\tau) = \text{máx}[\Delta T_n(t;\tau)]_{n\in\mathcal{N}_g} \tag{7.13}$$

Positive values indicate that the node has warmed during the preceding interval, whereas negative values indicate cooling. The separation between both curves shows whether the nodes of the selected element follow a similar evolution or experience different temperature changes.

The nominal example uses `BODY_PANEL` and requested windows of 10 and 30 minutes. This element contains several nodes with sufficiently different responses to make the maximum–minimum envelopes visible. The effective intervals obtained from the available output grid are approximately 10.084 and 30.251 minutes.

The longer window produces a wider envelope because it accumulates the temperature change over a greater part of the orbital cycle. However, the largest absolute variation does not by itself indicate how rapidly the change occurs. For this purpose, the application also calculates the average magnitude over the effective interval:

$$R_{T,n}(t;\tau) = \frac{|\Delta T_n(t;\tau)|}{\tau^*} \tag{7.14}$$

This metric represents an average change over the selected window rather than an instantaneous derivative. Its value therefore depends on the duration of the interval.

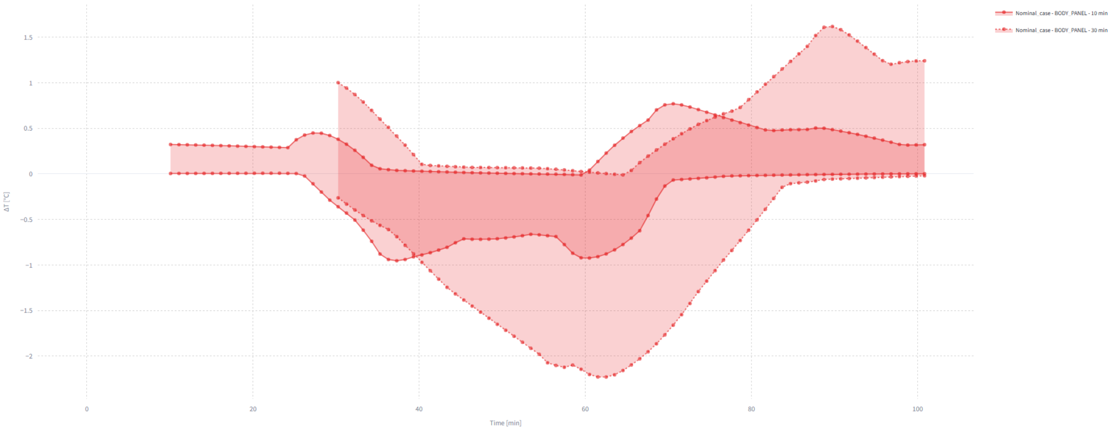


Figure 7.7: Maximum and minimum temperature variations of `BODY_PANEL` over 10- and 30-minute windows in the nominal transient case.

Table 7.2 summarises the critical results obtained for both windows. It includes the minimum and maximum temperature changes, their event times and nodes, the largest absolute variation and its corresponding average magnitude.

Table 7.2: Temperature variations of `BODY_PANEL` over the selected time windows in the nominal transient case.

Requested window (min)	**Effective window (min)**	$\Delta T_{\text{mín}}$ **(°C)**	$t_{\text{mín}}$ **(min)**	**Critical node**	$\Delta T_{\text{máx}}$ **(°C)**	$t_{\text{máx}}$ **(min)**	**Critical node**	$\lvert\Delta T\rvert_{\text{máx}}$ **(°C)**	$\lvert\Delta T\rvert_{\text{máx}}/\tau^*$ **(°C/min)**
10	10.084	-0.95	37.31	10602	0.77	70.59	10602	0.95	0.095
30	30.251	-2.23	62.52	10202	1.62	89.75	10602	2.23	0.074

The 30-minute interval reaches the largest absolute variation, 2.23 °C, whereas the 10-minute interval produces the higher average magnitude, 0.095 °C/min. The two results therefore describe different aspects of the transient response: the longer window identifies the greater accumulated change, while the shorter one is more sensitive to rapid variations.

The complete nodal values can also be displayed and exported from the interface. Optional limits for $|\Delta T|$ and $|\Delta T|/\tau^*$ allow margins and acceptance states to be calculated. A value of zero disables the corresponding criterion, as in the example presented here.

7.4. Transient Thermal Summary Table

The fourth area condenses the transient temperature results of the selected elements into a single tabular output. Instead of representing their complete evolution, it identifies the main temperature extrema and the largest and smallest spatial temperature ranges reached during the simulation.

This provides an initial overview of the most relevant values before examining a specific element in greater detail.

The configuration requires one or more cases and model elements. The case representation selector also enables the direct and difference-based options introduced in Section 6.4.2. The nominal example uses a single case, since this is sufficient to explain the content and organisation of the resulting table.

Figure 7.8 shows the selection area of this function.

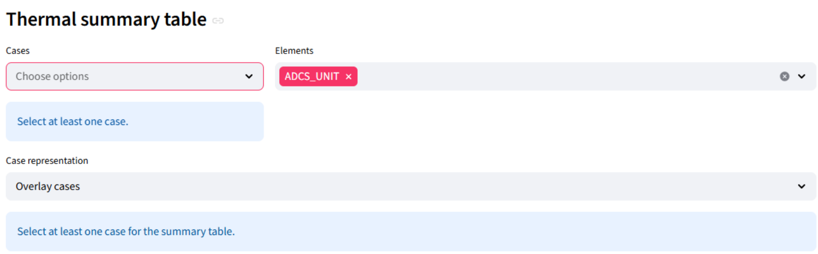


Figure 7.8: Configuration area for selecting the cases and model elements included in the transient thermal summary.

The set of output instants contained in the transient result file is denoted by $\mathcal{T}_{\text{out}}$. The global maximum and minimum temperatures of a group g are calculated as:

$$T_{g,\text{máx}}^{\text{glob}} = \text{máx}[T_n(t)]_{\substack{n\in\mathcal{N}_g \\ t\in\mathcal{T}_{\text{out}}}} \tag{7.15}$$

$$T_{g,\text{mín}}^{\text{glob}} = \text{mín}[T_n(t)]_{\substack{n\in\mathcal{N}_g \\ t\in\mathcal{T}_{\text{out}}}} \tag{7.16}$$

For each extreme, the output also identifies the node and simulation time at which it occurs. These data allow the critical value to be located without inspecting the temperature curves of all the nodes individually.

The table also evaluates the spatial temperature range of each group at every output instant:

$$\Delta T_g^{\text{sp}}(t) = T_{g,\text{máx}}(t) - T_{g,\text{mín}}(t) \tag{7.17}$$

Its maximum and minimum values throughout the simulation are then obtained from:

$$\Delta T_{g,\text{máx}}^{\text{sp}} = \text{máx}\left[\Delta T_g^{\text{sp}}(t)\right]_{t\in\mathcal{T}_{\text{out}}} \tag{7.18}$$

$$\Delta T_{g,\text{mín}}^{\text{sp}} = \text{mín}\left[\Delta T_g^{\text{sp}}(t)\right]_{t\in\mathcal{T}_{\text{out}}} \tag{7.19}$$

The corresponding times and hot–cold node pairs are retained in the output. This calculation differs from the time-window method of Section 7.3. There, $\Delta T_n(t;\tau)$ compares one node at two different instants, whereas $\Delta T_g^{\text{sp}}(t)$ compares different nodes of the same element at one instant.

The original output contains the temperature extrema and spatial ranges in a single table. For readability in the document, the nominal results are divided into two complementary tables. The first contains the global maximum and minimum temperatures, together with their event times and critical nodes, while the second contains the maximum and minimum spatial ranges reached by the same elements. The associated node pairs identify the locations that define each range.

Table 7.3: Global temperature extrema of the selected elements in the nominal transient case.

Element	$T_{\text{máx}}$ [°C]	$t_{T_{\text{máx}}}$ [s]	**Node**	$T_{\text{mín}}$ [°C]	$t_{T_{\text{mín}}}$ [s]	**Node**
`ADCS_UNIT`	17.06	2904.1	33002	16.29	4719.2	33005
`SOLAR_PANEL_2`	62.37	1452.1	73003	-17.31	3388.1	75201
`SOLAR_PANEL_1`	62.45	1452.1	70005	-17.25	3388.1	72501
`MLI_PANELS`	134.77	1452.1	50800	-109.03	4356.2	51003
`COMMS_BOX`	19.10	2057.1	32000	17.53	3630.1	32005
`OBC_BOX`	13.74	2238.6	31022	11.30	3388.1	31011
`RADIATOR`	0.05	4235.2	35504	-7.37	3388.1	34002
`BATTERY`	21.77	3388.1	30000	18.25	3932.7	30005
`CAMERA_CYL`	-2.15	1452.1	42500	-28.22	3388.1	43500
`CAMERA_ELEC`	11.07	2722.6	40501	10.49	0.0	40500
`CAMERA_BASE`	-7.64	1452.1	40404	-16.45	3569.6	40000
`BODY_PLATE_INF`	18.55	1694.1	12001	16.10	4477.2	12502
`BODY_PLATE_MED`	13.79	2299.1	13002	13.24	3690.7	12601
`BODY_PLATE_SUP`	11.03	2117.6	13600	10.60	4900.7	13103
`BODY_PANEL`	18.87	1452.1	10601	5.05	4053.7	11002

Table 7.4: Maximum and minimum spatial temperature ranges of the selected elements in the nominal transient case.

Element	$\Delta T^{\text{sp}}_{\text{máx}}$ (°C)	$t_{\Delta T_{\text{máx}}}$ [s]	**Hot–cold nodes**	$\Delta T^{\text{sp}}_{\text{mín}}$ (°C)	$t_{\Delta T_{\text{mín}}}$ [s]	**Hot–cold nodes**
`ADCS_UNIT`	0.62	3993.2	33002–33005	0.49	1694.1	33002–33005
`SOLAR_PANEL_2`	8.42	1512.6	73003–75202	0.29	2904.1	73000–75201
`SOLAR_PANEL_1`	8.52	1512.6	70005–72500	0.31	2904.1	70005–72501
`MLI_PANELS`	241.69	4053.7	50801–51003	111.33	2722.6	50500–51003
`COMMS_BOX`	1.19	3448.6	32000–32005	0.45	1452.1	32002–32005
`OBC_BOX`	2.36	3388.1	31022–31011	1.07	5445.2	31023–31011
`RADIATOR`	5.66	786.5	35504–34002	5.31	4235.2	35504–34002
`BATTERY`	2.98	3388.1	30000–30005	0.12	484.0	30002–30004
`CAMERA_CYL`	0.00	3448.6	42500–43500	0.00	1754.6	43500–42500
`CAMERA_ELEC`	0.43	1573.1	40501–40500	0.42	3751.2	40501–40500
`CAMERA_BASE`	7.44	1452.1	40404–40008	4.94	1996.6	40009–40008
`BODY_PLATE_INF`	2.23	1633.6	12001–12502	1.75	3630.1	12001–12502
`BODY_PLATE_MED`	0.44	3085.6	13002–12601	0.29	4356.2	13000–12601
`BODY_PLATE_SUP`	0.27	5203.2	13600–13103	0.18	3388.1	13600–13103
`BODY_PANEL`	13.63	1452.1	10601–11002	12.64	3388.1	10701–11002

The two outputs serve different purposes. Global extrema indicate the highest and lowest temperatures reached by an element during the complete simulation, while the spatial ranges describe its internal thermal non-uniformity at a given instant. Their combined use therefore distinguishes a general shift in component temperature from a large difference between its individual nodes.

The summary can be generated for any selection of elements and exported for later inspection. No separate graph is required in this area, since its main function is to provide a compact numerical screening of the transient solution.

7.5. Conductive and Radiative Heat-Flow Evolution

The fifth function examines the heat transferred between a selected source and target throughout the transient simulation. After choosing one or more cases, the user defines both elements and selects the conductive, radiative or total heat-flow components to be represented. The output can be expressed in watts or milliwatts, while the horizontal axis uses the selected elapsed-time unit.

Figure 7.9 shows the corresponding controls. The three available components are Q_{GL}, Q_{GR} and Q_{GT}, representing the linear conductive, radiative and combined exchanges, respectively. Their sign follows the source–target order defined in the interface.

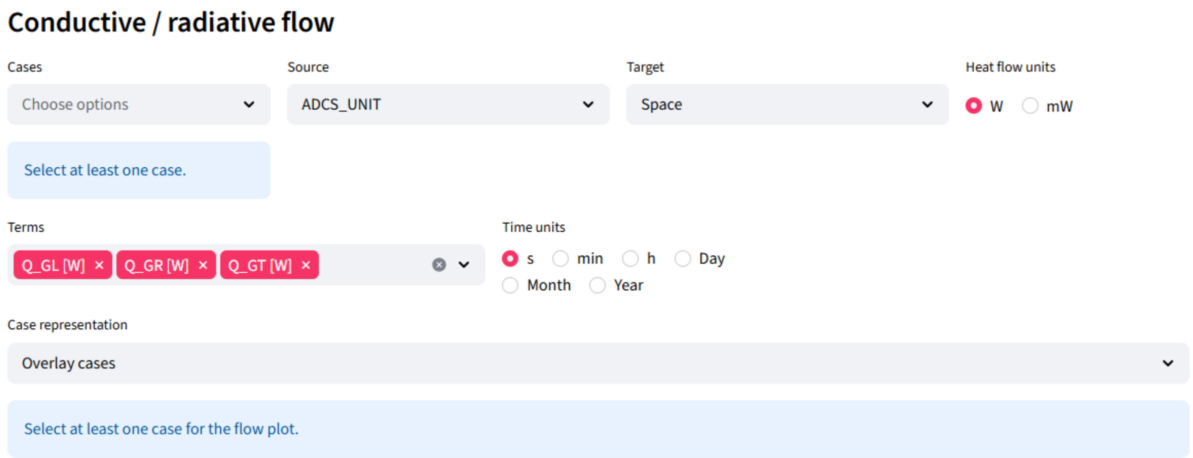


Figure 7.9: Configuration area for selecting the cases, source and target elements, heat-flow components and display units.

The set $\mathcal{P}_{AB}^{\mathrm{GL}}$ contains the node pairs connected by linear conductors between source A and target B. The conductive contribution is calculated as:

$$Q_{\mathrm{GL},A\leftarrow B}(t) = \sum_{(i,j)\in\mathcal{P}_{AB}^{\mathrm{GL}}} \mathrm{GL}_{ij}\left[T_j(t) - T_i(t)\right] \tag{7.20}$$

where node i belongs to the source and node j to the target. The radiative contribution is obtained from the pairs contained in $\mathcal{P}_{AB}^{\mathrm{GR}}$:

$$Q_{\mathrm{GR},A\leftarrow B}(t) = \sigma \sum_{(i,j)\in\mathcal{P}_{AB}^{\mathrm{GR}}} \mathrm{GR}_{ij} \left[T_{j,K}^4(t) - T_{i,K}^4(t)\right] \tag{7.21}$$

The temperatures are converted to kelvin before evaluating the fourth-power term. The total exchange combines both mechanisms:

$$Q_{\mathrm{GT},A\leftarrow B}(t) = Q_{\mathrm{GL},A\leftarrow B}(t) + Q_{\mathrm{GR},A\leftarrow B}(t) \tag{7.22}$$

With this ordering, a positive value represents heat entering the selected source from the target. A negative value indicates heat leaving the source towards the target. Reversing the source and target therefore reverses the sign of the represented exchange.

The nominal example uses `BATTERY` as the source and `BODY_PLATE_INF` as the target. The three heat-flow components are represented in watts over the complete simulation. Figure 7.10 shows that all values remain negative, indicating a continuous transfer from the battery towards its supporting plate.

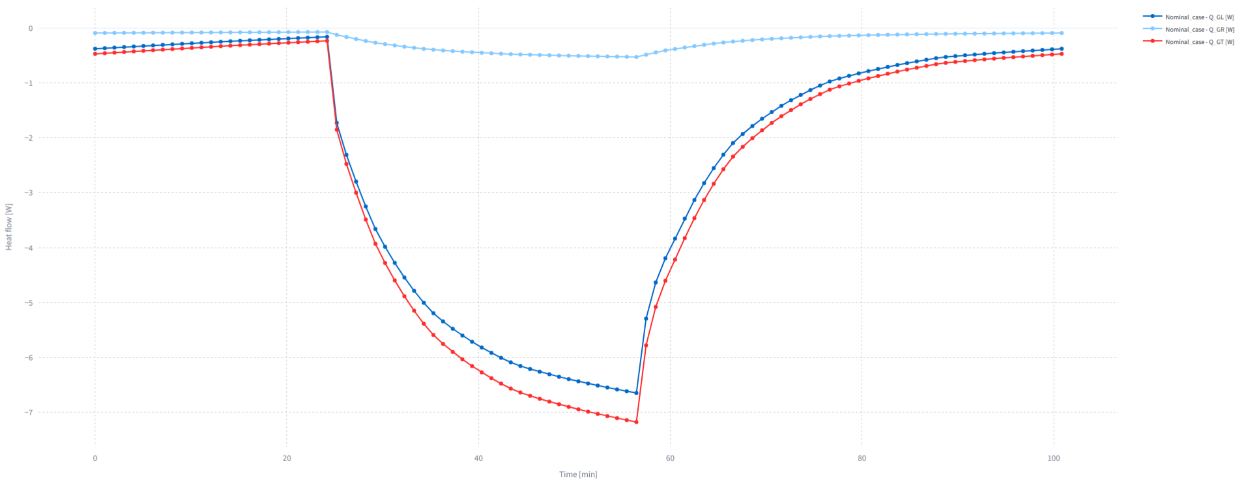


Figure 7.10: Conductive, radiative and total heat flows between `BATTERY` and `BODY_PLATE_INF` in the nominal transient case. Negative values indicate heat leaving the battery.

The conductive term provides the main contribution, while the radiative exchange remains comparatively small. Consequently, Q_{GT} follows the shape of Q_{GL}, with a slightly larger outward magnitude. The transfer increases during the eclipse interval and reaches its most negative value at approximately 3388.1 s.

The interface also generates a table containing the value of each heat-flow component at every output instant. Since the complete export would be unnecessarily long for the document, Table 7.5 summarises only the minimum and maximum values and their corresponding times.

Table 7.5: Extreme conductive, radiative and total heat-flow values for the `BATTERY—BODY_PLATE_INF` selection in the nominal transient case.

Term	Minimum [W]	$t_{\text{mín}}$ [s]	Maximum [W]	$t_{\text{máx}}$ [s]
Q_{GL}	-6.648	3388.1	-0.161	1452.1
Q_{GR}	-0.529	3388.1	-0.073	1452.1
Q_{GT}	-7.177	3388.1	-0.235	1452.1

The total heat flow ranges from approximately -0.235 W to -7.177 W. The negative sign is retained in the table because it forms part of the source–target convention, rather than indicating the magnitude alone.

7.6. Absorbed and Released Thermal Energy

The final area integrates the thermal-power terms associated with a selected element over time. Its configuration begins with the selection of one or more cases and a model element. The user then defines the power contributions to be included and chooses the time, power and energy units used in the representation.

The available power terms comprise Q_A, Q_E, Q_S, Q_I and Q_R, together with the net conductive and radiative exchanges with nodes outside the selected group, $Q_{\text{GL,ext}}$ and $Q_{\text{GR,ext}}$. A reconstructed thermal balance is also available and is calculated as:

$$Q_{\text{balance},g}(t) = Q_{A,g}(t) + Q_{E,g}(t) + Q_{S,g}(t) + Q_{I,g}(t) + Q_{R,g}(t) + Q_{\text{GL,ext},g}(t) + Q_{\text{GR,ext},g}(t) \quad (7.23)$$

where the external conductive and radiative terms contain only exchanges between nodes belonging to group g and nodes outside that group. Internal connections between two nodes of the selected element are excluded because they redistribute energy within the group without changing its total balance.

Figure 7.11 shows the controls used to select the cases, element, power terms and display units. The power curves may be expressed in watts or milliwatts, while the time-integrated energy is represented in joules or watt-hours.

For a selected power term $Q_{x,g}$, the net energy transferred from the start of the simulation to time t is

$$E_{x,g}(t) = \int_{t_0}^{t} Q_{x,g}(\xi)\,\mathrm{d}\xi \quad (7.24)$$

The numerical integration follows the output-time vector contained in the result file. Between two successive instants, the trapezoidal rule gives

$$E_{x,g}(t_k) = E_{x,g}(t_{k-1}) + \frac{Q_{x,g}(t_{k-1}) + Q_{x,g}(t_k)}{2}\,(t_k - t_{k-1}) \quad (7.25)$$

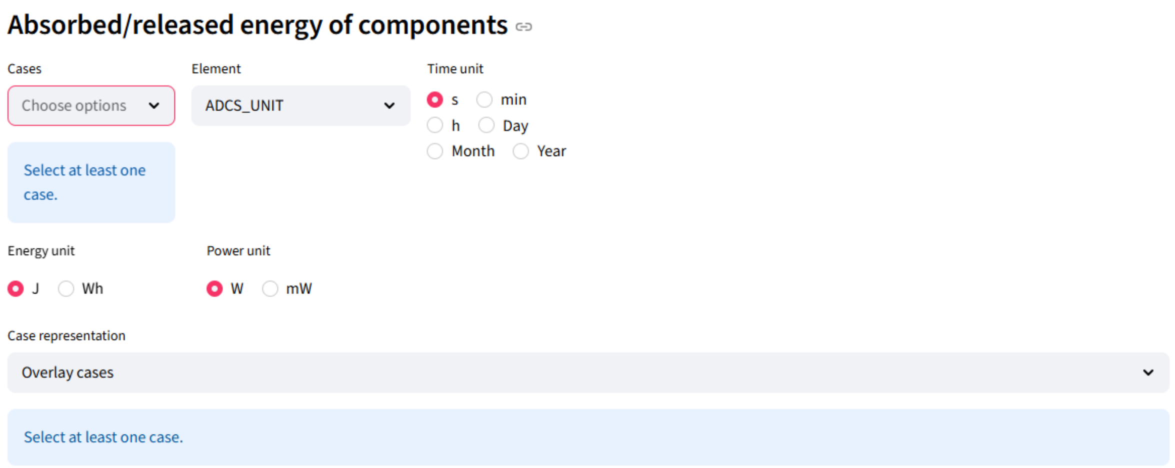


Figure 7.11: Configuration area for selecting the cases, model element, thermal-power terms and units used in the energy calculation.

with $E_{x,g}(t_0) = 0$. When time is expressed in seconds, the resulting energy is obtained directly in joules. Its conversion to watt-hours follows

$$E_{x,g}^{\mathrm{Wh}}(t) = \frac{E_{x,g}^{\mathrm{J}}(t)}{3600} \tag{7.26}$$

For the transient cases considered in this work, the nodal capacitances remain constant over the exported interval. The change in thermal energy stored by a selected group can therefore be evaluated independently from the nodal temperatures as

$$\Delta U_g(t) = \sum_{n \in \mathcal{N}_g} C_n \left[T_n(t) - T_n(t_0) \right] \tag{7.27}$$

This quantity provides an independent reference for assessing the post-processed balance reconstruction. The corresponding energy-closure residual is defined as

$$R_{E,g}(t) = E_{\mathrm{balance},g}(t) - \Delta U_g(t) \tag{7.28}$$

where $E_{\mathrm{balance},g}$ is obtained by integrating $Q_{\mathrm{balance},g}$. A non-negligible residual indicates that the reconstructed exchange terms should not be interpreted as a closed energy balance for that selection. The sign of the power term determines the evolution of its time-integrated energy. A positive value represents energy entering the selected group and increases the integrated value, whereas a negative value represents energy leaving the group and decreases it. When the power becomes zero, the corresponding value remains constant. The resulting curve is therefore not necessarily monotonic and represents the net energy transferred relative to the initial instant.

The nominal example uses the group `SOLAR_PANELS` and represents the albedo and direct-solar contributions Q_A and Q_S. Power is expressed in watts, energy in watt-hours and elapsed time in minutes. Both the instantaneous power curves and their time-integrated values are displayed in the same graph using separate vertical axes.

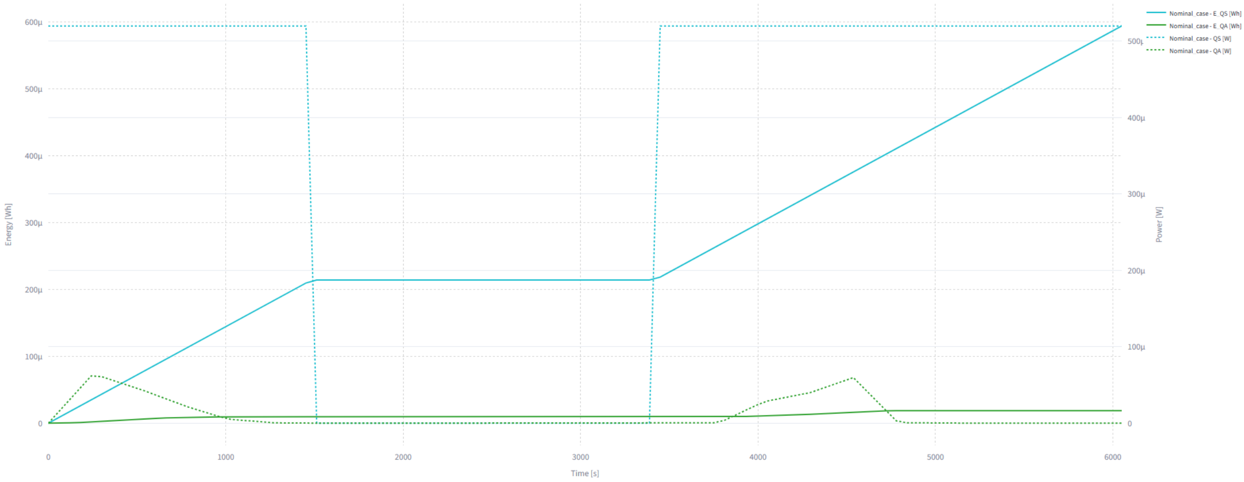


Figure 7.12: Albedo and direct-solar thermal powers and their time-integrated energy contributions for `SOLAR_PANELS` in the nominal transient case.

The direct-solar contribution dominates while the spacecraft remains illuminated and becomes zero during eclipse. Consequently, its integrated energy increases during the illuminated intervals and remains constant while direct solar radiation is absent. The albedo contribution is smaller and varies according to the orbital position, producing a more gradual change in its integrated value.

The graph illustrates the relation between an instantaneous thermal input and the net energy transferred over a finite interval.

The same integration can be applied to individual loads, external heat exchanges or the reconstructed balance Q_{balance}. The individual curves represent the time-integrated contribution of the corresponding exported or reconstructed power term. For Q_{balance}, its integral is compared with ΔU_g through the closure residual defined above. Only when this residual is sufficiently small can the integrated reconstructed balance be identified with the change in thermal energy stored by the selected element.

These results describe thermal power and thermal energy obtained from the ESATAN-TMS solution. They do not represent the electrical energy generated by the solar cells. The conversion of the incident solar load into electrical power and its use within the battery model are treated separately in the electrical-balance analysis.

8. Electrical Power Balance and Battery Modelling

The `Energy balance` tab estimates electrical generation, load demand and battery response from the transient thermal results. The absorbed solar load of the selected panel group is converted into electrical power, while the internal dissipations of the selected equipment provide an approximation of the electrical demand. Their difference determines the surplus available for battery charging or the deficit to be covered during discharge.

The calculation is performed after the ESATAN-TMS solution has been completed. The module reads Q_S, Q_I and the required temperature time series from the imported `.TMD` files, but it neither modifies the thermal results nor returns electrical values to the thermal model. The data flow is therefore one-way: the thermal simulation supplies the inputs used by the electrical balance and battery calculation.

The tab contains two connected areas. The first calculates solar generation, electrical demand and net power. The second uses the resulting surplus and deficit profiles, together with the selected battery temperature and model parameters, to determine stored energy, state of charge, terminal voltage, current, losses and operating restrictions. Solar-panel, load and battery groups are selected from the hierarchy of each imported model, so the calculation is not tied to fixed component names.

8.1. Solar Generation and Electrical Power Balance

8.1.1. Solar-to-Electrical Conversion and Load Demand

The selected solar-panel group defines the nodes considered in the conversion. When a thermo-optical property set is selected from the imported results, only the nodes shared by the group and that property set are retained. When manual thermo-optical values are introduced, the complete selected group is used. The absorbed direct-solar load $Q_S(t)$ is obtained from these nodes. If $Q_{S,\mathrm{inc}}(t)$ denotes their incident solar power and α is the corresponding solar absorptivity, the thermal result satisfies:

$$Q_S(t) = \alpha Q_{S,\mathrm{inc}}(t) \tag{8.1}$$

The application reconstructs the incident contribution as

$$Q_{S,\mathrm{inc}}(t) = \frac{Q_S(t)}{\alpha} \tag{8.2}$$

and estimates the electrical power generated by the cells from

$$Q_{S,\mathrm{elec}}(t) = \eta(t) Q_{S,\mathrm{inc}}(t) = \frac{\eta(t)}{\alpha} Q_S(t) \tag{8.3}$$

where $\eta(t)$ is the electrical conversion efficiency. A residual post-processed coefficient is also calculated as

$$\alpha_{\mathrm{th}}(t) = \alpha - \eta(t) \tag{8.4}$$

This coefficient is retained only to show the fraction remaining after the electrical estimate. It does not replace the absorptivity used by ESATAN-TMS, modify the solved value of Q_S or feed any result back into the thermal model. The admissible range is therefore restricted to $0 < \eta(t) < \alpha \leq 1$.

The efficiency can be introduced as a constant or as an expression of the temperature associated with the selected solar-cell nodes:

$$\eta(t) = f_\eta[T_\eta(t)] \tag{8.5}$$

Here, T_η is the arithmetic mean temperature of the solar-cell nodes retained for the conversion. With manual thermo-optical values, the complete selected solar-panel group is used instead. This option represents the temperature sensitivity of the conversion without introducing a complete current–voltage or maximum-power-point model.

The resulting electrical power is a simplified post-processing estimate based on the thermal nodes assigned to the solar-cell surfaces. It does not separately model cell packing factor, inactive panel area, coverglass and interconnection losses, string mismatch, maximum-power-point tracking, converter efficiency, harness losses, shadowing or end-of-life degradation. Consequently, the calculated generation should be interpreted as the electrical potential associated with the selected thermal representation, rather than as a fully qualified solar-array output.

The electrical demand is estimated from the positive internal dissipations of the selected load groups. For a set $\mathcal{G}_{\mathrm{load}}$, the application calculates

$$Q_{\mathrm{load}}(t) = \sum_{g \in \mathcal{G}_{\mathrm{load}}} \mathrm{máx}[Q_{I,g}(t), 0] \tag{8.6}$$

For the selected electronic units, Q_I represents the prescribed internal dissipation associated with operation and is therefore used as an estimate of the corresponding electrical demand. The result is not a complete spacecraft power budget and does not separately include converter losses, wiring losses or power exported from the selected units. A hierarchy parent and one of its children must not be selected simultaneously when both contain the same nodes, since their contributions would be added twice.

The balance before battery operation is

$$P_{\mathrm{net}}(t) = Q_{S,\mathrm{elec}}(t) - Q_{\mathrm{load}}(t) \tag{8.7}$$

The solar contribution supplies the load first. The remaining non-negative surplus and deficit are therefore:

$$P_{\text{sur}}(t) = \text{máx}[P_{\text{net}}(t), 0] \tag{8.8}$$

$$P_{\text{def}}(t) = \text{máx}[-P_{\text{net}}(t), 0] \tag{8.9}$$

Only P_{sur} can charge the battery, while P_{def} defines the demand to be covered during discharge.

8.1.2. Input Configuration and Results

The upper part of the tab groups the inputs required by the previous relations. The user selects the cases, the representation mode, the solar-panel group and the equipment included in Q_{load}. The conversion efficiency is entered as a constant or an equation, while the solar absorptivity is read from the thermo-optical property sets contained in the imported result files. Manual values may also be entered when a different definition is required.

Figure 8.1 shows the main configuration area. The selected thermo-optical set is identified by α, ε, the associated node count and the corresponding surface labels. The present conversion uses α; ε is displayed to preserve the complete surface-property reference selected by the user.

The interface provides pre-filled values to avoid re-entering a complete configuration whenever the application is opened. These values are editable starting points and do not restrict the analysis. The settings explicitly selected by the user, rather than the initially displayed convenience values, define the results obtained from the electrical balance.

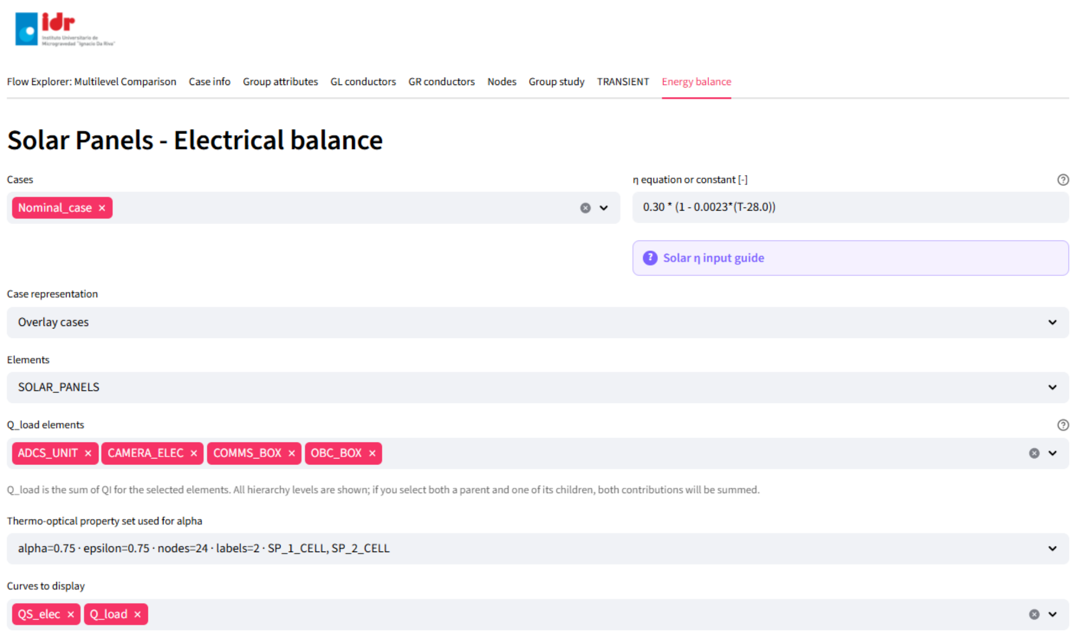


Figure 8.1: Configuration of the solar-panel electrical conversion, load selection and graphical output.

For the nominal calculation, the selected thermal group is `SOLAR_PANELS` and the solar-cell property set has $\alpha = 0.75$. The electrical efficiency is defined as

$$\eta(t) = \eta_{\text{ref}} \left[1 - k_\eta \left(T_\eta(t) - T_{\text{ref}}\right)\right] \tag{8.10}$$

where $\eta_{\text{ref}} = 0.30$ is the reference efficiency, $T_{\text{ref}} = 28\,^{\circ}\text{C}$ is the reference cell temperature at which this efficiency is defined, and $k_\eta = 2.30 \times 10^{-3}\,^{\circ}\text{C}^{-1}$ is the relative maximum-power temperature coefficient [19].

Thus, when $T_\eta(t) = T_{\text{ref}}$, the relation returns $\eta(t) = \eta_{\text{ref}}$. The value T_{ref} therefore provides the fixed baseline for the temperature correction, whereas $T_\eta(t)$ is the time-dependent mean temperature obtained from the selected solar-cell nodes. The coefficients apply from 28 to 140 °C; values below this range are therefore treated as an extrapolation of the simplified relation [19].

The demand includes the `ADCS_UNIT`, `CAMERA_ELEC`, `COMMS_BOX` and `OBC_BOX` groups. These are mutually disjoint Level-3 equipment groups in the hierarchy, so their internal dissipations can be added without repeating nodes. The `BATTERY` group is excluded because its prescribed Q_I represents thermal dissipation within the battery rather than an independent electrical load. The battery-model losses are calculated separately during post-processing.

The available thermo-optical definitions may be inspected directly before a property set is selected. Figure 8.2 shows this list for the imported model and allows the solar-cell definition to be identified from its labels and absorptivity.

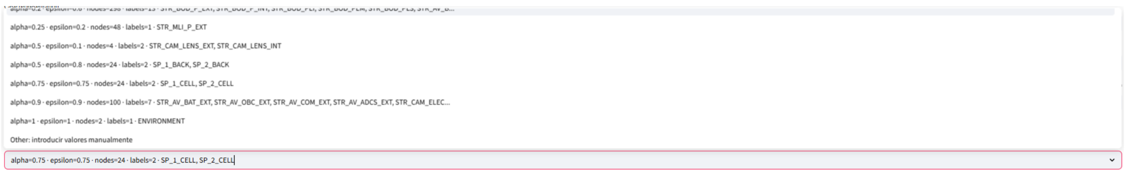


Figure 8.2: Selection of the thermo-optical property set associated with the solar-cell nodes.

The graphical output can include Q_S, $Q_{S,\text{elec}}$ and Q_{load}. Their representation identifies the illuminated and eclipse intervals and shows when electrical generation exceeds or falls below the selected demand. The reconstructed incident power $Q_{S,\text{inc}}$ remains available in the calculation table.

Figures 8.3 and 8.4 show the nominal electrical balance obtained using two alternative definitions of the solar-cell efficiency. In the first case, a constant value of $\eta = 0.30$ is applied. In the second, the temperature-dependent expression defined in Equation (8.10) is used.

With constant efficiency, the electrical generation remains approximately constant during each illuminated interval and becomes zero during eclipse. When the temperature-dependent expression is applied, the electrical power varies with the mean temperature T_η of the selected solar-cell nodes. The second formulation is used for the subsequent battery calculation because it preserves the connection between the thermal results and the electrical post-processing model. The comparison between the remaining environmental cases is reserved for Chapter 9.

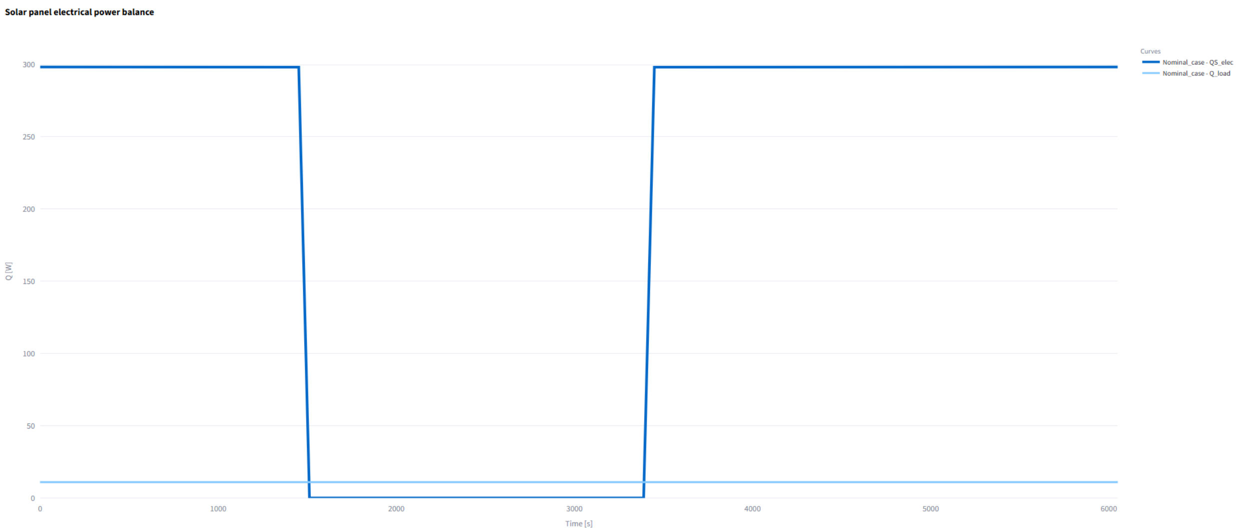


Figure 8.3: Solar electrical generation and selected load demand for the nominal transient case using a constant solar-cell efficiency of $\eta = 0.30$.

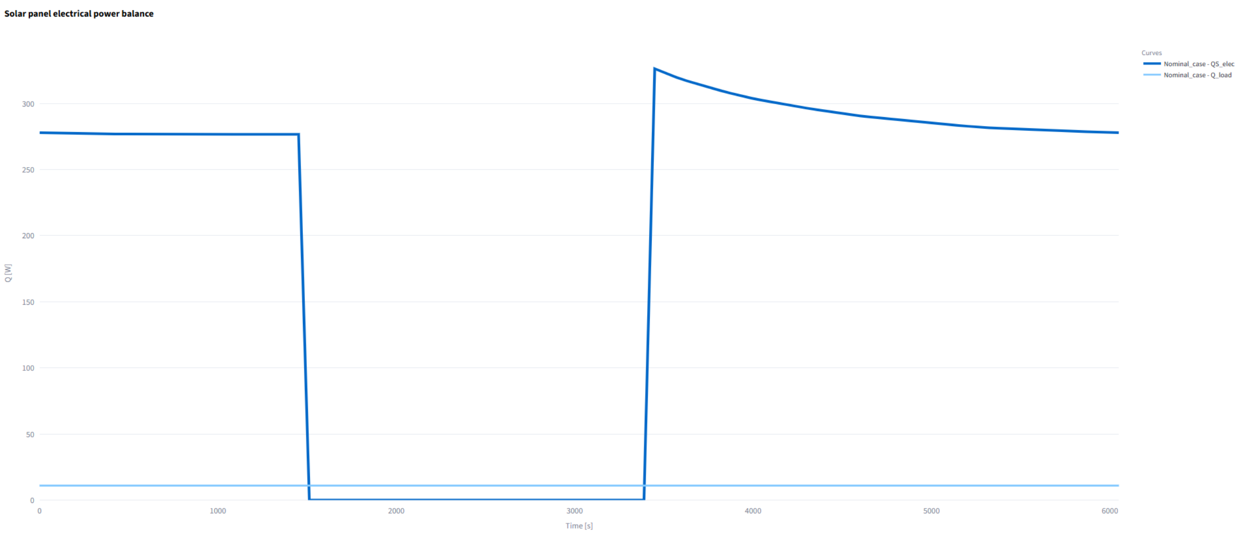


Figure 8.4: Solar electrical generation and selected load demand for the nominal transient case using the temperature-dependent solar-cell efficiency.

The associated calculation table retains the values obtained at every output instant. Its columns include Q_S, $Q_{S,\text{inc}}$, $Q_{S,\text{elec}}$, Q_{load}, α, α_{th} and η. When the efficiency depends on temperature, the value of T_η used in the expression is also included. These results provide the solar-generation and load-demand profiles required by the battery model.

8.2. Battery Model

The battery calculation starts from the solar-generation and load-demand profiles obtained in the first area of the tab. At each interval, solar power supplies the selected load first. Any remaining surplus is offered to the battery, whereas a remaining deficit requests battery discharge. The accepted power depends on stored energy, terminal-voltage limits, internal resistance, charge and discharge efficiencies, charge-control parameters and the battery temperature obtained from the thermal solution.

The battery is recalculated whenever an upstream electrical input or any battery parameter changes. Consequently, the same pack configuration may produce a different response after changing the selected panel group, the load elements, α, the solar-efficiency expression or the imported thermal case.

8.2.1. Energy-Based Formulation and Model Scope

The battery response is represented through an energy-based formulation combined with an equivalent-circuit description. The discharged-energy level ϕ represents the energy removed from a fully charged battery and is used as the main state variable, while separate charge and discharge resistances account for the corresponding internal electrical losses [20].

During discharge, the state evolves as

$$\phi(t) = \phi_0 + \frac{1}{3600} \int_{t_0}^{t} \left[V I_d + R_d I_d^2 \right] \mathrm{d}t \tag{8.11}$$

whereas charging reduces the discharged-energy level according to:

$$\phi(t) = \phi_0 - \frac{1}{3600} \int_{t_0}^{t} \left[V I_c - R_c I_c^2 \right] \mathrm{d}t \tag{8.12}$$

Here, V is the terminal voltage, I_d and I_c are the non-negative discharge and charge current magnitudes, and R_d and R_c are the corresponding internal resistances. Since the transient time vector is expressed in seconds while ϕ is evaluated in watt-hours, the factor $1/3600$ provides the required time conversion.

The corresponding terminal voltages are:

$$V_d(\phi, I_d) = E_d(\phi) - R_d I_d \tag{8.13}$$

$$V_c(\phi, I_c) = E_c(\phi) + R_c I_c \tag{8.14}$$

where $E_d(\phi)$ and $E_c(\phi)$ represent the internal-voltage functions associated with discharge and charge. These functions may be obtained from measured voltage curves, while additional dynamic branches can be introduced when a more detailed transient electrical response is required [20].

The application adopts a simplified implementation of this formulation. Separate charge and discharge resistances and the energy-based state variable are retained, while a common piecewise-linear internal-voltage curve $E(\phi)$ is used for both operating directions. The curve is defined from the selected cut-off, nominal and charge voltage limits together with intermediate interpolation points. No dynamic resistor–capacitor branches are included.

The nominal energy capacity is denoted by $C_{\text{pack,Wh}}$, while $E_{\text{st}}(t)$ represents the stored energy. The internal state used by the application is

$$\phi(t) = C_{\text{pack,Wh}} - E_{\text{st}}(t) \tag{8.15}$$

and the reported state of charge is

$$\text{SOC}(t) = 100\,\frac{E_{\text{st}}(t)}{C_{\text{pack,Wh}}} \tag{8.16}$$

At each numerical substep, the current is obtained from $P = VI$ using the current value of ϕ and the corresponding terminal-voltage relation. The stored energy is updated from the internal power $E(\phi)I$, while the difference between internal and terminal power is accounted for as resistive loss. Separate conversion efficiencies are applied during charge and discharge.

The model is intended to provide a configurable engineering estimate of the battery response for the electrical profiles generated by the application. The selected pack parameters therefore define the study configuration and do not represent a cell-specific calibrated battery model.

8.2.2. Pack Configuration and Charge–Discharge Control

The first battery block connects the model with the imported thermal result and defines the pack configuration. The *Battery TMD element* supplies the mean battery temperature. This temperature is used by the charge-derating function and by efficiency expressions containing T.

All battery parameters shown in the tab are editable. The pre-filled values provide a complete initial configuration for convenient use of the application, while the user can adapt the thermal limits, cell arrangement, capacity, voltage limits, initial SOC, efficiencies, terminal-power limits, internal resistances and CC–CV controls to the requirements of each imported model. Any change triggers a new battery calculation using the selected solar generation, electrical load and battery-temperature inputs.

Figure 8.5 shows the corresponding controls. The lower and upper charge-temperature limits define the interval in which charging is permitted, while the ramp margin introduces a linear transition near each limit.

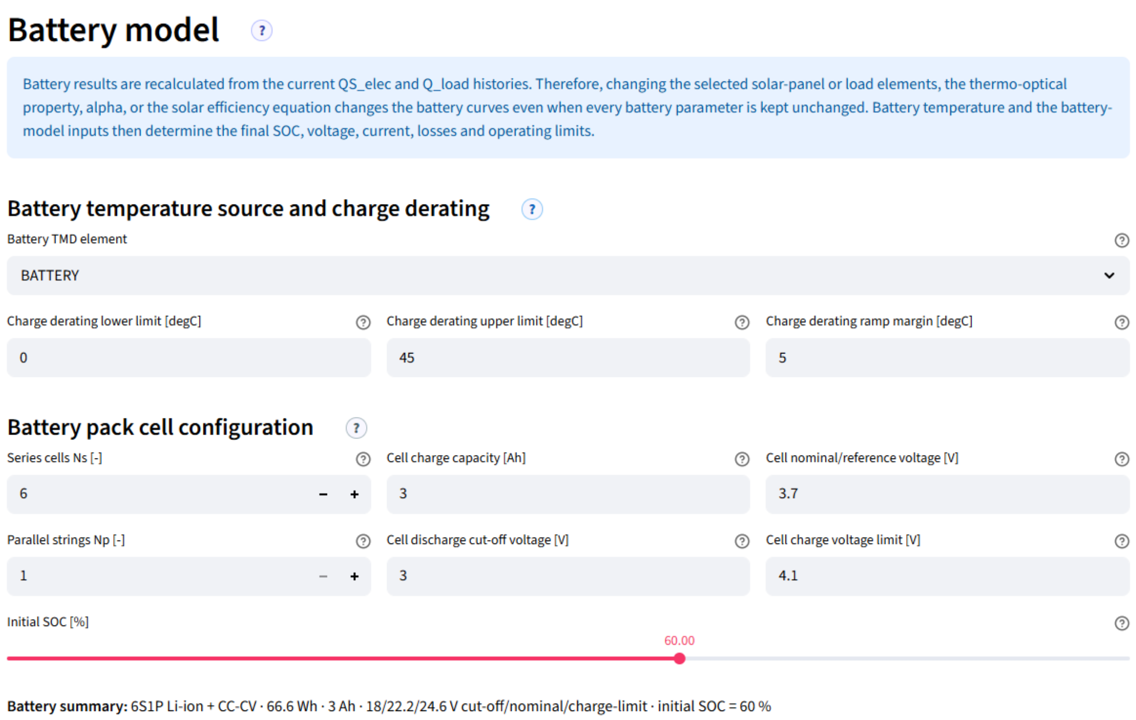


Figure 8.5: Battery-temperature source, charge derating and cell-pack configuration.

The values used below define the common study configuration selected for the case comparison. They are configurable model inputs and, unless explicitly stated otherwise, should not be interpreted as experimentally identified parameters or validated operating limits for a particular flight battery. The selected configuration uses the `BATTERY` thermal group, charge limits of 0 and 45 °C, and a 5 °C derating margin. The pack contains six cells in series and one parallel string. Each cell has a charge capacity of 3 Ah, a cut-off voltage of 3.0 V, a nominal voltage of 3.7 V and a charge limit of 4.1 V. This gives pack limits of 18.0, 22.2 and 24.6 V, a nominal energy capacity of 66.6 Wh, and an initial SOC of 60 %.

For a ramp margin m_T that does not cause the lower and upper transitions to overlap, the charge-derating factor is:

$$f_T(T) = \begin{cases} 0, & T \leq T_{\text{low}}, \\ \dfrac{T - T_{\text{low}}}{m_T}, & T_{\text{low}} < T < T_{\text{low}} + m_T, \\ 1, & T_{\text{low}} + m_T \leq T \leq T_{\text{high}} - m_T, \\ \dfrac{T_{\text{high}} - T}{m_T}, & T_{\text{high}} - m_T < T < T_{\text{high}}, \\ 0, & T \geq T_{\text{high}} \end{cases} \tag{8.17}$$

The factor multiplies both the maximum charge current and the maximum charge power. A value of one permits normal charging, a value of zero blocks it, and intermediate values reduce the accepted surplus. Discharge is not derated by this factor.

The cell arrangement contains N_s cells in series and N_p parallel strings. The pack voltages are

$$V_{x,\text{pack}} = N_s V_{x,\text{cell}}, \qquad x \in \{\text{cut}, \text{nom}, \text{chg}\} \tag{8.18}$$

while the pack capacities become

$$C_{\text{pack,Ah}} = N_p C_{\text{cell,Ah}} \tag{8.19}$$

$$C_{\text{pack,Wh}} = V_{\text{nom,pack}} C_{\text{pack,Ah}} \tag{8.20}$$

The equivalent charge and discharge resistances follow

$$R_{x,\text{pack}} = R_{x,\text{cell}} \frac{N_s}{N_p}, \qquad x \in \{c, d\} \tag{8.21}$$

The initial SOC sets the stored energy at the first output instant:

$$E_{\text{st},0} = \frac{\text{SOC}_0}{100} C_{\text{pack,Wh}} \tag{8.22}$$

The second block defines the electrical efficiencies, terminal-power limits, internal resistances and simplified CC–CV control. Figure 8.6 shows this area of the interface.

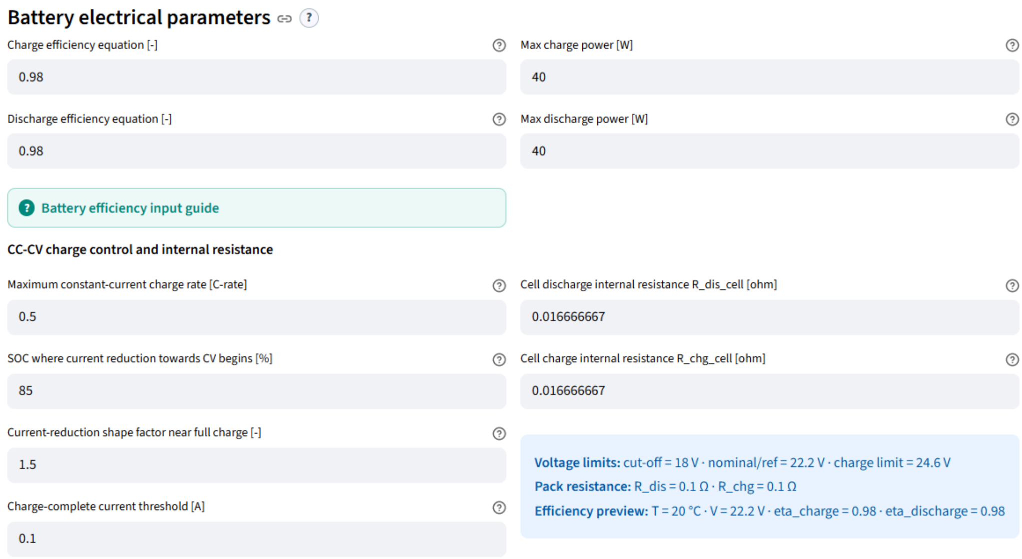


Figure 8.6: Battery efficiencies, power limits, internal resistances and simplified CC–CV charge-control parameters.

The nominal calculation uses constant charge and discharge efficiencies of 0.98, maximum charge and discharge powers of 40 W, and cell resistances of 0.01667 Ω in both operating directions.

The resulting pack resistances are 0.10 Ω. The CC–CV parameters are a maximum charge rate of $0.5C$, a current-reduction transition at 85 % SOC, a taper exponent of 1.5, and a charge-complete current threshold of 0.10 A.

The selected common configuration is numerically compatible with the electrical-demand profiles reconstructed for the three simulated cases. The largest calculated battery discharge power is 16.84 W, below the selected 40 W limit, and no relevant unserved-load energy is reported. The SOC also remains at or above its initial 60 % value in all three cases. These results support the use of the configuration for the multi-case comparison performed in this work, but do not constitute an independent flight-battery qualification or mission-level sizing.

Charge and discharge efficiencies may be constants or functions of battery temperature and terminal voltage:

$$\eta_c(t) = f_c[T_{\text{batt}}(t), V(t)] \qquad \eta_d(t) = f_d[T_{\text{batt}}(t), V(t)] \tag{8.23}$$

During charge, η_c determines the fraction of the available solar surplus that reaches the battery terminals. During discharge, η_d determines the fraction of terminal discharge power delivered to the load. The remaining contribution is reported as conversion loss. The internal resistive loss for each operating direction is

$$P_{\text{loss,int}}(t) = R_x I_x^2(t), \qquad x \in \{c, d\} \tag{8.24}$$

The maximum charge and discharge powers provide independent terminal-power limits. During the constant-current part of charging, the current ceiling is

$$I_{\mathrm{CC,max}} = C_{\mathrm{rate}} C_{\mathrm{pack,Ah}} \tag{8.25}$$

and is reduced by the thermal factor from Equation (8.17). The terminal voltage must also satisfy

$$E(\phi) + R_c I_c \leq V_{\mathrm{chg,pack}} \tag{8.26}$$

Above the selected transition value $\mathrm{SOC}_{\mathrm{CV}}$, an additional taper factor reduces the charge-current limit:

$$f_{\mathrm{CV}}(\mathrm{SOC}) = \left[\frac{100 - \mathrm{SOC}}{100 - \mathrm{SOC}_{\mathrm{CV}}}\right]^{\gamma}, \qquad \mathrm{SOC} \geq \mathrm{SOC}_{\mathrm{CV}} \tag{8.27}$$

where γ controls the reduction shape. Charging is considered complete when the resulting current falls below the selected threshold in the upper SOC region. Together with the terminal-voltage limit, this SOC-based taper provides a simplified approximation of constant-current and constant-voltage charging; it does not represent a dedicated charger circuit [21]. Any surplus that cannot be accepted because of power, current, voltage, temperature or capacity restrictions is reported as curtailed solar power.

During discharge, the current is limited by the available stored energy, the maximum discharge power and the pack cut-off voltage. Any remaining deficit is reported as unserved load. These variables make the consequences of an undersized battery or restrictive operating limit visible without modifying the upstream thermal results.

8.2.3. Battery Outputs and Summary Results

The first battery graph represents the evolution of state of charge. Rising segments indicate net energy storage, falling segments indicate battery support of the load, and horizontal segments correspond to no net exchange or to an active operating restriction.

The second output combines terminal voltage with separate non-negative charge and discharge currents. Voltage uses the left vertical axis, while both currents use the right axis. The detailed output table also retains a signed battery current, defined as positive during discharge and negative during charge.

The third graph combines the mean battery temperature with the charge-derating factor and the charge and discharge efficiencies. It distinguishes a thermal charging restriction from variations introduced by the selected efficiency expressions.

For the nominal case, the battery temperature remains inside the unrestricted charging interval. The charge-derating factor therefore remains equal to one, while the selected constant charge and discharge efficiencies remain equal to 0.98.

The detailed battery table contains the complete result at every output instant. Its main columns in-

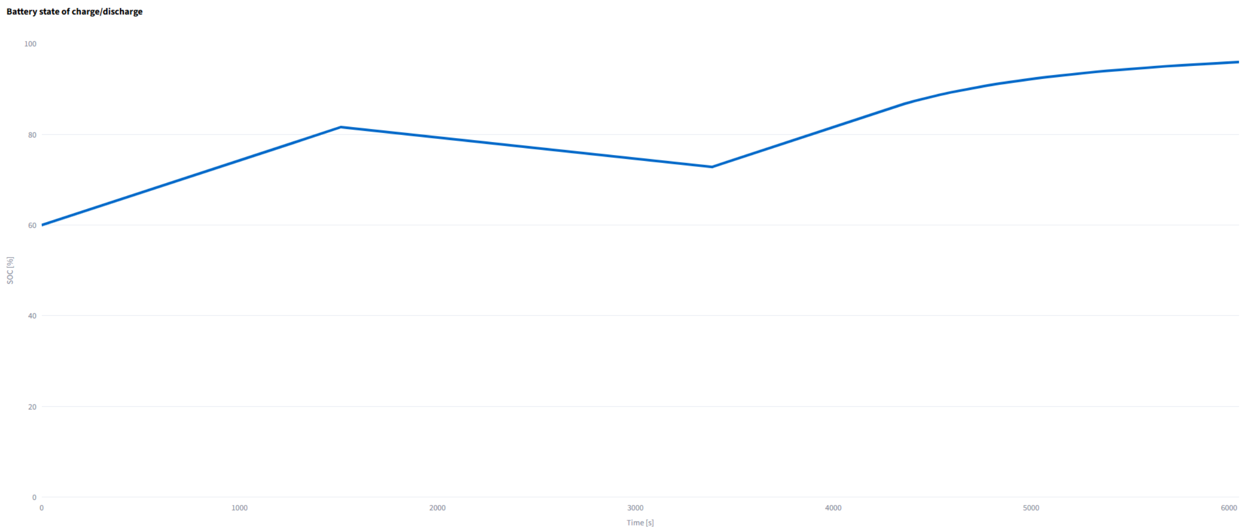


Figure 8.7: Battery state-of-charge evolution for the nominal transient case.

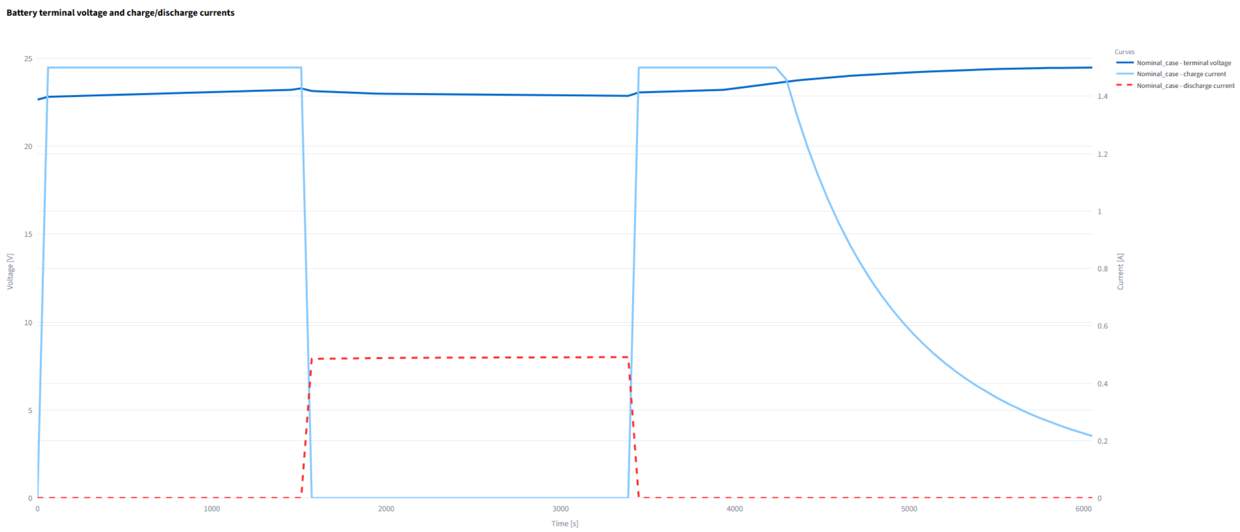


Figure 8.8: Battery terminal voltage and charge/discharge currents for the nominal transient case.

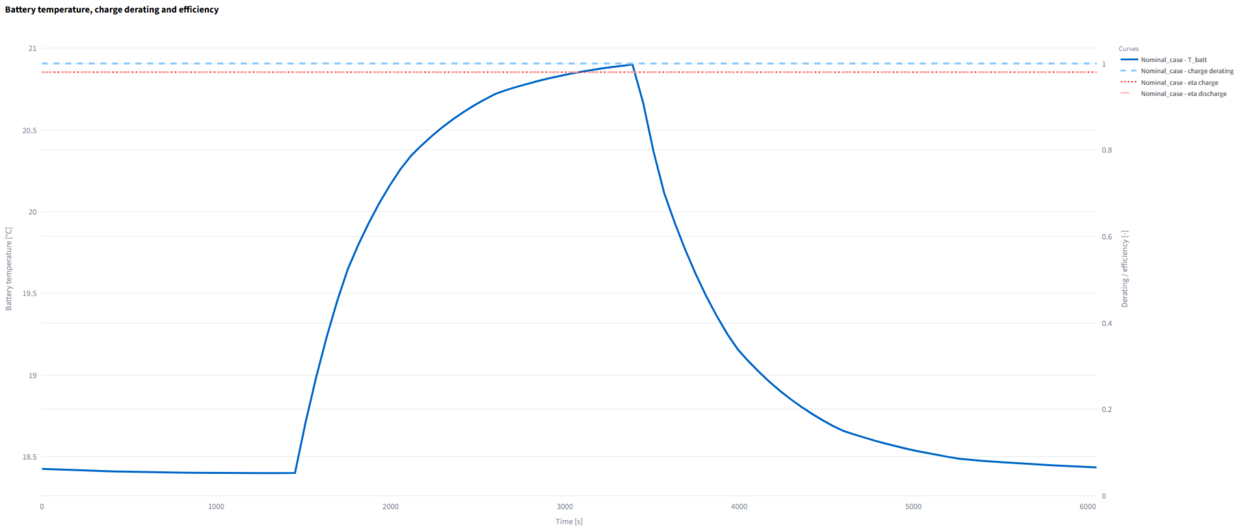


Figure 8.9: Battery temperature, charge-derating factor and charge/discharge efficiencies for the nominal transient case.

clude $Q_{S,\mathrm{elec}}$, Q_{load}, direct solar supply, surplus, deficit, charge and discharge powers, signed battery power, terminal and internal voltage, charge and discharge currents, SOC, ϕ, stored energy, efficiencies, curtailed solar power, unserved load, resistive and conversion losses, battery temperature, derating factor and operating phase.

The summary table provided by the tab reports the final, minimum and maximum SOC, maximum charge and discharge powers, curtailed and unserved energy, integrated internal and conversion losses, battery-temperature range, minimum derating, efficiency ranges and the time spent in the principal charging and discharging phases. Further values, such as voltage and current ranges and final stored energy, can be obtained from the detailed calculation table.

Table 8.1 combines the initial state of charge with the principal results obtained for the nominal case using the selected battery configuration. The same results are used to compare the nominal, hot and cold cases in Chapter 9.

Table 8.1: Initial state and main battery results for the nominal transient case using the selected $6S1P$ configuration.

Item	**Value**	**Unit**
Initial state of charge	60.00	%
Final state of charge	95.98	%
Stored energy at the final instant	63.921	Wh
Terminal-voltage range	22.648–24.463	V
Maximum charge current	1.500	A
Maximum discharge current	0.491	A
Maximum charge power	35.387	W
Maximum discharge power	11.224	W
Battery-temperature range	18.400–20.900	°C
Minimum charge-derating factor	1.000	–
Curtailed solar energy	284.708	Wh
Unserved-load energy	≈ 0	Wh
Integrated internal resistive loss	0.184	Wh
Integrated conversion loss	0.729	Wh

The curtailed, unserved and loss powers stored by the battery calculation are interval-average quantities associated with the preceding numerical interval. Their accumulated energies are therefore obtained as:

$$E = \sum_i \overline{P}_i \, \Delta t_i \tag{8.28}$$

where $\overline{P}_i$ is the average power calculated for interval i. A second trapezoidal integration is not applied to these interval-average values.

The complete calculation therefore provides a continuous post-processing sequence from the absorbed solar load to the resulting battery operating state. The electrical balance determines the power available after supplying the selected loads, while the battery model evaluates how the remaining surplus or deficit affects stored energy, SOC, terminal voltage, current and losses under the imposed operating limits. Battery temperature remains an input obtained from the thermal solution, and the electrical results are not returned to the ESATAN-TMS model. Applying the same electrical and battery configuration to all compatible transient cases provides a common basis for the subsequent multi-case comparison.

9. Multi-Case Results and Discussion

The nominal, hot and cold transient solutions are compared using the common thermal model defined in Chapter 5. Geometry, nodal hierarchy, materials and thermal couplings remain unchanged, while the radiative environment and internal dissipations vary between cases. The common output-time vector permits a direct comparison throughout the complete simulated interval.

Absolute curves preserve the temperature, power, heat-flow and energy levels of each solution. The *Case difference* representation is added when a direct subtraction provides a clearer measure of the separation from the nominal case. The electrical conversion and battery model retain the inputs defined in Chapter 8; consequently, their variations originate from the imported thermal results and the case-dependent load profiles.

9.1. Cases and Comparison Criteria

The three simulations correspond to the cases listed in Table 5.5. The hot case combines greater external heat input, higher internal dissipations and a shorter eclipse. The cold case applies lower external input, reduced dissipations and a longer eclipse. No heater operation is included. The same groups, attributes, source–target ordering and units are retained whenever the cases are superposed.

For a result $x(t)$, the difference between the nominal case N and a comparison case j is defined as

$$\Delta x_{N-j}(t) = x_N(t) - x_j(t), \qquad j \in \{H, C\} \tag{9.1}$$

where H and C denote the hot and cold cases. A negative value of Δx_{N-H} therefore indicates that the hot-case result exceeds the nominal value. This convention is retained in every plot and table generated with *Case difference*. Absolute curves are preferred when the operating level is relevant, whereas differences are used to quantify the separation between solutions or to expose changes hidden by overlapping curves.

9.2. Thermal Response

Applied loads establish the heating and cooling intervals, while the resulting temperatures determine spatial differences, interface gradients and heat exchanges.

9.2.1. Applied Loads and Temperatures

Figure 9.1 combines the direct-solar load and mean temperature of `SOLAR_PANEL_1`. The hot case receives the largest solar input and remains illuminated until the late part of the simulation. Its eclipse covers only approximately the interval from 84 to 96 min. The nominal eclipse extends from about 25 to 56 min, while the cold case remains without direct solar input from approximately 42 to 77 min.

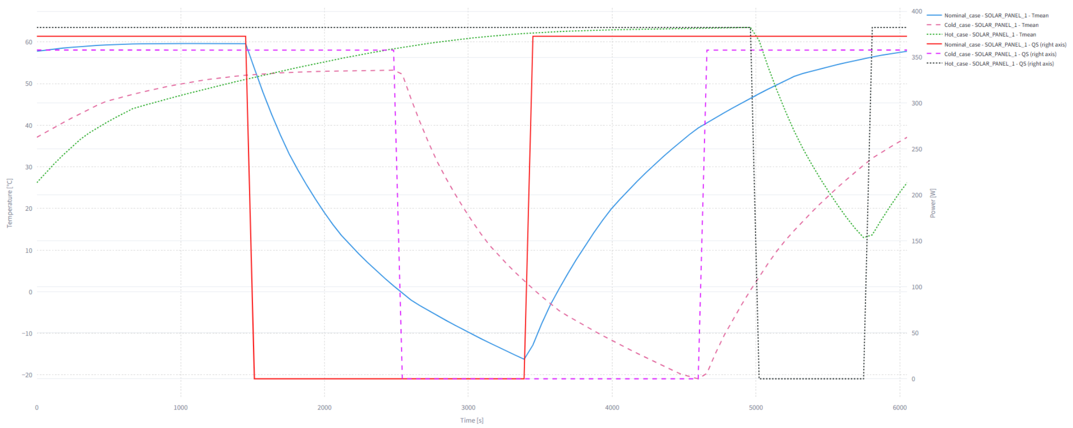


Figure 9.1: Direct-solar thermal load and mean temperature of `SOLAR_PANEL_1` in the nominal, hot and cold cases.

The panel temperature follows the illumination sequence with a delayed response during each transition. The hot case reaches the highest nodal temperature, 65.42 °C, and remains above 12.15 °C because of its short eclipse. The nominal and cold cases cool to −17.25 °C and −21.31 °C, respectively. The maximum spatial range inside `SOLAR_PANEL_1` increases from 5.78 °C in the hot case to 8.52 °C in the nominal case and 10.14 °C in the cold case. The lower overall temperature of the cold solution is therefore accompanied by the largest internal non-uniformity of the selected panel group.

The internal battery dissipation and mean temperature are compared in Figure 9.2. The prescribed load is largest in the hot case and smallest in the cold case, both inside and outside eclipse. The temperature response preserves the same ordering throughout the simulation: the hot battery remains near 36–38 °C, the nominal battery near 18–21 °C and the cold battery near 4 °C.

The nodal extrema in Table 9.1 confirm the general displacement of the thermal response. The hot case produces the highest temperatures for all selected elements, whereas the cold case produces the lowest. The shift is particularly clear in the battery, OBC and radiator. Battery temperatures range from 35.43 to 38.97 °C in the hot case, from 18.25 to 21.77 °C in the nominal case and from 3.50 to 4.60 °C in the cold case. The radiator follows the same trend but remains substantially colder, reaching minima of 5.56 °C, −7.37 °C and −17.26 °C, respectively.

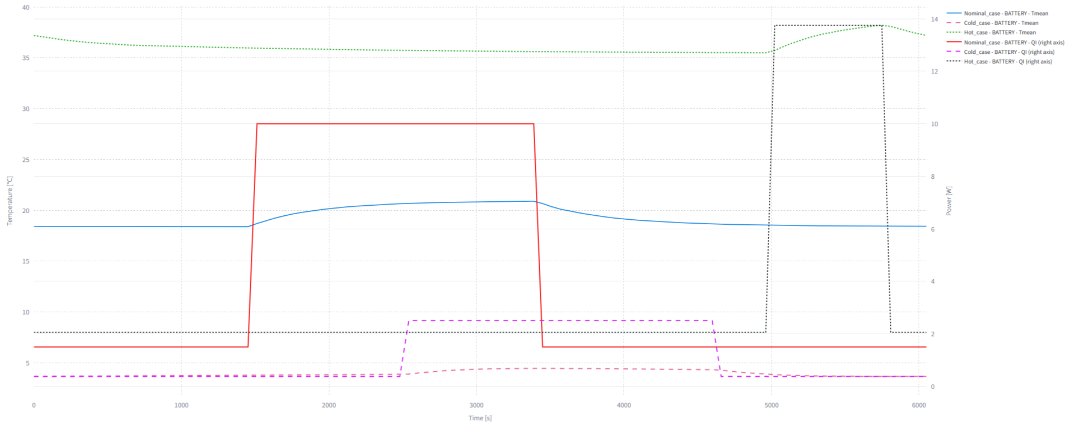


Figure 9.2: Internal heat load and mean temperature of `BATTERY` in the three transient cases.

Table 9.1: Temperature extrema and maximum spatial temperature ranges of the selected elements in the nominal, hot and cold transient cases.

Element	**Case**	$T_{\text{máx}}$ [°C]	$T_{\text{mín}}$ [°C]	$\Delta T^{\text{sp}}_{\text{máx}}$ [°C]
`SOLAR_PANEL_1`	Nominal	62.45	-17.25	8.52
	Hot	65.42	12.15	5.78
	Cold	56.63	-21.31	10.14
`BODY_PANEL`	Nominal	18.87	5.05	13.63
	Hot	35.76	20.90	14.72
	Cold	5.63	-5.55	10.99
`OBC_BOX`	Nominal	13.74	11.30	2.36
	Hot	30.43	27.83	2.59
	Cold	0.85	-1.28	2.05
`RADIATOR`	Nominal	0.05	-7.37	5.66
	Hot	14.49	5.56	6.68
	Cold	-10.85	-17.26	4.88
`BATTERY`	Nominal	21.77	18.25	2.98
	Hot	38.97	35.43	2.90
	Cold	4.60	3.50	0.84

The absolute temperature levels do not determine the spatial ranges by themselves. The hot `BODY_PANEL` reaches the largest internal range, 14.72 °C, but the cold battery presents the smallest battery range, only 0.84 °C. The common network therefore transfers the environmental shift differently according to the geometry, connections and local dissipations of each group.

The separation of the mean battery temperatures is isolated in Figure 9.3. The nominal-minus-cold curve remains positive, between approximately 14 and 17 °C, whereas the nominal-minus-hot curve remains negative and reaches −19.75 °C at 95.80 min. The signs agree with the reference-minus-comparison convention and make the thermal displacement more visible than the superposed absolute curves.

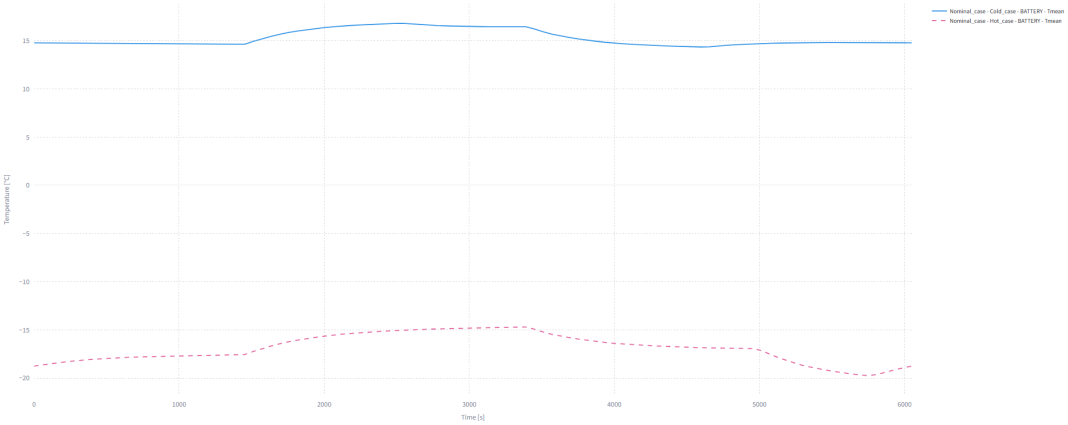


Figure 9.3: Nominal-minus-hot and nominal-minus-cold differences in mean battery temperature.

Table 9.2 extends the same representation to several groups and results at 95.80 min. At this instant, the nominal battery is 19.75 °C colder than the hot battery and 14.79 °C warmer than the cold battery. The large 42.96 °C nominal-minus-hot difference in `SOLAR_PANEL_1` occurs because the nominal panel is illuminated while the hot case is still in eclipse. The table also separates this orbital-phase effect from the prescribed internal dissipations: the battery load difference is −12.25 W for nominal minus hot and 1.13 W for nominal minus cold.

Table 9.2: Case differences at $t = 95.80$ min, corresponding to the maximum absolute nominal–hot difference in mean battery temperature.

	Nominal − Hot			**Nominal − Cold**		
Group	ΔT_{mean} **[°C]**	ΔQ_S **[W]**	ΔQ_I **[W]**	ΔT_{mean} **[°C]**	ΔQ_S **[W]**	ΔQ_I **[W]**
`SOLAR_PANEL_1`	42.96	372.87	0.00	25.57	15.07	0.00
`OBC_BOX`	-16.33	4.54	-1.00	12.89	0.18	2.00
`RADIATOR`	-11.98	12.65	0.00	9.54	0.51	0.00
`BATTERY`	-19.75	0.002	-12.25	14.79	< 0.001	1.13

Note: Differences follow the reference-minus-comparison convention defined in Equation (9.1).

9.2.2. Temperature Variations and Interface Differences

The temperature-change envelopes of `BODY_PANEL` over a nominal 30 min interval are represented in Figure 9.4. The effective interval obtained from the output-time vector is 30.25 min. Negative values identify cooling over the selected window and positive values identify heating.

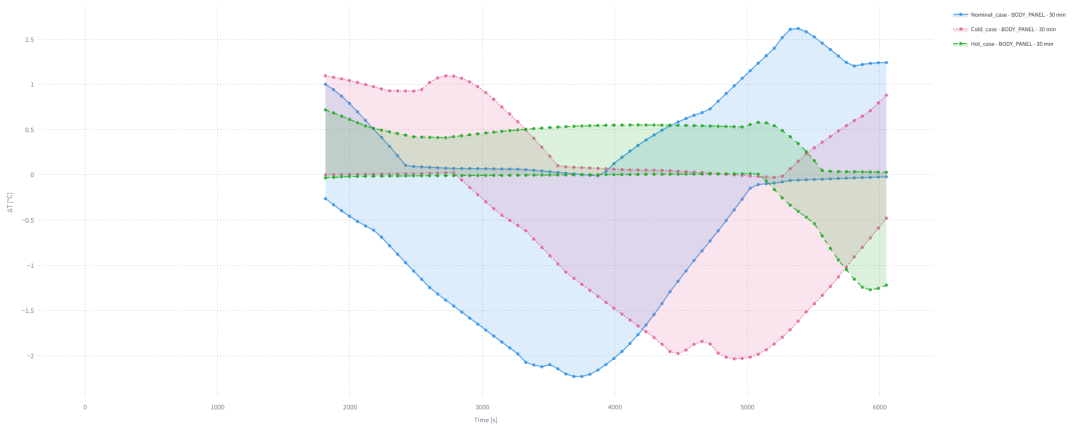


Figure 9.4: Temperature-change envelopes of `BODY_PANEL` over a 30 min window.

The largest absolute variation occurs in the nominal case, with a decrease of 2.23 °C at 62.52 min. The cold case reaches 2.03 °C at 81.68 min, while the hot case remains below 1.28 °C. The highest absolute temperature case is therefore not the one with the fastest change over this interval. The nominal and cold envelopes widen after their respective eclipse transitions, whereas the shorter hot eclipse limits the magnitude of the delayed cooling and reheating response.

The connected-node differences between `OBC_BOX` and `RADIATOR` are shown in Figure 9.5. At each instant, the upper and lower curves correspond to the connected pair that produces the largest absolute difference for that case. The pair can change during the simulation, so the figure is interpreted together with the maximum-value summary in Table 9.3. The maximum interface difference increases from 17.78 °C in the cold case to 20.81 °C in the nominal case and 24.67 °C in the hot case. The nominal and hot maxima are formed by nodes 31023 and 35506, while the cold maximum involves nodes 31017 and 35502. In all three cases, the critical node pair is connected through GR, although other coupling types may be present elsewhere in the interface. The hot environment produces the largest separation between the selected OBC and radiator nodes.

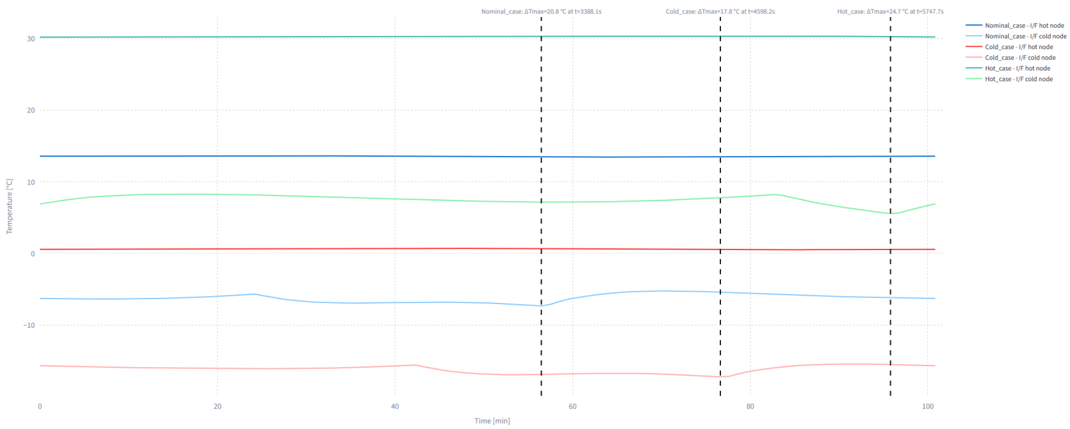


Figure 9.5: Critical connected-node temperatures between `OBC_BOX` and `RADIATOR`.

Table 9.3: Maximum connected-node temperature difference between `OBC_BOX` and `RADIATOR` in the three transient cases.

Case	$\lvert\Delta T\rvert_{\text{máx}}$ [°C]	**Time [min]**	T_{hot} [°C]	T_{cold} [°C]	**Hot–cold nodes**	**Coupling**
Nominal	20.81	56.47	13.44	-7.37	31023–35506	GR
Hot	24.67	95.80	30.22	5.56	31023–35506	GR
Cold	17.78	76.64	0.52	-17.25	31017–35502	GR

9.2.3. Heat Flows and Net Energy Transfer

The total heat flow for the source–target selection `BATTERY`– `BODY_PLATE_INF` follows the convention established in Section 7.5. Positive values indicate heat entering the battery from the plate, whereas negative values indicate heat leaving the battery. Figure 9.6 compares Q_{GT} for the three cases.

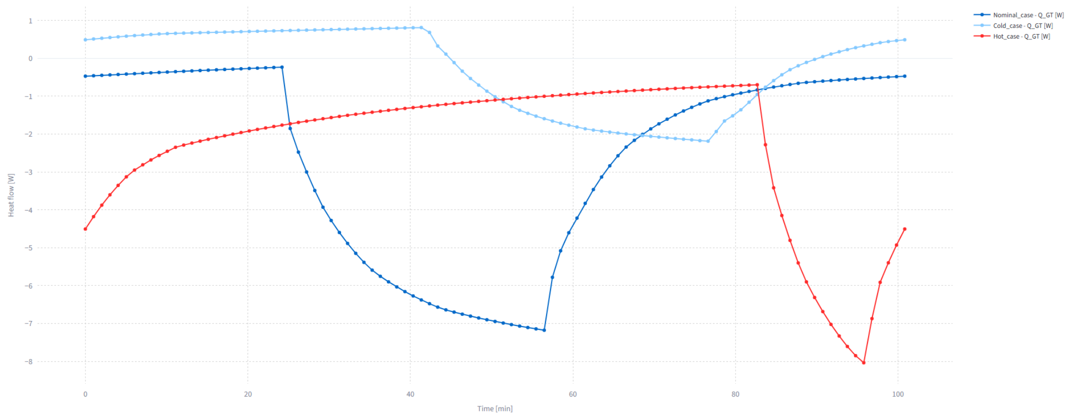


Figure 9.6: Total heat flow between `BATTERY` and `BODY_PLATE_INF`. Negative values indicate heat leaving the battery.

The nominal and hot flows remain negative throughout the interval. Their most negative values are −7.18 W and −8.04 W, respectively, showing that the hotter battery continuously rejects heat to its supporting plate. The cold case ranges from 0.81 W to −2.19 W and changes direction. During part of the orbit the plate warms the battery, while the sign reverses when the battery becomes the warmer side of the connection. In all three cases, the conductive contribution is dominant and the radiative term only modifies the total magnitude.

The corresponding *Case difference* curves are represented in Figure 9.7. Nominal minus hot ranges from −6.17 to 7.50 W, while nominal minus cold ranges from −7.19 to 1.06 W. The sign changes show that a single constant offset cannot describe the effect of the environmental cases. The relative heat transfer depends on the different eclipse intervals and on the case-dependent battery dissipation.

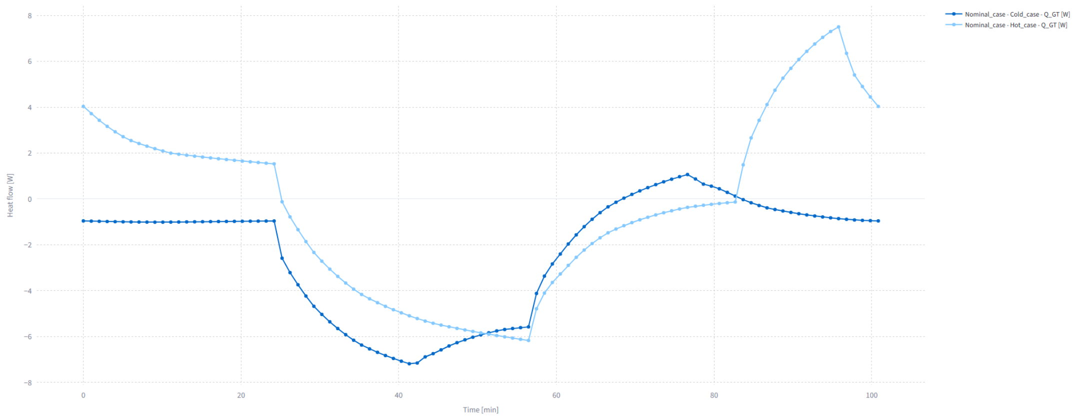


Figure 9.7: Nominal-minus-hot and nominal-minus-cold differences in total battery-to-plate heat flow.

The absorbed direct-solar and albedo contributions of `SOLAR_PANELS` are integrated over time to compare the thermal energy received in the three environments. These contributions are selected because they are directly determined by the orbital radiative conditions and clearly show how the instantaneous input produces a time-integrated energy transfer.

Figure 9.8 represents the direct-solar power Q_S and its time integral E_{Q_S}. The power becomes zero during eclipse, so the corresponding integrated-energy curve remains constant until direct illumination resumes. The different eclipse intervals therefore produce distinct plateaus in the three cases.

The hot case receives the largest direct-solar energy over the complete simulation because its illuminated intervals are longer. The nominal and cold curves contain longer constant segments during eclipse, although their instantaneous power levels remain comparable while illumination is present. The final value of each curve represents the thermal energy transferred by direct solar radiation from the initial instant, rather than energy stored within the panels.

The albedo contribution in Figure 9.9 varies more gradually with orbital position. Unlike Q_S, it does

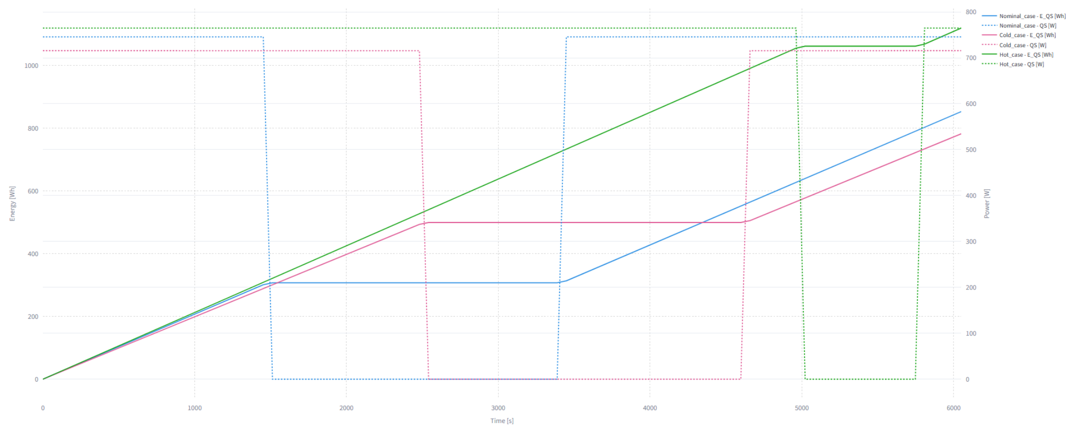


Figure 9.8: Direct-solar thermal power and its time-integrated energy contribution for `SOLAR_PANELS` in the nominal, hot and cold cases.

not follow a two-level illuminated and eclipse pattern. Its integration therefore produces smoother curves and smaller final energy values.

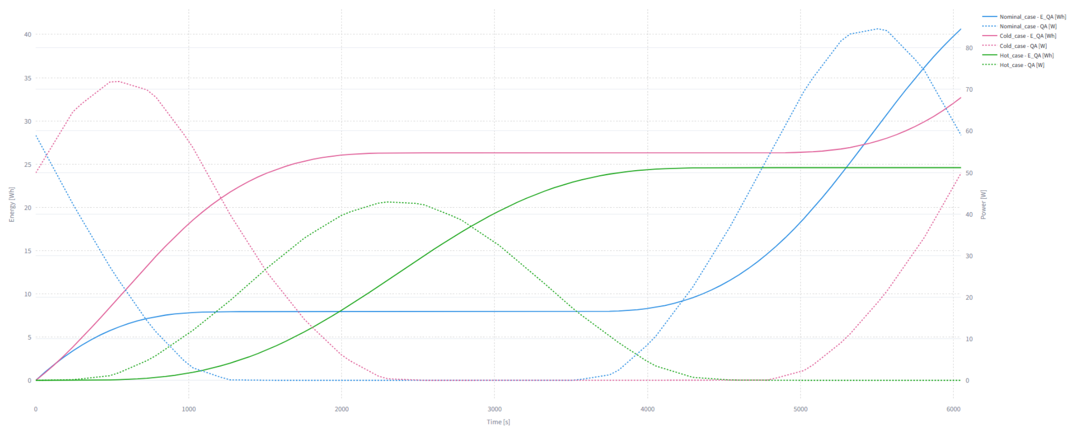


Figure 9.9: Albedo thermal power and its time-integrated energy contribution for `SOLAR_PANELS` in the nominal, hot and cold cases.

The albedo profiles reach their maxima at different orbital positions and become negligible during intervals in which the corresponding planetary reflection is not visible. The final integrated values show that this contribution remains considerably smaller than the direct-solar input, while still differing between the three radiative environments.

The *Case difference* representation in Figure 9.10 isolates the effect of the non-coincident eclipse intervals. The nominal-minus-hot and nominal-minus-cold power differences change abruptly whenever one case is illuminated and the other remains in eclipse. Their integrated curves evolve linearly during these intervals and remain constant whenever the instantaneous difference becomes zero.

This comparison demonstrates that the difference between two integrated energy curves is governed

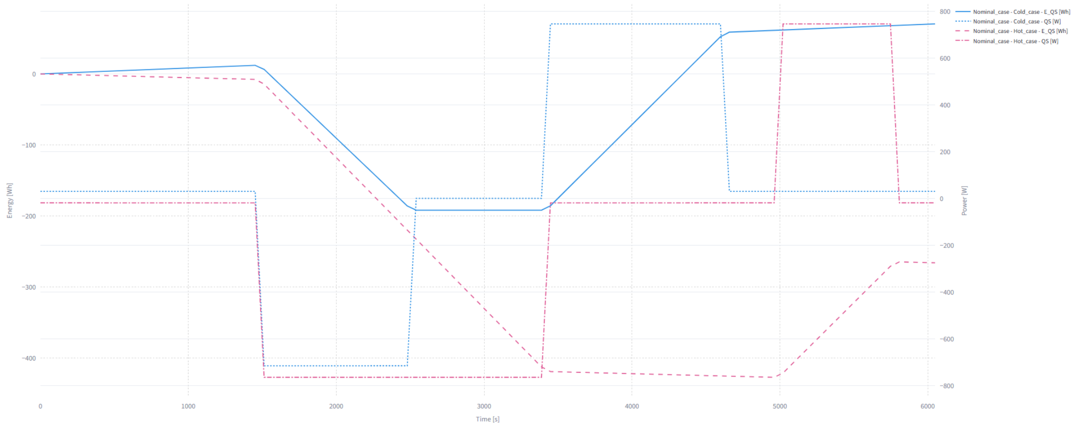


Figure 9.10: Nominal-minus-hot and nominal-minus-cold differences in direct-solar thermal power and its time-integrated energy contribution for `SOLAR_PANELS`.

by both the magnitude of the instantaneous power separation and the duration for which it is maintained. The result is not an absolute stored-energy difference, but the difference between the time-integrated direct-solar contributions of the two cases.

9.3. Electrical and Battery Response

The same solar group, load groups, thermo-optical definition, efficiency expression and battery parameters are used for all three cases. The electrical comparison therefore reflects the case-dependent absorbed solar load, panel temperature, internal dissipations and battery temperature.

9.3.1. Solar Generation and Load Demand

Figure 9.11 compares the estimated solar electrical generation $Q_{S,\mathrm{elec}}$ with the demand formed by the selected equipment groups. Generation becomes zero during the eclipse of each case, while the load changes according to the dissipations defined in Table 5.4. The hot case therefore combines the largest load with the shortest generation interruption; the cold case combines the smallest load with the longest eclipse.

The maximum instantaneous electrical generation is 326.25 W in the nominal case, 316.03 W in the hot case and 317.53 W in the cold case. The hot case receives the largest absorbed solar load, but its higher panel temperature reduces the conversion efficiency and prevents it from producing the largest instantaneous value. The total generated energy is nevertheless highest in the hot case because of its much longer illuminated interval.

The difference curves in Figure 9.12 are dominated by the non-coincident eclipse intervals. Nominal minus hot falls to approximately -289 W when the nominal case enters eclipse while the hot case

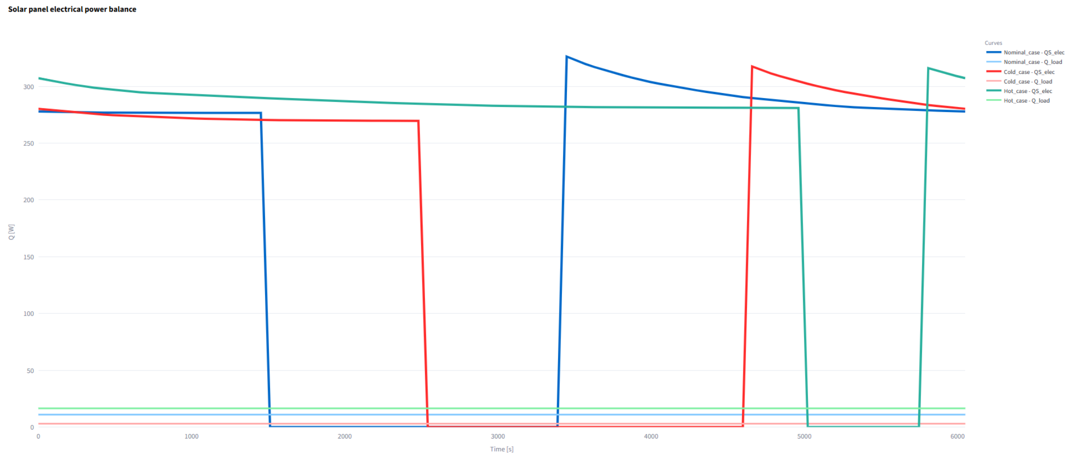


Figure 9.11: Estimated solar electrical generation and selected load demand in the three cases.

remains illuminated, and later rises to about 285 W when the hot case enters its late eclipse. Nominal minus cold reaches 326 W after nominal illumination resumes while the cold case remains in eclipse. When both cases share the same illumination state, the remaining separation is associated with the different incident loads and panel-temperature-dependent efficiencies.

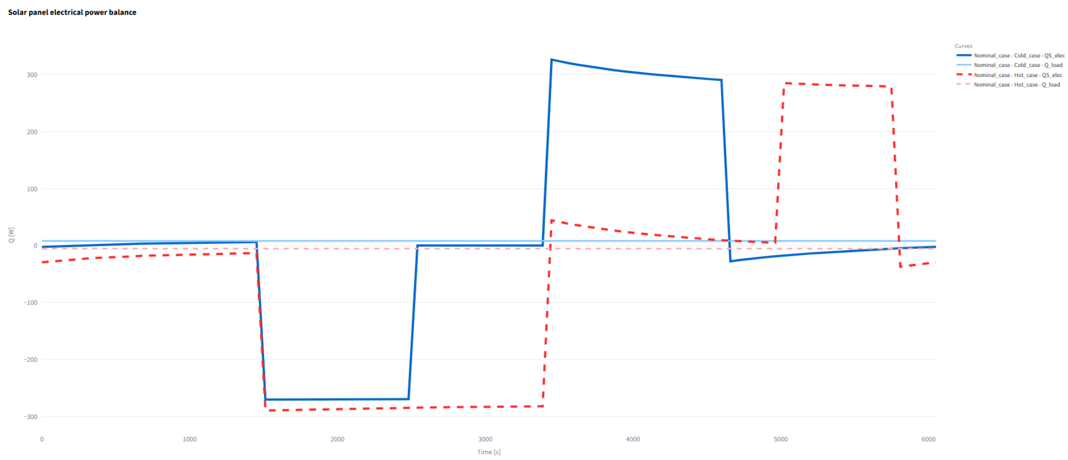


Figure 9.12: Nominal-minus-hot and nominal-minus-cold differences in estimated solar electrical generation.

The integrated results are collected in Table 9.4. The hot case generates 421.60 Wh and demands 27.73 Wh, whereas the nominal case generates 328.07 Wh and demands 18.49 Wh. The cold case has the lowest generation, 306.41 Wh, but its reduced internal loads limit the demand to 5.04 Wh. All three direct net balances remain positive over the complete interval. The direct deficits are restricted to eclipse and equal 5.92 Wh, 3.60 Wh and 1.76 Wh for the nominal, hot and cold cases, respectively. These values are evaluated before the battery restrictions are applied.

Table 9.4: Integrated solar generation and selected load demand for the nominal, hot and cold transient cases.

Case	Generated energy [Wh]	Load energy [Wh]	Direct surplus [Wh]	Direct deficit [Wh]	Net balance [Wh]
Nominal	328.07	18.49	315.50	5.92	309.58
Hot	421.60	27.73	397.47	3.60	393.86
Cold	306.41	5.04	303.14	1.76	301.37

Note: Direct surplus and deficit are the time integrals of the positive and negative parts, respectively, of $Q_{S,\mathrm{elec}} - Q_{\mathrm{load}}$, before applying the battery operating limits.

9.3.2. Battery State and Operating Limits

Battery operation is calculated independently for each physical case. Direct subtraction is not used for SOC, voltage or current because the difference between two battery states does not describe an independent operating state. The absolute SOC curves in Figure 9.13 provide the main comparison.

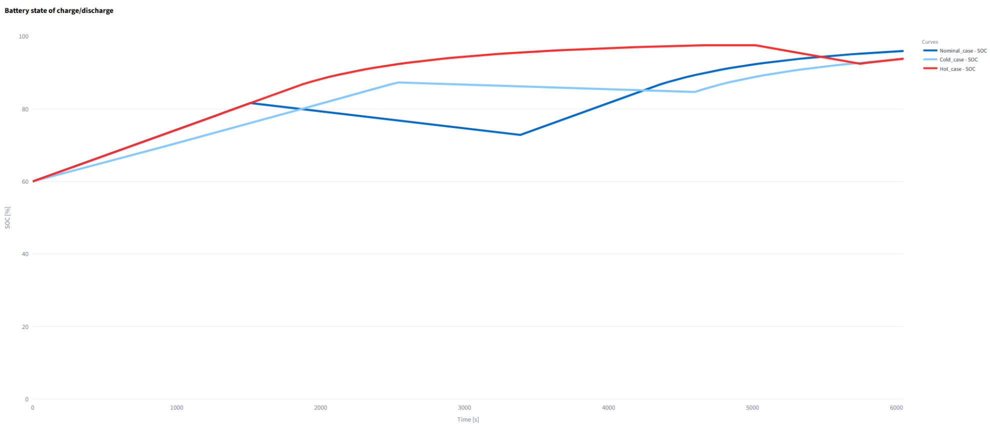


Figure 9.13: Battery state of charge obtained with the common battery configuration.

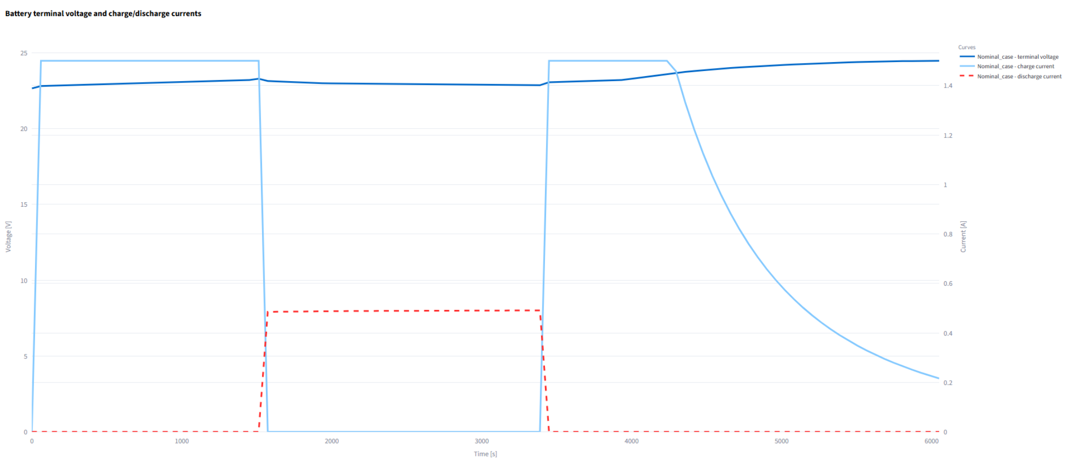
All cases begin at 60 % SOC. The nominal battery charges before eclipse, discharges while solar generation is absent and subsequently recovers to a final value of 95.98 %. The cold battery follows the same sequence with a longer discharge interval but a substantially lower demand, ending at 93.75 %. The hot battery reaches the largest maximum SOC, 97.56 %, before its late eclipse. Its higher eclipse load then causes the strongest discharge, and the remaining illuminated interval restores the SOC to 93.87 %. Final SOC therefore depends on the timing of generation and demand as well as on their total integrated values.

The battery temperatures and thermal charging restrictions are compared in Figure 9.14. Charge and discharge efficiencies remain fixed at 0.98 in all cases. The nominal and hot battery temperatures remain inside the unrestricted charging interval, so their charge-derating factors stay equal to one. The cold battery remains between 3.66 and 4.46 °C, inside the lower 5 °C derating ramp. Its factor varies from 0.733 to 0.892 and restricts charging throughout the complete 100.84 min simulation.

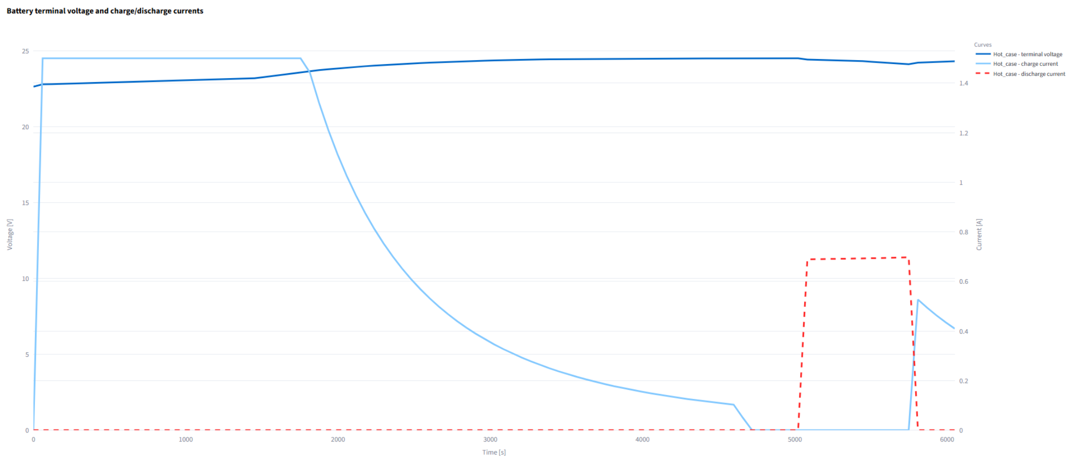
Figure 9.14: Battery temperature, charge-derating factor and electrical efficiencies in the three cases.

The active thermal derating limits the cold-case maximum charge power to 30.42 W, compared with approximately 35.4 W in the nominal and hot cases. Despite this restriction, the available generation and stored energy remain sufficient to supply the complete selected load. The unserved-load energy is negligible in every simulation.

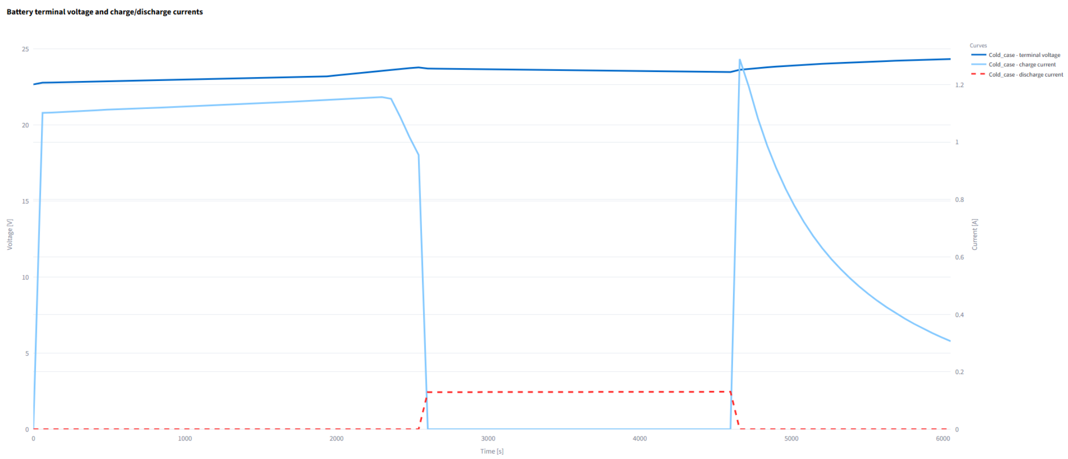
Terminal voltage and charge/discharge currents are shown separately in Figure 9.15 to avoid overlapping nine curves in a single graph. The same battery parameters and axis definitions are retained in the three subfigures.



(a) Nominal case.



(b) Hot case.



(c) Cold case.

Figure 9.15: Battery terminal voltage and charge/discharge currents for the nominal, hot and cold transient cases.

Table 9.5: Main battery results for the nominal, hot and cold transient cases using the common battery configuration.

Result	Nominal	Hot	Cold	Unit
Final state of charge	95.98	93.87	93.75	%
Minimum state of charge	60.00	60.00	60.00	%
Maximum state of charge	95.98	97.56	93.75	%
Maximum charge power	35.39	35.40	30.42	W
Maximum discharge power	11.22	16.84	3.06	W
Battery-temperature range	18.40–20.90	35.50–38.20	3.66–4.46	°C
Minimum charge-derating factor	1.000	1.000	0.733	–
Charge-derated duration	0.00	0.00	100.84	min
Curtailed solar energy	284.71	370.56	278.26	Wh
Unserved-load energy	≈ 0	≈ 0	≈ 0	Wh

The nominal voltage ranges from 22.65 to 24.46 V, with maximum charge and discharge currents of 1.50 and 0.49 A, respectively. The hot case reaches 24.52 V and presents the largest discharge current, 0.70 A, consistently with its 16.84 W maximum discharge power. The cold case reaches 24.31 V, while its maximum charge and discharge currents are 1.29 and 0.13 A, respectively. The reduced cold-case charge current is consistent with the active thermal derating, whereas its smaller discharge current follows from the lower selected load.

The summary confirms that no case reaches the lower SOC limit or leaves electrical demand unserved. The hot case produces the largest curtailed solar energy, 370.56 Wh, because its high generation frequently exceeds the charging capability of the battery. The nominal and cold values are 284.71 and 278.26 Wh, respectively. The nominal battery is unrestricted thermally, whereas the cold battery operates with a reduced charge capability because of the active thermal derating.

9.4. Overall Comparison

The selected cases provide contrasting hot and cold thermal environments without changing the common model architecture. The hot case produces the highest equipment temperatures and the largest OBC–radiator interface difference, while the cold case produces the lowest temperatures and activates the battery charge derating. The nominal solution generally remains between both case responses, although differences in eclipse timing can temporarily reverse the ordering of solar-panel temperatures, loads and heat flows.

The transient comparisons also show that absolute temperature, temporal variation and spatial non-uniformity are separate results. The hot case has the highest temperatures but the smallest 30 min variation of `BODY_PANEL`; the cold battery has the lowest temperature but also the smallest internal battery range. Interface and flow results similarly depend on the local connection and can change node pair, magnitude or direction between cases.

The electrical response is governed by both illumination and temperature. Higher panel temperature reduces instantaneous conversion efficiency, whereas eclipse duration controls the total generated energy and the discharge interval. The hot case consequently generates the most energy, while its higher load and late eclipse produce the strongest discharge near the end of the simulation. The cold case receives the least solar energy and remains thermally derated, although its reduced demand prevents unserved load. Both cases finish above 93 % SOC, with the cold case reaching the lowest final value by a small margin.

The combined results confirm the value of retaining both superposed curves and *Case difference*. Absolute representations preserve the operating levels required to assess temperatures, flows and battery limits, while the difference mode quantifies the separation from the nominal solution and makes the influence of non-coincident orbital events directly visible.

10. Conclusions and Future Work

10.1. Conclusions

The developed application meets the main objective of processing, analysing and comparing several transient cases associated with the same ESATAN-TMS thermal model. The initial steady-state tool has been extended without removing its previous inspection capabilities, providing a common environment for examining complete time series and results at selected instants.

A representative spacecraft thermal model has been developed to support the implementation and evaluation of these functions. The nominal, hot and cold cases retain the same geometry, node hierarchy, materials and thermal connections, while the environmental conditions and internal dissipations are modified. This common basis allows equivalent groups, nodes and exported result fields to be compared directly throughout the simulated interval.

The application can load several compatible result files, preserve their case identity and apply the same selection to all of them. Transient temperatures, thermal loads, conductive and radiative heat flows, interface temperature differences and time-integrated energy transfers can be represented using superposed curves. The *Case difference* option complements these absolute results by calculating the separation between a reference case and one or more comparison cases. Together, both representations preserve the physical operating levels and make smaller differences or non-coincident orbital events easier to identify.

Additional calculations extend the imported ESATAN-TMS results. Temperature changes can be evaluated over selected time windows, critical interfaces are identified from explicit thermal connections and summary tables condense the main extrema and their occurrence times. Analyses and their configurations can also be saved, restored and exported, reducing the need to repeat the complete selection process when the same result must be examined again.

The thermal post-processing is complemented by an electrical analysis based on the selected solar-panel and internal-load results. The absorbed solar load is used to estimate incident and converted electrical power, while the selected internal dissipations provide an approximation of the electrical load demand. These profiles supply a separate battery model that calculates state of charge, terminal voltage, current, accepted and curtailed power, conversion and resistive losses, and temperature-dependent operating restrictions. All these results are obtained during post-processing and do not modify the original thermal solution.

The comparison of the three cases confirms that the highest absolute temperature does not necessarily coincide with the largest temporal variation or the greatest local gradient. The hot case produces the highest equipment temperatures and the largest OBC–radiator interface difference, whereas the cold case produces the lowest battery temperature and activates the charging derating. Differences in eclipse timing also affect solar generation, load support and the final battery state.

The hot case generates the greatest electrical energy and experiences the strongest final discharge because of its late eclipse and higher demand. The cold case finishes with the lowest state of charge by a small margin, despite its lower demand, while remaining thermally restricted throughout the simulation. None of the cases produces unserved load. These results show that battery operation cannot be determined from the generated or demanded energy alone, since their temporal distribution and the active operating limits are also relevant.

The resulting tool therefore connects transient thermal inspection, multi-case comparison and a simplified electrical interpretation within the same workflow. It does not replace the thermal solver or constitute a complete spacecraft electrical-power-system model. Its purpose is to organise the exported results, reproduce consistent analyses and reveal the relations between environmental conditions, thermal response, electrical generation and battery operation. The modular structure also provides a suitable basis for further extensions without requiring the existing analysis areas to be redesigned.

10.2. Future Work

The following lines extend the current physical models, analysis capabilities and software architecture without changing the scope of the completed implementation.

- **Battery-model extension.** The current implementation is a configurable post-processing approximation based on a generic piecewise voltage curve, separate charge and discharge resistances and a simplified CC–CV strategy. Greater physical fidelity would require voltage–SOC curves calibrated for a specific cell, different charge and discharge characteristics, temperature-dependent resistance and usable capacity, hysteresis and dynamic RC branches. Ageing, cycle-dependent capacity loss and validation against experimental or manufacturer data would also allow the calculated voltage, current and state of charge to be evaluated for a defined battery technology.

- **Bidirectional thermal–electrical coupling.** The thermal and electrical calculations currently follow a one-directional sequence: thermal results determine solar generation, load demand and battery operation, but the calculated electrical losses are not returned to the thermal model. A bidirectional procedure could convert battery resistive losses, conversion losses and operating-mode changes into time-dependent heat loads for a subsequent ESATAN-TMS solution. An iterative exchange between both models would provide a more consistent thermo-electrical response without requiring the post-processing application to replace the thermal solver.

- **Comparison between different thermal-model architectures.** Multi-case comparison is presently restricted to result files that share a common node hierarchy. A further extension could compare successive versions or different architectures of the same thermal model by defining correspondences between equivalent groups and components. The comparison would then identify common elements, warn about unmatched regions and normalise selected results by properties such as area, mass or thermal capacitance when a direct nodal subtraction is not meaningful.

- **Alignment by orbital events.** Temporal alignment by eclipse entry, eclipse exit or orbital phase would permit comparisons between simulations with different starting instants or output-time vectors. This option would separate the effect of the physical event from a displacement in the simulation timeline and facilitate the comparison of cases with different eclipse durations.

- **Sensitivity and uncertainty analysis.** The existing multi-case framework provides a basis for sensitivity and uncertainty studies. Sets of cases could be generated by varying contact conductances, thermo-optical properties, internal dissipations, orbital conditions or electrical parameters. Automated envelopes, uncertainty bands and rankings would identify the inputs with the greatest influence on each component.

- **Automatic critical-event detection.** The same process could detect temperature-limit violations, maximum case differences, rapid thermal variations, changes in heat-flow direction, minimum state of charge, active derating and periods of unserved load. A common event table could then identify the affected case, element, time and result without requiring every curve to be inspected individually.

- **Performance and scalability.** Larger thermal models and broader collections of transient cases would benefit from selective loading of required result fields, deferred processing, reuse of intermediate calculations and parallel execution between cases. These improvements could reduce memory use and response time when processing long simulations, finer temporal discretisations or a large number of result files.

- **Batch processing and automatic report generation.** A batch mode could apply a saved analysis configuration to an entire folder of cases and export the corresponding plots, tables and numerical summaries. Report templates for LaTeX, PDF or spreadsheet formats would extend this process towards reproducible and partially automated result documentation.

- **Persistent-storage improvements.** Metadata, widget configurations and generated files are currently coordinated between a local database and output folders. A unified transactional layer could prevent incomplete records, detect orphaned files and avoid unnecessary duplication. Portable analysis packages would allow complete configurations to be exported, transferred and restored on another installation. Version information and migration rules would preserve compatibility when the application structure or available controls change.

- **Dedicated web architecture.** The interface may evolve from Streamlit towards a separate frontend based on HTML, CSS and JavaScript, connected to a Python backend through an application programming interface. This architecture would offer greater control over layout, asynchronous calculations, user sessions and remote deployment while retaining the calculation modules as an independent backend.

- **Automated testing and validation.** Unit, integration and regression tests could check file loading, sign conventions, numerical integration, multi-case differences, battery calculations and saved-analysis restoration after each software change. These tests would also provide a stable basis for introducing new result formats, analysis modules and physical models.

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Evolution of the Inherited Inspection Modules

The inherited inspection modules retain their original purpose while using the shared hierarchy, case collection and transient controls of the final application. Equivalent physical selections are shown whenever both versions are compared. The differences therefore correspond to the interface and its operating logic rather than to unrelated thermal results.

The comparison follows the order of the common tabs. Each block first identifies the original inspection task and then shows how the same task is integrated into the transient and multi-case workflow. The figures are intentionally presented at a large scale because the location of the selectors, case controls and result areas forms part of the comparison.

A.1. Flow Explorer

The original Flow Explorer processes one active dataset and provides a direct source–destination heat-flow query. The source group, destination group, thermal quantity and unit are selected before the result is generated. This arrangement is sufficient when only one steady-state dataset is available, because neither the case identity nor the output instant needs to be selected explicitly.

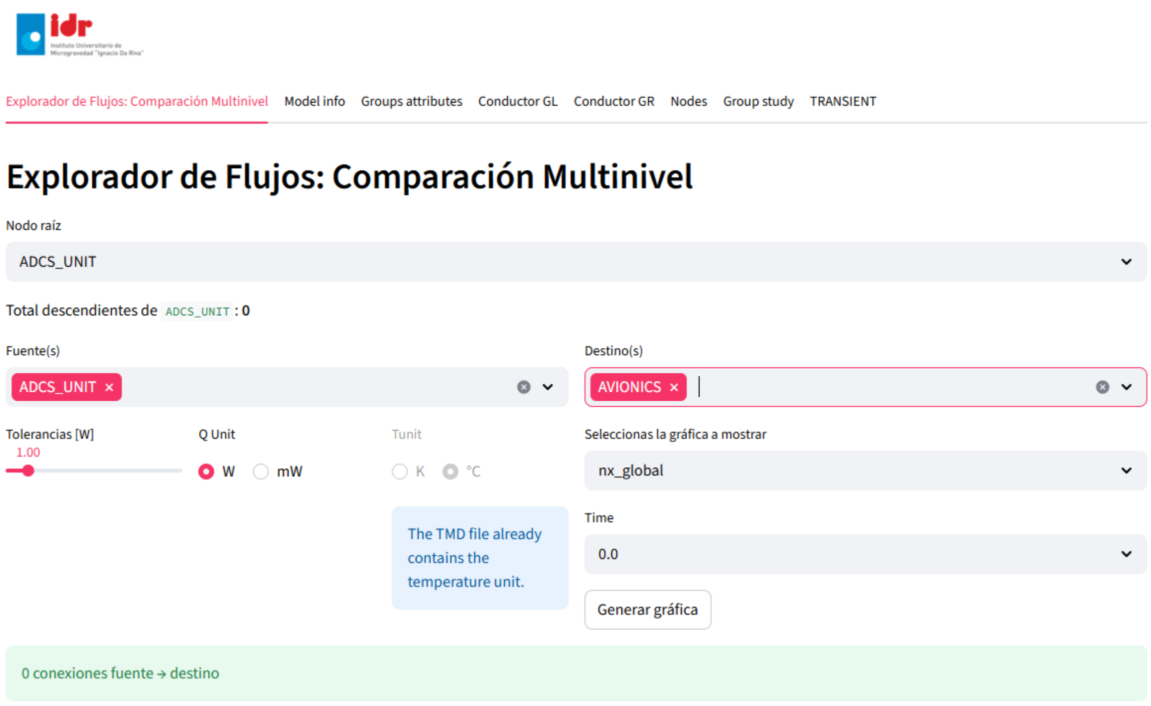


Figure A.1: Original Flow Explorer interface.

The original layout already defines the physical workflow retained by the final application: a source element is related to a destination element and the corresponding exchanged quantity is represented.

The developed Flow Explorer retains this sequence while making the complete analysis context visible before the result is generated. Several compatible cases can be selected, transient results are evaluated at a chosen output time and the representation can show either the absolute values or the difference between a reference case and one or more comparison cases.

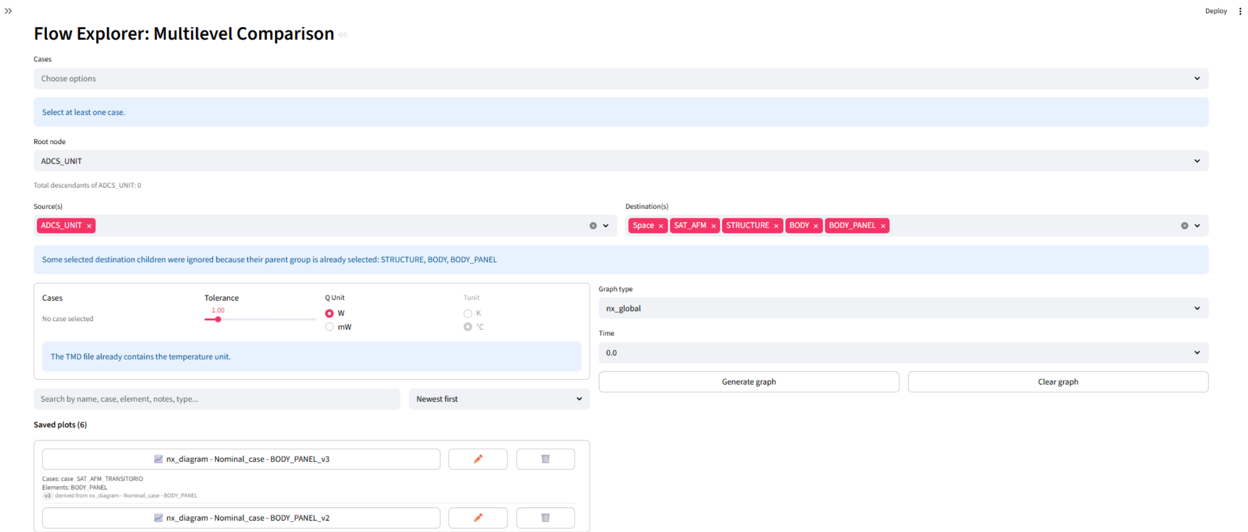


Figure A.2: Developed Flow Explorer interface with transient and multi-case controls.

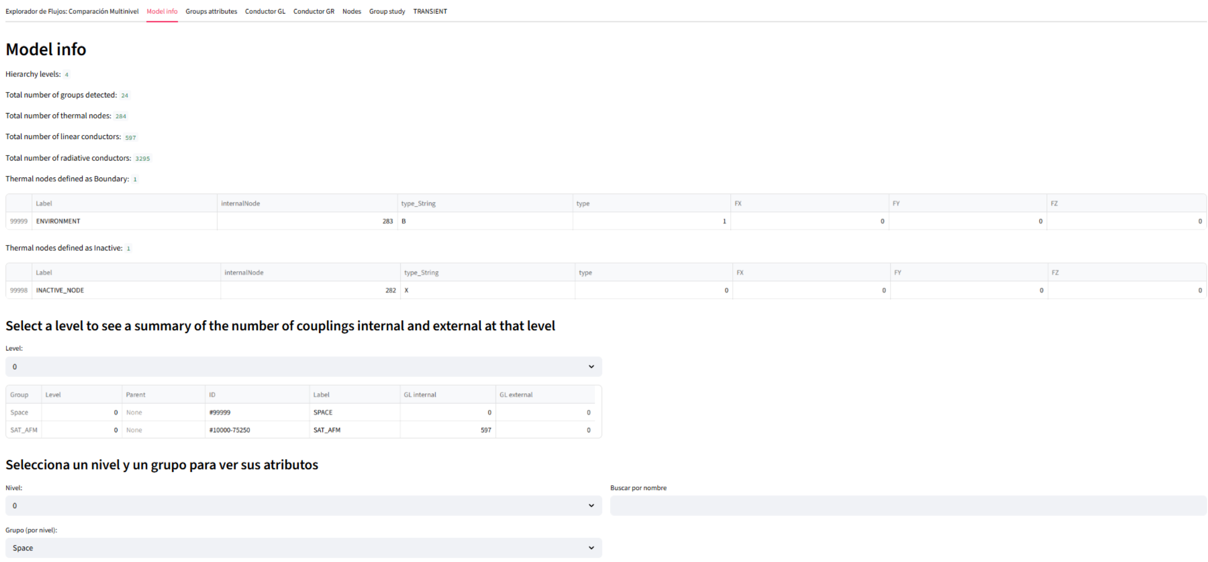
The model hierarchy is also used to order the available groups and to remove redundant descendants when a parent branch is already selected. This prevents the same part of the model from being included twice in an aggregated result. The generated representation can subsequently be exported or stored together with the case set, physical selection, units, time and graphical options.

Although the developed tab contains more controls, the original source–destination structure remains recognisable. Case management and transient evaluation are placed around the inherited query rather than being implemented as a separate analysis procedure.

A.2. Case Information

The original *Model info* tab reports the structural and coupling data of the active result file. The developed *Case info* tab retains these summaries for every loaded case, preserves the visible case name and adds level-based coupling inspection at a selected output time.

Original version



Developed version

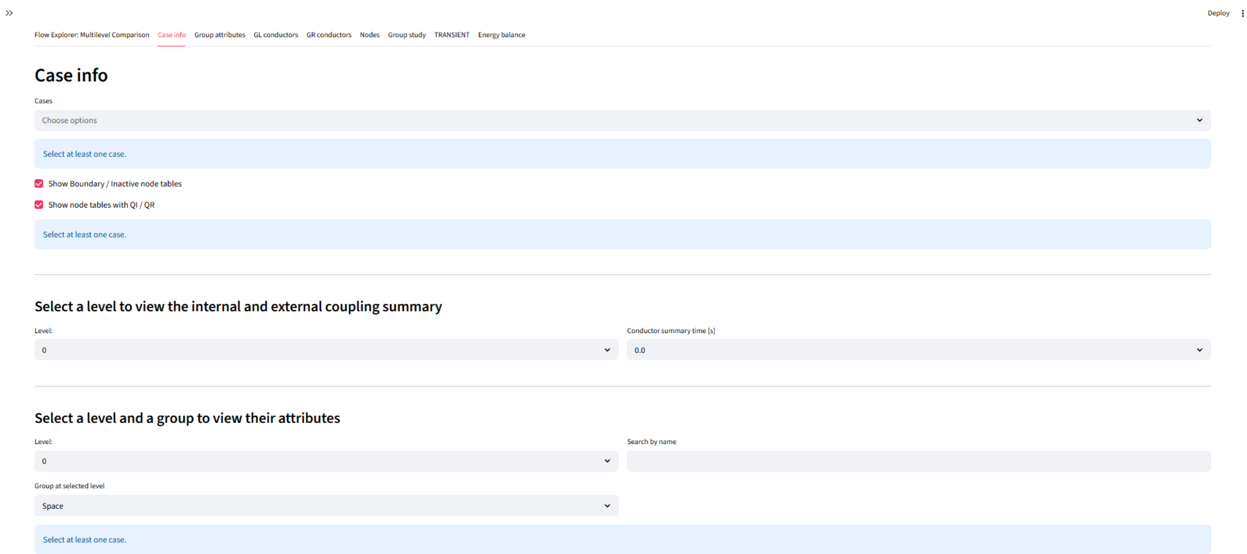


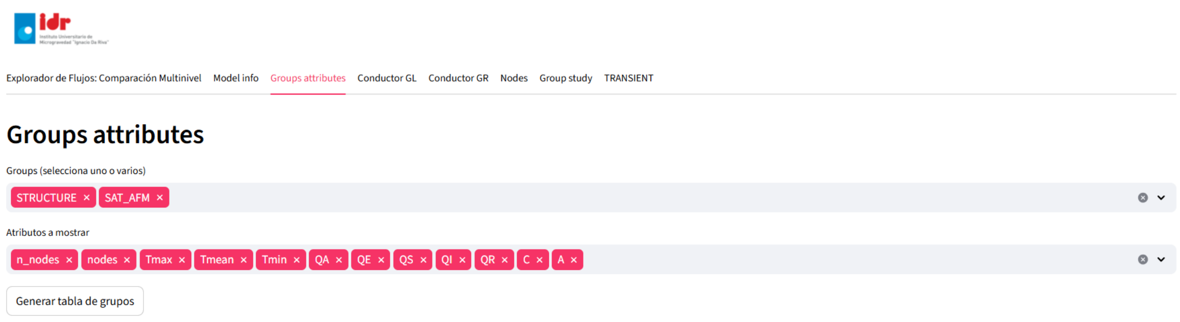
Figure A.3: Evolution from active-model information to case-specific inspection.

Both versions preserve access to the model size, hierarchy levels and coupling summaries. The final interface removes the dependence on a single active file: the requested information is associated with the selected case, while the common hierarchy continues to provide the engineering structure used to interpret the identifiers. The temporal selector is only relevant when the reported values depend on the transient output instant.

A.3. Group Attributes

The original tab obtains the requested group attributes from one active case and, for transient datasets, from the first available result row. The developed module separates case selection, group selection and attribute selection, then evaluates the request at the chosen output time. When several cases are active, the same physical and numerical selection is applied to each of them.

Original version



Developed version

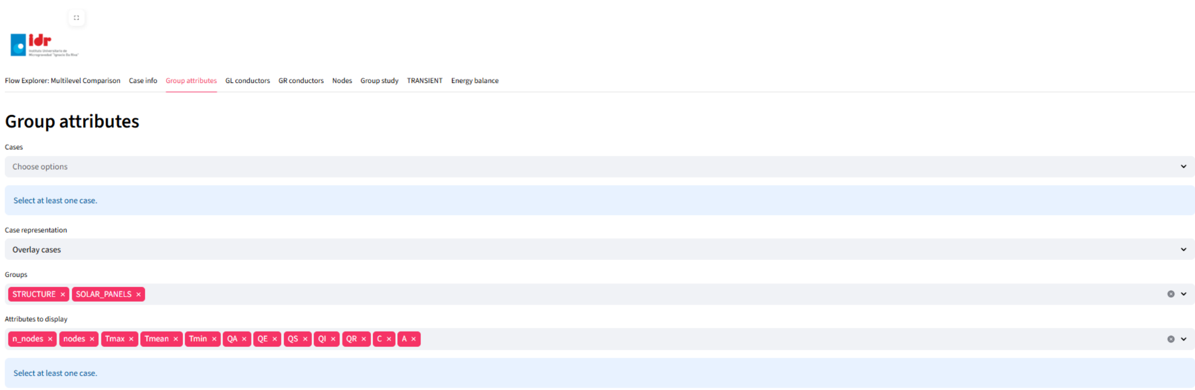


Figure A.4: Evolution of the group-attribute inspection area.

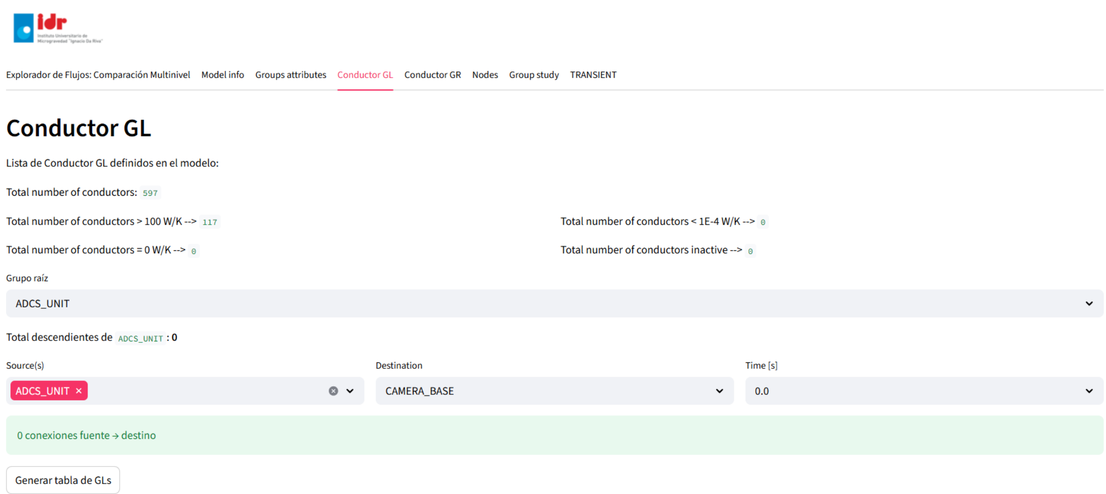
The resulting tables remain case-specific, which avoids merging values that belong to different simulations. A difference representation can instead be requested explicitly when the compared attributes are compatible. The final layout therefore preserves the original group-based inspection while making the case and temporal basis of every table unambiguous.

A.4. Linear and Radiative Conductors

The conductor inspection remains divided into linear and radiative matrices. This separation is important because GL and GR describe different coupling terms even when they connect the same physical groups. The developed modules apply a common source–target selection to the loaded cases and evaluate the corresponding values at the selected output time.

Linear conductors. The GL tab retains the original source and destination selectors while adding case selection, explicit time control and a signed source–target table. The number of detected couplings and their total conductance are reported together with the detailed entries.

Original GL module



Developed GL module

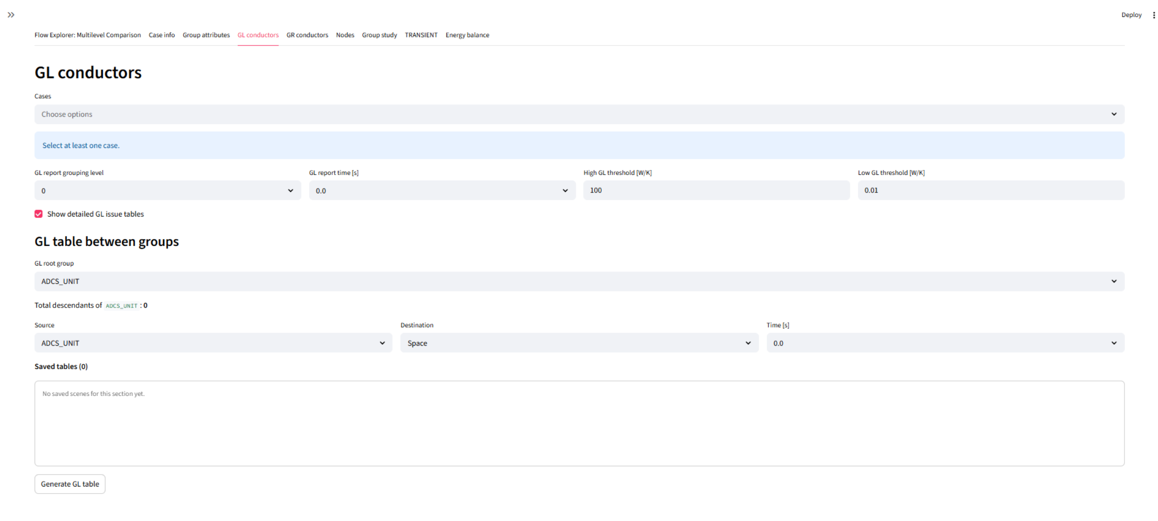
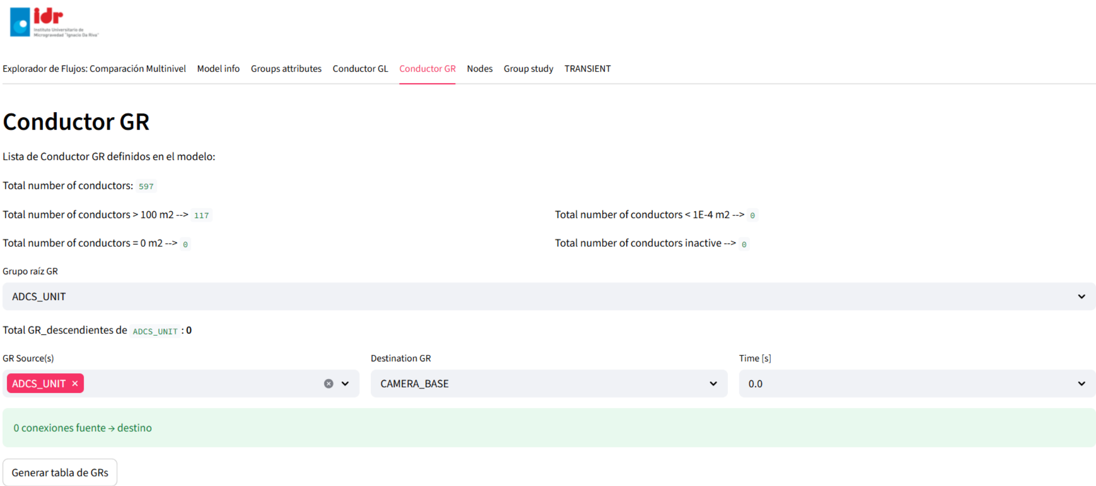


Figure A.5: Evolution of the linear-conductor inspection area.

The table can be exported and its selection can be restored later. These functions do not change the underlying GL data; they preserve the context required to reproduce the same inspection for one or several cases.

Radiative conductors. The GR module follows the same organisation, but the radiative coefficients remain independent from the linear conductances. The common arrangement allows the user to move between GL and GR inspection without redefining the case, hierarchy or temporal context.

Original GR module



Developed GR module

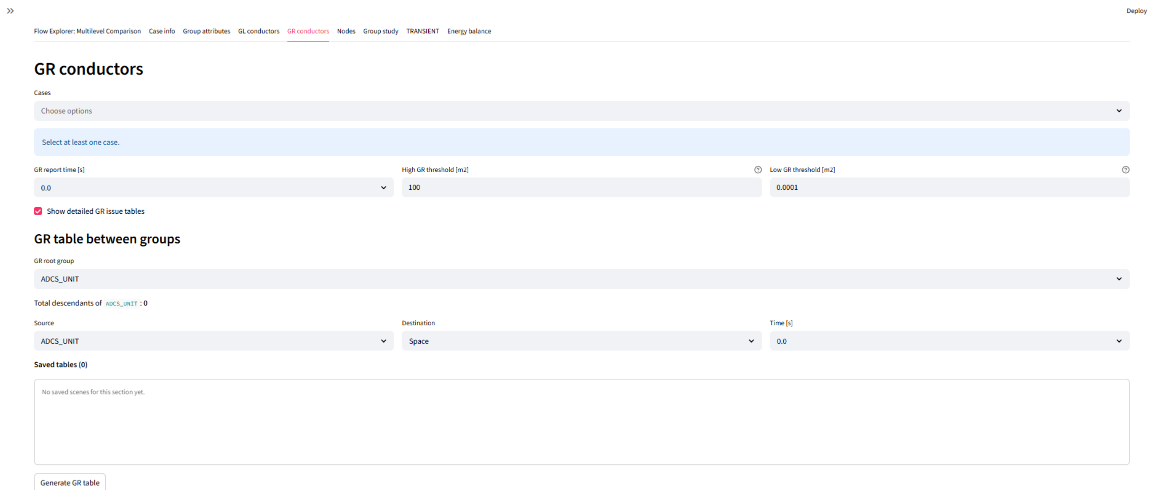


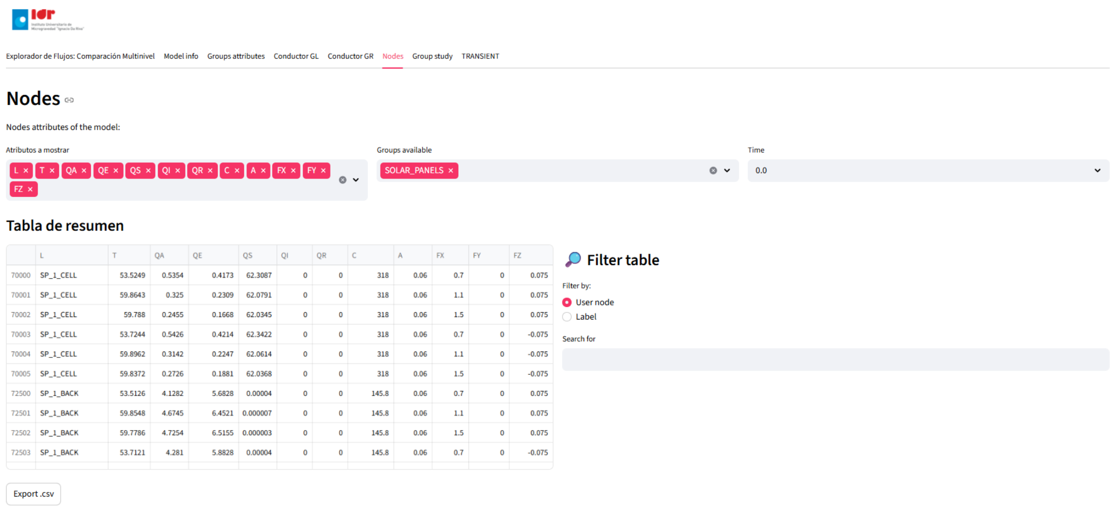
Figure A.6: Evolution of the radiative-conductor inspection area.

The final GL and GR tabs therefore provide parallel procedures while preserving the physical distinction between both matrices. Each exported table identifies the corresponding coupling type and the selected direction between groups.

A.5. Nodes and Group Study

The nodal table retains attribute and group selection while adding an explicit filter action, output-time selection and case-specific or difference tables. The filtering controls are separated from the imported result table, so the visible subset can be changed without modifying the stored data.

Original Nodes module



Developed Nodes module

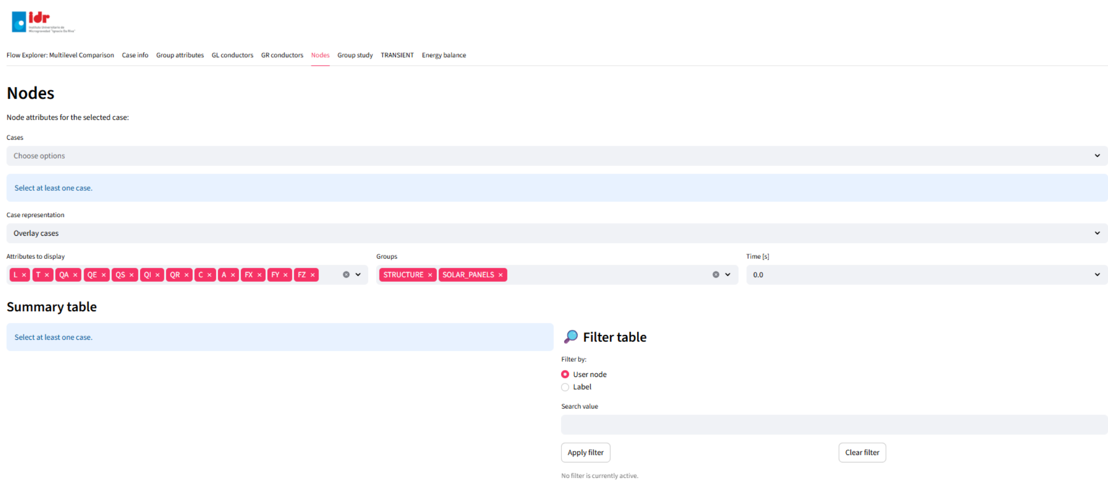
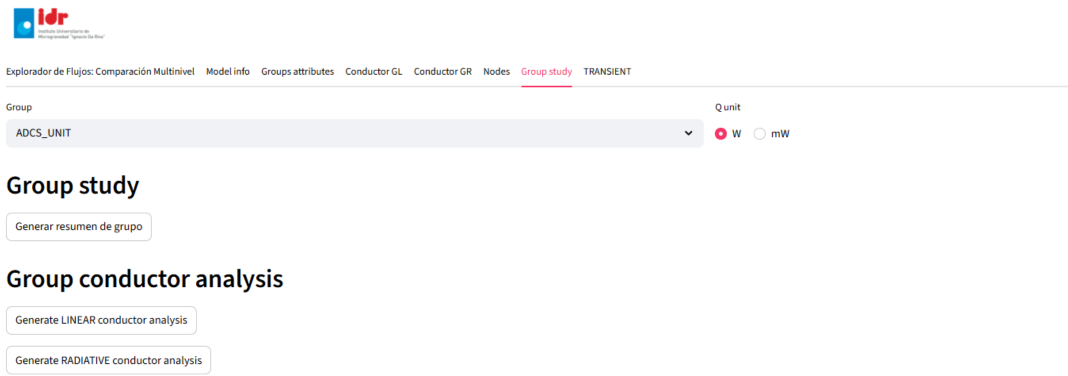


Figure A.7: Evolution of the nodal-inspection interface.

The developed version can apply the same nodal request to several compatible cases and can export the resulting table. The explicit application of the filter also makes the visible selection reproducible when the analysis is restored.

The *Group study* tab combines a component summary with the inspection of internal and external GL and GR connections. The original interface already organises these operations around one selected group. The developed version retains this component-centred structure while introducing the shared case, time and export controls used by the rest of the application.

Original Group Study module



Developed Group Study module

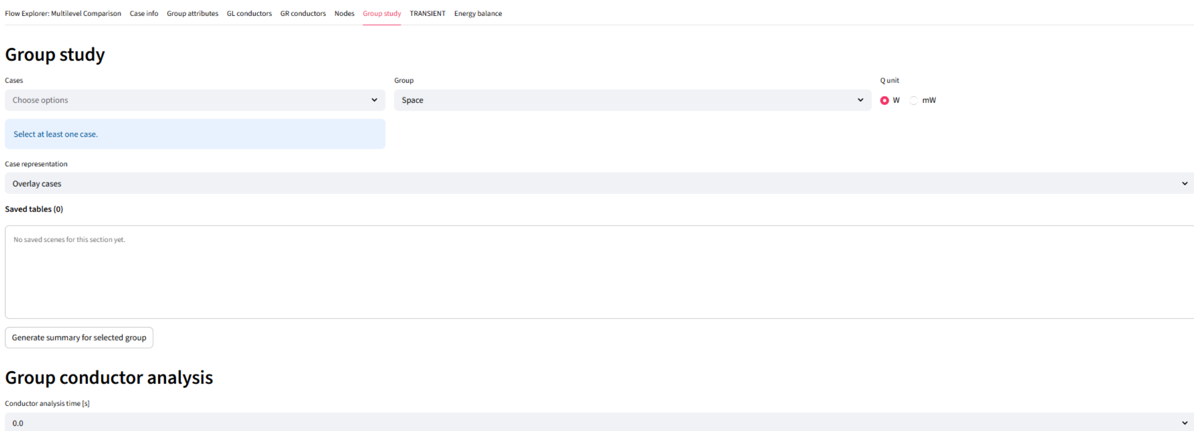


Figure A.8: Evolution of the group-study interface.

The component summary and the conductor analyses can be generated and restored independently. They nevertheless use the same loaded cases and hierarchy, which keeps the selected component consistent throughout the complete inspection.

A.6. Summary of the Inherited Modules

Table A.1 condenses the adaptations shown in the preceding figures. The initial inspection functions remain available, while the final implementation extends their case organisation, transient evaluation, selection consistency and reproducibility. The practical effect column distinguishes the visible interface change from its consequence during repeated analysis.

Table A.1: Adaptation of the inherited inspection modules.

Module	**Initial function**	**Final adaptation**
Flow Explorer	Hierarchical heat-flow inspection for one active dataset.	Hierarchy-ordered selection, removal of redundant descendants, several cases, selected output time, overlay or difference representation, export and restoration.
Case information	Structural and coupling data from the active model.	Independent summaries, visible case names and level-based coupling inspection at a selected time.
Group attributes	Attribute table from the first or only stored result row.	Time-dependent tables for several cases and reference-minus-comparison results.
GL conductors	Conductive-coupling inspection for the active case.	Signed source–target table, selected time, coupling count, total GL, export and restoration.
GR conductors	Radiative-coupling inspection for the active case.	Signed source–target table, selected time, coupling count, total GR, export and restoration.
Nodes	Nodal table with immediate text filtering.	Multi-case and time-dependent tables, difference mode, explicit filtering and export.
Group study	Component summary and internal or external conductor inspection.	Common case and time controls, case-specific tables and independently restorable GL and GR analyses.

The inherited inspection scope is therefore retained within a common transient and multi-case workflow, allowing equivalent selections to be applied, compared and restored consistently.

Secondary Implementation Improvements

Several adaptations introduced during the development are secondary to the transient and electrical methods described in the main chapters. They improve selection consistency, temporal evaluation, conductor inspection, user feedback, persistent storage and code organisation. The following sections compare the relevant behaviour of the initial and final implementations and document only functions present in the final application.

B.1. Comparison of Initial and Final Behaviour

Table B.1 summarises the selected adaptations. The comparison is based on the operation of both application versions and on the final source-code organisation. It is not intended as a complete change log, since the transient and electrical developments are already described in Chapters 6, 7 and 8.

Table B.1: Comparison of selected secondary implementation improvements.

Initial behaviour	**Final behaviour**	**Main location**
Group selectors were not consistently ordered according to the model structure.	Available groups follow the parent–child order defined by the Excel hierarchy, with fallback ordering for malformed or disconnected branches.	Common hierarchy-selection helpers and inherited group selectors.
Parent and descendant groups could be selected together and counted twice in aggregated results.	Descendants already included in a selected parent are removed and a visible message identifies the corrected selection.	Flow Explorer and compatible hierarchical selectors.
Several inherited modules evaluated only the active or first stored output state.	The selected value from the result-file time vector is applied to each compatible calculation and retained when the analysis is restored.	Flow Explorer, Group Attributes, GL/GR Conductors, Nodes and Group Study.
GL and GR inspection focused on individual source–target tables.	The final tabs also provide matrix-quality summaries, configurable high and low thresholds and detailed issue tables.	GL Conductors and GR Conductors.
Nodal filtering was applied while the filter text was being entered.	Separate *Apply filter* and *Clear filter* controls preserve an explicit active-filter state.	Nodes.

Table B.1 – continued

Initial behaviour	**Final behaviour**	**Main location**
Interface candidates could be selected from hierarchy groups without confirming an explicit thermal connection.	GL and GR definitions are scanned to identify connected target groups and node pairs before the interface calculation.	Transient interface analysis.
Temperature changes over prescribed time intervals were not available as a complete analysis area.	Maximum and minimum temperature changes are calculated over selectable 10-, 15-, 30- and 60-minute windows and summarised by group.	Transient temperature-variation analysis.
Validation and feedback differed between individual interface blocks.	Missing cases, empty selections and unavailable data produce visible messages before the calculation continues.	All final analysis modules.
Saving logic was distributed among individual output areas.	Saved analyses share a common database, file store, restoration recipe and editing workflow.	Scene-management and restoration modules.
Most interface areas were concentrated in the main application script.	The launcher, analysis tabs, shared helpers and persistence functions are separated into dedicated modules.	Application architecture.

B.2. Hierarchy Safeguards and Time-Dependent Inspection

Hierarchy-aware selection prevents the same physical branch from being included more than once in an aggregated result. Figure B.1 shows the message produced when a parent group and one or more of its descendants are selected simultaneously. The descendants are removed because their nodes are already contained in the parent selection.

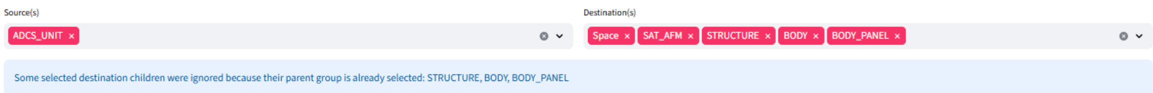


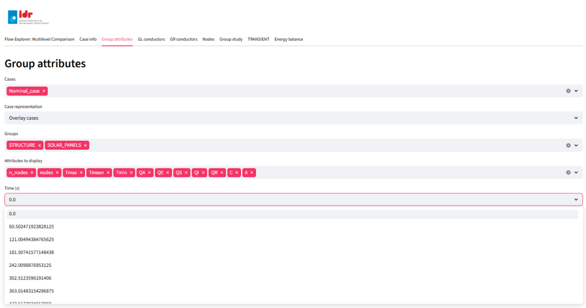
Figure B.1: Automatic removal of redundant descendant groups.

The inherited modules use the output-time vector stored in each transient result file. The selected instant is evaluated independently in each tab so that its configuration can be saved and restored together with the remaining controls. Table B.2 identifies the result evaluated at that instant.

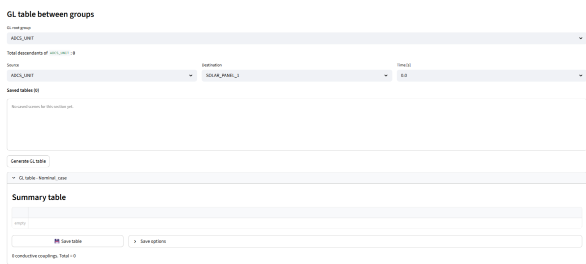
Table B.2: Use of the selected output time in the inherited modules.

Module	**Time-dependent output**	**Stored setting**
Flow Explorer	Instantaneous source–target heat flows used by the selected graph.	Output time, groups, units, tolerance and graph type.
Group Attributes	Group temperatures, thermal loads and selected metadata.	Output time, cases, groups, attributes and representation mode.
GL Conductors	Linear-conductor values between the selected source and target.	Output time, groups and case selection.
GR Conductors	Radiative-conductor values between the selected source and target.	Output time, groups and case selection.
Nodes	Nodal temperatures, loads and selected node attributes.	Output time, cases, groups, attributes and active filter.
Group Study	Group summary and internal or external conductor values.	Output time, case, group, conductor type and units.

Figure B.2 provides two representative examples. The Group Attributes selector is populated from the transient result vector, whereas the GL tab uses the chosen time when generating the conductor table. Equivalent controls are used by the remaining compatible modules.



(a) Group Attributes.



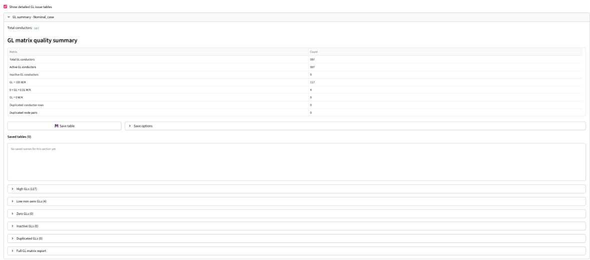
(b) GL Conductors.

Figure B.2: Representative output-time controls in inherited modules.

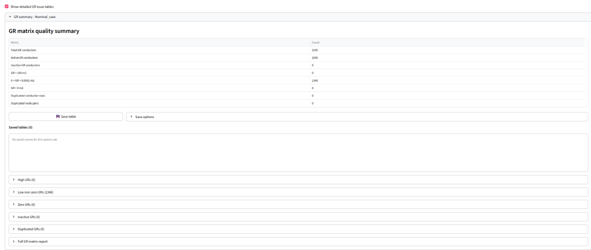
B.3. Conductor Inspection and Interface Detection

The GL and GR tabs include matrix-level diagnostics in addition to the source–target tables described in Appendix A. The summaries report the number of couplings and classify values according to configurable high and low thresholds. Detailed issue tables can then isolate the affected pairs and be saved independently.

The transient interface analysis uses the same explicit connection data to restrict the destination selector when the conductor filter is active. Only node pairs connected through GL or GR definitions are evaluated. The hierarchy remains available as a fallback when no explicit target is detected.



(a) GL matrix summary.



(b) GR matrix summary.

Figure B.3: Matrix-quality summaries available in the conductor tabs.

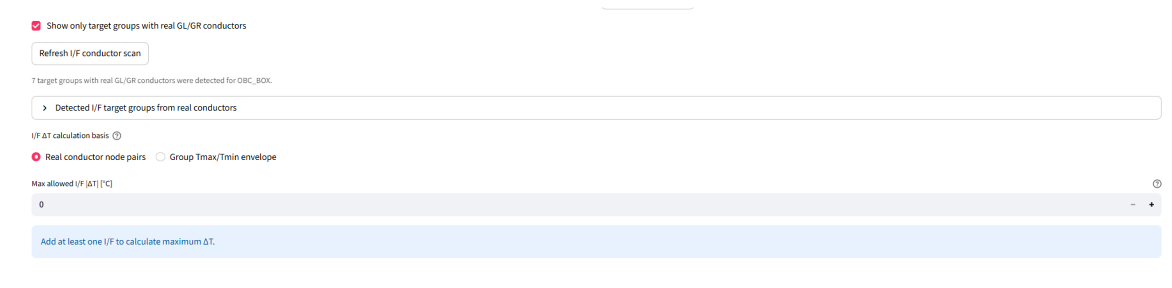


Figure B.4: Conductor-based filtering of candidate interface groups.

Table B.3 provides a numerical example for `OBC_BOX`. The target groups and pair counts are obtained from the same connection scan used to populate the interface selector. The table documents the connection basis applied before the temperature differences are calculated.

Table B.3: Target groups detected from the explicit GL and GR connections to `OBC_BOX` in the nominal case.

Target group	GL pairs	$\sum \lvert GL \rvert$ [W/K]	GR pairs	$\sum \lvert GR \rvert$ [m^2]	Total pairs
`BODY`	4	22.68	559	0.355732	563
`RADIATOR`	4	0.40	181	0.020584	185
`BODY_PLATE_MED`	4	22.68	96	0.115546	100
`BODY_PANEL`	0	0.00	364	0.147134	364
`BODY_PLATE_SUP`	0	0.00	91	0.092968	91
`Space`	0	0.00	24	0.001042	24
`BODY_PLATE_INF`	0	0.00	8	0.000083	8

B.4. Input Validation and User Feedback

The final interface checks the information required by each calculation before processing begins. Missing cases, empty group selections, unsupported combinations and unavailable conductor data produce visible messages instead of incomplete outputs or uncaught errors. Figure B.5 shows the response when no case has been selected. Equivalent checks are applied locally by the remaining modules.

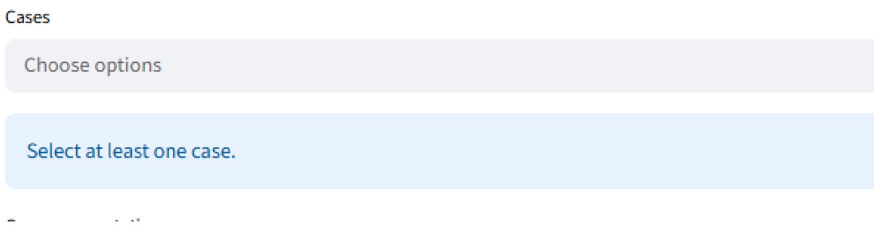


Figure B.5: Validation message displayed when no case is selected.

B.5. Persistent Analysis Storage

The saving workflow described in Section 6.5 uses an SQLite database together with a structured file store. The database relates each saved analysis to its model, cases, source tab, model element and stored recipe. The file directory contains the generated representations and a JSON description of the settings required for restoration.

A saved entry follows the general structure below. The exact intermediate folders depend on whether the output belongs to one case, several cases or a case-difference comparison.

```
scene_store/
+-- <model>/
    +-- <scope_and_cases>/
        +-- <element_type>_<element>/
            +-- scene_<timestamp>_<tab>_<name>/
                +-- scene.json
                +-- figure.plotly.json
                +-- figure.html
                +-- table.csv
                +-- table.xlsx
                +-- figure.png
                +-- figure.svg
```

Only the files associated with the generated output are created. Plotly figures produce JSON and HTML files, data tables produce CSV and spreadsheet files, and Matplotlib figures produce PNG and SVG images. The metadata file is always written.

Consistency checks remove database entries whose folders no longer exist and scene folders that are no longer referenced by the database. An auxiliary cleanup script creates backups before removing

Table B.4: Files and records associated with a saved analysis.

Stored item	Purpose
SQLite model, case and analysis records	Relate the saved output to its source files, visible case names, element, tab and creation metadata.
Recipe JSON in the database	Stores widget values required to restore the analysis configuration.
`scene.json`	Provides a readable file copy of the saved metadata and recipe.
`figure.plotly.json` and `figure.html`	Preserve an interactive Plotly figure and provide an independent browser representation.
`table.csv` and `table.xlsx`	Preserve a generated numerical table in two conventional formats.
`figure.png` and `figure.svg`	Preserve a Matplotlib figure for reports and scalable editing.

saved analyses unrelated to a defined set of retained model files. These functions reduce stale references without changing the original `.TMD` or hierarchy files.

B.6. Source-Code Reorganisation

The initial interface concentrates the inherited tabs in a single application script. The final organisation separates each analysis area from the launcher and isolates saving and restoration functions in dedicated modules. This structure allows one tab to be modified without rewriting the complete interface and supports the addition of the transient and electrical modules.

Table B.5: Correspondence between the initial interface blocks and the final modules.

Initial block	Final module
Flow Explorer block in `app_interactiva.py`	`tabs/tab_flow_explorer.py`
Model-information block	`tabs/tab_case_info.py`
Group-attribute block	`tabs/tab_group_attributes.py`
GL and GR blocks	`tabs/tab_gl_conductors.py` and `tabs/tab_gr_conductors.py`
Nodes and group-study blocks	`tabs/tab_nodes.py` and `tabs/tab_group_study.py`
Initial transient prototype	`tabs/tab_transient.py`
Solar and battery calculations	`tabs/tab_sp.py`
Saving and restoration functions	`app_thermalviewer/scene_manager.py`, `scene_ui.py`, `restored_scenes.py` and `scene_editor.py`
Common widget, hierarchy and export routines	`app_thermalviewer/app_helpers.py` and `viewer_utils.py`

The reorganisation changes only the software structure; all modules continue to share the same imported case and hierarchy objects.